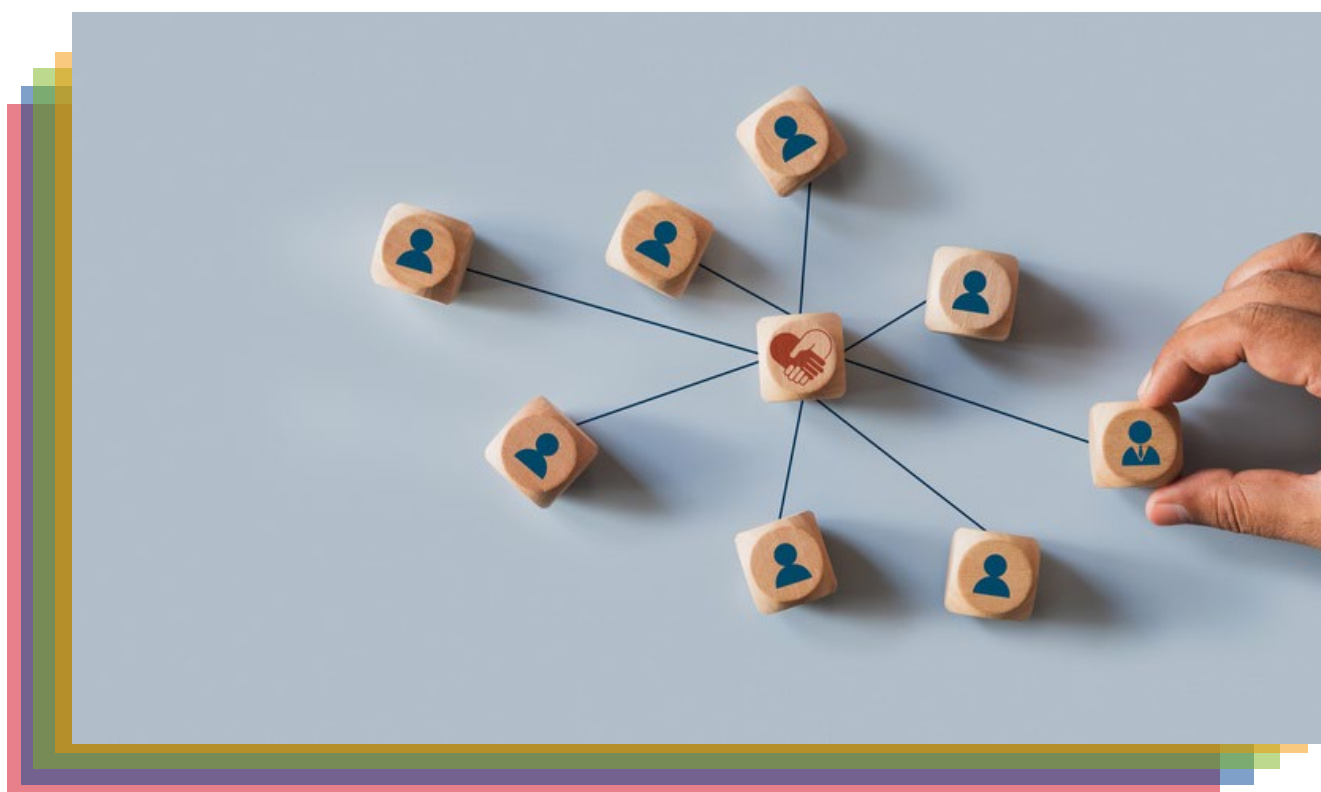


MANAGEMENT OF MYALGIC ENCEPHALOMYELITIS/CHRONIC FATIGUE SYNDROME IN BELGIUM: AN ANALYSIS BASED ON PATIENT NEEDS



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Reported interests:

All experts and stakeholders consulted within this report were selected because of their involvement in this topic. Therefore, by definition, each of them might have a certain degree of conflict of interest to the main topic of this report.

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LIST OF ABBREVIATIONS

ABBREVIATION	DEFINITION
AGENAS	Agenzia Nazionale per i Servizi Sanitari Regionali
AMSTAR	A MeaSurement Tool to Assess systematic Reviews
BACME	British Association of Clinicians in ME/CFS
CADTH	Practical tool for searching health-related grey literature
CBT	Cognitive Behavioural Therapy
CDC	Centers for Disease Control and Prevention
EU	European Union
EUROMENE	European Network on ME/CFS
GET	Graded Exercise Therapy
GP	General Practitioner
HTA	Health and Technology Assessment
HRQoL	Health-related quality of life
HSR	Health and Services Research
INAMI	Institut national d'assurance maladie-invalidité
INESSS	Institut national d'excellence en santé et en services sociaux
IOM	Institute of Medicine
ME/CFS	Myalgic encephalomyelitis/Chronic Fatigue Syndrome
MeSH	Medical Subject Headings
NFU	Nederlandse Federatie van Universitaire Medische Centra
NHS	National Health Service
NICE	National institute for Health and Care and Excellence
NIHD	National Insurance for Health and Disability Insurance
NIHR	National Institute for Health and Care Research
OECD	Organisation for Economic Co-operation and Development
PAIS	Post-acute infection syndromes
PCC	Post-COVID condition
PEM	Post-exertional malaise
PICO	Participants-Intervention-Comparator-Outcomes
POTS	Postural orthostatic tachycardia syndrome
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-analyses
PROMs	Patient reported outcome measures
RCT	Randomised Controlled Trial
RIZIV	Rijksinstituut voor Ziekte- en Invaliditeitsverzekering



SR	Systematic review
UK	United Kingdom
US	United States of America
VWS	Ministerie van Volksgezondheid, Welzijn en Sport
ZonMW	ZorgOnderzoek Nederland Medische Wetenschappen

■ SCIENTIFIC REPORT

◆ INTRODUCTION

1 SCOPE AND OBJECTIVE OF THIS REPORT

Myalgic encephalomyelitis, also commonly reported as myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a chronic, multisystem, and disabling disorder that affects millions of individuals worldwide. It markedly compromises quality of life and, in its most critical forms, can lead to permanent dependence on caregiving.¹

This study aimed to identify and describe the health-related unmet needs of people living with ME/CFS, and to build an evidence base to inform the organisation of health services. We conducted a direct assessment of patient-reported needs across health, healthcare and social domains among people with ME/CFS in Belgium. In addition, we examined potential care models to inform how ME/CFS health services could be organised and adapted.

The study was initiated following a request from an ME/CFS patient association submitted through the KCE annual call for research proposals. Patient groups report that many needs remain unmet and call for changes in healthcare organisation. In parallel, they wish to identify opportunities for research and innovation in treatments beyond those currently available.

This project aims to inform how ME/CFS clinical services should be adapted to better reflect patient needs. It aligns with the Belgian Superior Health Council's recommendations for multidisciplinary care pathways and a network-based model of service delivery.² The findings are intended to support policymakers in setting priorities and improving care provision, with relevance for healthcare providers and researchers.

The introduction chapter of this report was not developed as a formal systematic review, nor does it constitute a clinical guideline. Instead, it is based on a comprehensive and pragmatic overview of the relevant literature and expert views, with the aim of reflecting key evidence and influential documents in the field. A targeted review of the scientific literature was conducted, consulting key publications including recent clinical guidelines or guidance documents (mainly the 2021 NICE guideline-NG206³, the 2022 CDC review⁴), and peer-reviewed narrative or systematic reviews^a. While efforts were made to include representative sources, the selection process did not follow a predefined systematic methodology and may therefore not be exhaustive. In particular, the section addressing the management of ME/CFS reflects the heterogeneity of approaches within the field. Given the ongoing debate and heterogeneity of approaches, this section is intended to provide contextual background rather than a definitive or consensus-based appraisal of therapeutic options.

^a The evidence base was identified through structured searches in PubMed and Google Scholar using the keywords "ME/CFS" and "chronic fatigue syndrome".

2 DISEASE DESCRIPTION

The clinical entity currently designated as ME/CFS was first documented in the early 20th century and subsequently characterised by different terminology and diagnostic criteria.⁵ ME/CFS has garnered increasing attention in recent years due to accumulating evidence of its striking similarities to post-COVID condition (PCC, also termed 'long COVID').⁶ These parallels extend to both core clinical features and potential underlying pathophysiological mechanisms, including immune dysregulation triggered by viral infection.⁷⁻⁹ Individuals affected by either condition typically exhibit a heterogeneous and fluctuating symptom profile, often characterised by alternating periods of persistence and fluctuation.¹⁰

Both disorders exert a profound negative impact on quality of life and impose a substantial societal burden, reflected in elevated healthcare costs, diminished productivity, and increased unemployment.¹¹

In this report, although the two conditions share areas of overlap and a subset of individuals with long COVID meet the diagnostic criteria for ME/CFS, we specifically address ME/CFS. Nevertheless, relevant connections will be highlighted between both conditions, given their involvement of multiple organ systems, and their need for multidisciplinary care approaches. To ensure clarity and consistency, the term “ME/CFS” will be used throughout this report, as it remains the most commonly adopted designation in the scientific literature.

Finally, this study includes only individuals aged 16 years and older; therefore, ME/CFS in paediatric populations is beyond its scope.

2.1 Symptoms and diagnostic criteria

2.1.1 Clinical spectrum

Overall, ME/CFS is primarily characterised by abnormal, persistent fatigue that does not improve with rest, and is associated with a marked reduction of stamina.^{1, 7, 12, 13} Notably, a defining feature of ME/CFS is post-exertional malaise (PEM). PEM describes a worsening of some or all symptoms, along with a further decline in functioning, following physical, cognitive, orthostatic, or emotional exertion that had previously caused no problem. Its onset can occur immediately, or be delayed by several hours to days. PEM is now recognised as a hallmark symptom of ME/CFS (see diagnostic criteria 2.1.2).^{5, 7} This characteristic has even prompted proposals to rename the condition as “*systemic exertion intolerance disease*”. PEM is typically prolonged, often requiring extended recovery periods for patients to return to their baseline functional status. Moreover, its severity is frequently disproportionate to the intensity of the precipitating activity.⁷

The clinical spectrum of ME/CFS also encompasses a wide range of symptoms involving multiple organ systems. The most prominent include non-restorative sleep, cognitive disorders, musculoskeletal pain, headache, flu-like symptoms, orthostatic intolerance, gastro-intestinal disturbances, and hypersensitivity to light, noise or temperatures changes. The severity of ME/CFS varies, ranging from mild to severe presentations in which patients are unable to perform most activities of daily living independently.^{1, 14, 15}

Due to the complexity of symptoms, their overlap with other health conditions, and the absence of diagnostic tests, the diagnosis of ME/CFS is challenging and is frequently subject to significant delays.⁷



2.1.2 Diagnosis

In the presence of a clinical picture suggestive of ME/CFS, it is critical to acknowledge that hallmark symptoms, especially fatigue and pain, are non-specific and shared by numerous other medical disorders. Therefore, a thorough and systematic diagnostic assessment should be undertaken initially to rule out other potential medical conditions.^{15, 16} Diagnostic criteria for ME/CFS have been proposed and are primarily symptom-based. Over time, various sets of clinical criteria have been introduced and progressively refined, as earlier definitions did not consistently include PEM (e.g. Oxford, or Fukuda criteria).¹² These disparities have contributed to a considerable heterogeneity of included study populations among research.

Progressively, expert consensus statements introduced more comprehensive frameworks. The Canadian Consensus Criteria require the presence of a greater number of symptoms and explicitly incorporate PEM. They were first published in 2003¹⁷ and revised in 2010.¹⁸

In 2015, the *Institute of Medicine* (IOM), now called the *National Academy of Medicine*, updated the diagnostic criteria to promote earlier diagnosis and improve awareness of ME/CFS among healthcare professionals and the society.⁵ The criteria are presented in Textbox 1.

Experts recommend early diagnosis to reduce the risk of long-term complications as much as possible.¹⁹ In line with this, the *National Institute for Health and Care Excellence* (NICE) guidelines, published in 2021, advise in their diagnostic criteria reducing the diagnostic latency from 6 months (as proposed by the IOM) to 3 months.³ This change aims to enable earlier intervention and improve long-term outcomes. Likewise, the guidelines recommend referral to a multidisciplinary ME/CFS specialist team at 3 months after symptom onset to support timely individualised care and potentially limit disease progression.

Textbox 1 – Diagnostic criteria from the *Institute of Medicine* (*National Academy of Medicine*)

Required symptoms

- **A substantial reduction or impairment in ability to engage in pre-illness levels of activity (occupational, educational, social, or personal life) that lasts for more than 6 months**, and is accompanied by fatigue that is: profound, new-onset, not the result of ongoing or unusual excessive exertion and not substantially alleviated by rest.
- **Post-exertional malaise:** worsening of symptoms after physical, mental, or emotional exertion that would not have caused a problem before the illness.
- **Unrefreshing sleep**

Additional symptoms: at least one of the symptoms below

- **Cognitive impairment:** problems with thinking, memory, executive function, and information processing, attention deficit and impaired psychomotor functions. All can be exacerbated by exertion, effort, prolonged upright posture, stress, or time pressure and may have serious consequences on a patient's ability to maintain a job or attend school full time.
- **Orthostatic intolerance:** a worsening of symptoms upon assuming and maintaining upright posture. This is measured by heart rate and blood pressure abnormalities during standing, bedside orthostatic vital signs, or head-up tilt testing.

Source: *National Academy of Medicine*⁵

2.1.3 Evolution

The proportion of adults who achieve full recovery remains poorly studied, and recent data on recovery or improvement rates are lacking.²⁰ In addition, there is no standardised definition for these outcomes, and their assessment is complicated by the fluctuating nature of the disease and the need for long-term follow-up. Clinical experience suggests that early diagnosis combined with timely and appropriate management may improve outcomes.^{1, 19} Supporting this, a small single-centre study, conducted in France, recently confirmed an inverse relationship between diagnostic delay and the likelihood of recovery or improvement.²¹

Complete recovery is uncommon; partial improvement may occur, but is often followed by relapses. As a result, many patients do not regain their pre-illness level of functioning and must adapt to living with the condition.¹

2.2 Co-existing conditions and overlap

Several conditions are commonly observed in patients with ME/CFS and may share overlapping clinical signs and symptoms. These co-existing disorders do not represent exclusion criteria for the diagnosis and may include neurological conditions such as small fibre neuropathy, gastrointestinal disorders (e.g. irritable bowel syndrome), or immunological disorders (e.g. mast cell activation syndrome).¹³ The purpose of this report is not to provide an exhaustive list, but two conditions deserve special attention:

- Fibromyalgia is one of the most frequent co-occurring conditions.²²⁻²⁵ It is a syndrome primarily characterised by chronic widespread pain^b resulting from central sensitisation and neuroinflammatory processes, and it is considerably more common than ME/CFS^c. Fibromyalgia and ME/CFS share overlapping symptoms, leading to debate over whether they represent two phenotypes of the same disorder or separate entities. A recent review revealed substantial clinical overlap since an important proportion of patients with fibromyalgia meet the criteria of ME/CFS.²⁵ However, the review noted high heterogeneity and reliance on outdated definitions. This could suggest that both conditions share similar underlying pathophysiological mechanisms but some differences have nonetheless been described.^{26, 27} Importantly, it is worth pointing out that diagnostic criteria for fibromyalgia are the definitions from the *American Rheumatology College* criteria, which do not include the presence of PEM. Current best practices recommend considering the predominant clinical phenotype. As such, if pain is the predominant symptoms and if fatigue or PEM seems secondary symptoms, it may be more appropriate to consider and offer a treatment for fibromyalgia.¹²
- In recent years, ME/CFS has gained increased interest due to its clinical similarities to long COVID.⁶ Numerous studies have reported that both conditions share similar symptom profiles, reinforcing the hypothesis of a post-infectious aetiology.^{8, 28-30} For example, a recent large-scale survey reported PEM in 89.7% of ME/CFS patients and in 79.4% of those with long COVID, with similar rates for neurocognitive disorders or non-restorative sleep in both conditions.³¹ Despite these similarities, differences exist: in the same study, few ME/CFS patients reported respiratory symptoms such as dyspnoea or chest pain, while these symptoms were highly prevalent in long COVID patients.

Both conditions are characterised by multisystem involvement and complex pathophysiology, reflecting the absence of a single, identifiable cause that could be addressed by targeting a specific dysfunction. The shared manifestation of PEM in both conditions remains a central element to guide therapeutic strategies and research priorities.

^b Diagnostic criteria were defined by the American College of Rheumatology (<https://rheumatology.org/>).

^c Prevalence of fibromyalgia ranges between 2 to 4% (<https://rheumatology.org/>).



While further studies are necessary to clarify the similarities and distinctions between these two clinical entities, it is evident that research on long COVID will continue to inform the understanding of ME/CFS, and reciprocally.^{32, 33}

2.3 Prevalence

Estimating the prevalence of ME/CFS is challenging due to several methodological issues among epidemiological studies. Case definitions vary widely across studies, making comparisons difficult (see 2.1). Diagnostic approaches also differ (e.g. patient interviews *versus* full clinical assessment), and there is considerable heterogeneity in study populations and severity classifications. Furthermore, the proportion of undiagnosed cases remains unknown, limiting reliable prevalence estimates.^{34, 35}

A meta-analysis published in 2020, included studies conducted between 1990 and 2018. It reported a pooled prevalence of 0.89% (95% CI: 0.60-1.33), based on the Fukuda criteria, which was the most commonly used case definition.³⁶ Substantial heterogeneity was noted across studies. Another recent study sought to identify prevalence studies at the European level and identified only three, conducted in the United Kingdom and Iceland, with an important estimates variations (ranging from 0.1% to 2.2%).³⁷

More recent primary studies have reported similar figures. One analysis of national primary care data in Australia estimated a prevalence ranging from 0.95% to 1.04% between 2015 and 2019^{d, 34} Another U.S. study, used data from the *National Health Interview Survey* (NHIS) compiled by the CDC^e. Prevalence was 1.3%, with higher prevalence proportions among women (1.7%) compared to men (0.89%). Ethnic disparities were also observed, with ME/CFS being more common among non-Hispanic White adults.³⁸

Overall, evidence consistently shows that ME/CFS is more prevalent in women than men, with onset most commonly occurring in adolescence or young adulthood.^{13, 39}

Despite uncertainty about the exact prevalence of ME/CFS, available data and modelling indicate that millions of people are affected in the world, and this number is likely to rise due to COVID-19^{40, 41} For instance, a recent observational study conducted by the RECOVER Consortium (*Researching COVID to Enhance Recovery*) identified an increased risk of developing ME/CFS^f following SARS-CoV-2 infection^g, compared with non-infected individuals.⁴²

2.4 Hypothesis on aetiology and pathophysiology

Although research in recent years has shed light on several pathophysiological processes, the aetiology of ME/CFS remains poorly understood. Increased interest, driven by research on long COVID, has pointed toward common potential underlying mechanisms including dysregulation across multiple biological systems (i.e. neurological, immunological, endocrine, vascular, and metabolic pathways). However, current knowledge does not yet clarify how these abnormalities interact or explain the disease as a whole. Indeed, biological disorders should not be interpreted in isolation because they are most likely intertwined.^{8, 32}

ME/CFS frequently arises following an acute infectious episode, but in many cases, the condition remains undiagnosed because the initial infection appears self-resolving and clinical documentation typically ceases once resolution is presumed.⁴³ As a result, ME/CFS has been proposed to be

^d The most commonly used case definition criteria in Australia are the Fukuda and the Canadian Consensus Criteria

^e <https://www.cdc.gov/nchs/nhis/index.html>

^f IOM diagnostic criteria were used.

^g Hazard ratio: 4.93 (95% CI: 3.62- 6.71)

classified (in a large number of cases) into the post-acute infection syndromes (PAIS).^{44, 45} These syndromes are characterised by persistent symptoms (including fatigue, PEM, cognitive impairments,...) that develop after an acute infection, and persist after the pathogen is no longer detectable. Current hypotheses suggest that infectious trigger(s) initiate(s) a cascade of pathological processes leading to the symptomatology⁸, although no consensus has yet been reached on this hypothesis.^{46, 47} Non-infectious triggers have also been suggested, including chronic stress exposure^h, potentially involving dysregulation of the hypothalamic-pituitary-adrenal axis and the autonomic nervous system.^{48, 49} Ultimately, the underlying pathophysiology is probably not limited to a single direct mechanism but could reflect complex interactions between biological processes and individual genetic predisposing factors.^{50, 51}

As such, multiple pathophysiological mechanisms have been proposed to explain both the occurrence of symptoms and their persistence. However, these are thought to result from complex, interacting processes that vary between individuals, rather than from a single pathway.

- Among the main aetiological mechanisms proposed, key hypotheses include immune and inflammatory dysfunction, mitochondrial and metabolic dysfunctions, neuroinflammation with autonomic nervous system impairment, and endothelial abnormalities. Since the objective of this report is not to provide an exhaustive inventory of research hypotheses nor to detail studies that have demonstrated biological abnormalities related to these mechanisms in patients, readers are referred to specialised reviews on the subject for a more comprehensive analysis.^{8, 9, 48, 50, 52, 53}
- Potential mechanisms that may help explain why symptoms become chronic have recently been summarised.⁸ These include, for instance, the occult persistence of an infectious agent (or their molecular remnant); tissue injury sustaining chronic inflammation, or gut microbiota alterations inducing a low-grade inflammation.
- Research initiatives have also focused on potential biomarkers such as indicators of immune dysfunction, genetic profiles, and metabolic or vascular markers.⁵⁴ Although no validated biomarker currently exists, recent advances including genomic analysisⁱ or the identification of cell free RNA identification, offer promising results.^{55, 56}

It is also relevant to consider other additional hypotheses that explore the interactions between biological systems and psychological factors. Although any chronic disease entails long-term psychological consequences, these may exert an aggravating role. In line with this, distinguishing between predisposing, precipitating, and perpetuating factors is essential.⁵⁷ This perspective underscores the complexity and heterogeneity of the condition, reinforcing the need for a comprehensive and a holistic approach that addresses the multiple dimensions involved in the symptomatology and its consequences. In some settings, the biopsychosocial model has been proposed, as for other chronic diseases since it considers the reciprocal influence of biological, psychological, and social factors. However, this has raised concerns among patients. These concerns stem for the perception that the model downplays ongoing biological processes, and place undue emphasis on psychological management.^{2, 58-60}

^h Either long-term exposure to stress, or a short exposure to severe stress.

ⁱ Microarray-based gene expression profiling.



3 MANAGEMENT OF ME/CFS

ME/CFS represents a considerable therapeutic challenge, given the considerable heterogeneity in symptomatology, severity, and disease course across individuals.^{1, 61} At present, no curative treatment exists, and no pharmacological therapies have been specifically approved for its management.

The primary aim of management is to alleviate symptoms, and improve functional capacity, ultimately reducing disease burden and enhancing quality of life.⁷ Central to current management strategies are principles of energy management, including pacing (see 3.1.2), the structured alternation of rest and activity, sleep management, and patient education regarding individual energy thresholds (for example, through identifying triggers associated with symptom fluctuation).⁶²

A person-centered approach is recommended, combining multidimensional interventions that are individually tailored to each patient. This approach is typically implemented through a multidisciplinary team comprising various healthcare professionals (e.g. physicians, physiotherapists, occupational therapists, psychologists, and nurses).^{3, 12} Expert consensus further underscores the importance of early implementation of these strategies to mitigate the risk of long-term morbidity.^{2, 19, 63} Nevertheless, these approaches do not demonstrate uniform effectiveness across all patients, and therapeutic options remain particularly limited for individuals with severe forms of disease.^{1, 62}

3.1 Management approach

The evidence base supporting therapeutic interventions for ME/CFS remains generally limited, as highlighted by the 2021 review conducted by the *National Institute for Health and Care Excellence* (NICE)³, and by a recent systematic review on ME/CFS management conducted by the *Centers for Disease Control and Prevention* (CDC).⁴ Particularly, two interventions (graded exercise therapy (GET) and cognitive behavioural therapy (CBT)) warrant specific consideration, as updated guidance from these institutions has introduced substantial revisions to previously endorsed recommendations. These changes have generated considerable debate and controversy within both the scientific community and patient organisations, particularly with regard to GET (3.1.1) and, to a lesser extent, CBT (3.1.4). While NICE (2021) was used as the principal guideline-level reference for this chapter, it should be noted that approaches to ME/CFS management remain the subject of ongoing scientific and clinical debate.

3.1.1 Controversy about graded exercise therapy

GET is an intervention in which an individual's achievable baseline level of physical activity is first established, followed by scheduled, incremental increases in activity duration. The intended aim is to improve day-to-day functioning by enhancing physical capacity and reducing symptoms, largely based on the premise that fatigue arises, at least in part, from physical deconditioning associated with maladaptive coping behaviours such as activity avoidance.^{12, 64}

GET was recommended for many years on the basis of studies reporting improvements in outcomes, such as fatigue, pain, and physical functioning.⁶⁵⁻⁶⁸ However, in ME/CFS, physical exertion may exacerbate symptoms particularly through the induction of PEM. As a result, the need to balance gradual increases in activity with the avoidance of symptom flare-ups was emphasised.⁶⁹

In 2011, the PACE trial (*Pacing, graded Activity, and Cognitive Behaviour Therapy; a randomised Evaluation*), a large randomised controlled trial (RCT), conducted in the UK, reported high rates of improvement and recovery.⁷⁰ However, the PACE trial has since been subject to substantial methodological criticism, including concerns regarding the definition of recovery, post hoc changes to outcome criteria, the use of the Oxford diagnostic criteria, and reliance on self-reported outcomes.^{7, 71} A subsequent re-analysis, by independent researchers using the original outcome thresholds did not confirm the initially reported effectiveness.⁷²

A Cochrane review^j published in 2019 continued to suggest potential benefits compared with passive approaches although acknowledging important limitations among the available evidence on exercise therapy including a lack of long-term outcome data, uncertainty about safety, and heterogeneity in diagnostic criteria.⁷³ In parallel, qualitative studies based on patient experience have raised concerns about potential harms associated with GET, including worsening symptoms and the induction of PEM.⁷⁴⁻⁷⁸ Despite plans for an update, the project was ultimately discontinued, and the decision generated substantial debate.⁷⁹

In light of these concerns regarding both effectiveness and safety, updated guidance from NICE (2021), CDC (2022) and the *Belgian Health Superior Council* (2020)² no longer recommends GET as a standard therapeutic approach. Both NICE and CDC highlighted that the evidence base supporting GET is of low to very low quality (emphasising concerns relating to inconsistent intervention definitions, unclear components, variability in outcome measures, potential recruitment bias, and insufficient safety reporting).^{3, 4}

These shifts in recommendations, where previously endorsed interventions are now discouraged, have generated substantial debate and criticism, including the evaluation processes and methodological approaches underpinning recent guideline updates.⁸⁰⁻⁸⁸

Nevertheless, both NICE and CDC indicate that carefully individualised and supervised physical activity programmes may be considered on a case-by-case basis, for patients who feel able to increase their activity levels, provided that these remain within individual tolerance limits.^{3 1} However, the threshold for what constitutes “tolerable” activity is not defined, and there is currently no robust evidence base to support a generalisable standard across patients.

Importantly, guidelines and expert consensus emphasise that management of ME/CFS should be delivered by clinicians with appropriate expertise in the condition, to ensure individualised care and prioritisation of patient safety.

3.1.2 Pacing

Although not curative, pacing aims to support patients maintain activities of daily living by optimising energy use. Generally recommended for all patients, it is a self-management strategy designed to conserve energy and prevent or mitigate PEM by limiting overall energy expenditure across physical, cognitive, and social activities to improve the quality of life.^{1, 7, 13}

Pacing is based on the concept of the individual ‘energy envelope’, which reflects a patient’s personal energy limit (i.e. the level of activity that can be sustained without exacerbating symptoms and avoiding PEM)^k.¹² As energy thresholds may fluctuate over time, patients are encouraged to continuously adjust activity levels in accordance with their current capacity, set appropriate limits, and prioritise tasks, while recognising that setbacks are an inherent part of the condition.⁷

The evidence base supporting pacing remains limited. The literature review underpinning the updated guidance from NICE highlighted substantial heterogeneity in the delivery of pacing interventions, including face-to-face sessions, video-based formats, telephone counselling, and written booklets. Similar variability had previously been documented in earlier consensus statements.⁸⁹ The NICE committee also noted that pacing is widely used by patients as a primary self-management strategy and is frequently incorporated as an initial component of multi-component interventions such as CBT to stabilise activity levels.³

A systematic review published in 2022 identified RCTs including pacing interventions and found that pacing is most often incorporated within multi-component programmes primarily CBT.⁹⁰ Some interventions also embedded elements encouraging gradual increases in physical and cognitive

^j Literature search up to May 2014.

^k The person plans activities and divides them up with short recovery periods in order to conserve energy.



activity^l. Despite substantial heterogeneity across studies, pooled analyses of this systematic review suggested improvements in outcomes such as fatigue and physical functioning, with larger effects observed in programmes that included progressive activity components. However, pacing was rarely evaluated as a stand-alone intervention, leaving a gap in the evidence base regarding its independent effectiveness.⁹⁰ A 2023 scoping review, which included a broader range of study designs similarly emphasised the heterogeneity of pacing interventions and the generally limited methodological quality of the available evidence. The authors concluded that high-quality studies are required to better assess the effectiveness of pacing and to ensure more complete and transparent reporting of the interventions applied.⁹¹

3.1.3 Management of symptoms

Overall, there is no scientific evidence on the use of pharmacological interventions to manage symptoms of ME/CFS and medicines are not recommended.³ For example, with respect to pain management, NICE highlighted an absence of evidence sufficient to support the use of ME/CFS-specific analgesic pharmacotherapies^m.³ Despite this lack of evidence, some expert groups note that symptom-targeted medications may be considered on a case-by-case basis to relieve selected manifestations.^{7, 16} For instance, sleep disturbances may be addressed through sleep management strategies, but some guidances suggest pharmacological interventions in a second step.^{7, 16} On the other hand, it should be noted that patients sometimes used medications off-label, and that some may already been taking medications before reaching ME/CFS specialists.^{12, 16, 92, 93}

Regarding complex autonomic manifestations such as orthostatic intolerance, some expert guidances suggests supportive measures including increased fluid and sodium intake, the use of compression garments, and in selected cases the administration of mineralocorticoids. However, there is no evidence supporting these approaches.^{7, 16} The NICE guidelines did not make any recommendations on this topic, but advised that any medicines for orthostatic intolerance should be prescribed or supervised only by clinicians with expertise in orthostatic intolerance because commonly used medications may exacerbate other ME/CFS symptoms.³

Finally, adaptation to a chronic and disabling illness may give rise to comorbid psychological conditions, warranting referral for appropriate assessment and treatment.^{1, 12}

3.1.4 Cognitive-behavioural therapy

Psychosocial interventions such as CBT may also be offered to patients with ME/CFS, and have been extensively studied.^{3, 12} CBT aims to improve symptoms and functional capacity by addressing maladaptive behaviours and beliefs that may contribute to fatigue and functional impairment.⁹⁴ It also targets factors that can exacerbate symptoms or reduce functioning (e.g. disrupted sleep, mood disorders).⁹⁰ It incorporates a range of techniques, including symptom management and coping strategies, cognitive restructuring, sleep management, psychoeducation and self-monitoring.⁹⁵ Delivery formats vary and may include individual face-to-face therapy, group sessions, or guided web-based programmes with therapist support.⁹⁴

Several systematic reviews and meta-analyses have reported that CBT is associated with improvements in outcomes such as fatigue and physical functioning in individuals with ME/CFS.^{82, 96-99}

In the 2007 guideline, NICE recommended CBT as treatment option.⁶⁸ However, in its 2021 update NICE concluded that the evidence supporting CBT was limited, with inconsistent effects across most

^l The concept of “graded activity” involves first stabilising a baseline level of activity to minimise symptom variability, and then progressively increasing both physical and cognitive activities over time.

^m Guidelines from NICE recommend to manage the person's pain according to best practice, referring to specialist pain services if appropriate.

outcomes^{n,3} The overall quality of evidence was rated as low to very low quality, and the committee expressed limited confidence in the observed effects, citing concerns related to heterogeneity in intervention components, variability in outcome measures, and methodological limitations that complicated evidence synthesis.³ Similarly, the CDC noted that although CBT may be associated with improvements in certain outcomes, the magnitude of these effects is generally small to moderate, and the evidence base is characterised by heterogeneity, variable case definitions, and methodological constraints.⁴

Consequently, NICE recommends CBT not as a curative treatment, but as a supportive intervention to assist patients in coping with and managing symptoms of chronic illness, and specifies that it should be delivered by healthcare professionals with expertise in ME/CFS.³ As with GET, these changes in recommendations have generated substantial debate within the scientific community, including criticism regarding the interpretation of the evidence, and the methodology underpinning the guideline revision.⁸⁸

Nevertheless, the role of CBT in ME/CFS remains an active area of scientific investigation.⁹⁸ This underscores the need for a nuanced and evolving assessment of its potential contribution to management strategies, particularly as understanding of the condition advances (for example, through improved characterisation of PEM, greater diagnostic standardisation^o, descriptions of practices^p).^{4, 12, 100, 101} Importantly, CBT does not negate the biological basis of ME/CFS but rather acknowledges the complex interplay between cognitive, emotional, behavioural, and physiological processes. A recent review has further outlined key considerations for its appropriate implementation (see Textbox 2).¹⁰⁰

Textbox 2 – Prerequisite for the implementation of cognitive behavioural therapy

Provider competency: understanding of the complexity and multifaceted nature of ME/CFS

Personalised approach:

- Interventions tailored to patient-specific needs and respect energy limitations
- Continuous monitoring and adjustment of interventions required to prevent post-exertional malaise manifestations

Objectives and limitations: CBT targets cognitive, emotional, and behavioral processes within disorders characterised by biological foundations

Source: Takamatsu N et al, 2025¹⁰⁰

3.1.5 Information and social support

Management also encompasses the contribution of social care professionals. Patients and families should receive comprehensive up-to-date information on disease characteristics (including severity, chronicity, fluctuations,...) and access to support services, national resources, financial aid, and social services. Patients may be informed about the existence of peer-support resources. Whether patients want to make use of these depends on individual preferences and needs, acknowledging the heterogeneous experiences reported in the literature. In some cases, environmental adaptations, such as wheelchairs, disability parking, are necessary to support autonomy. Individual needs should be assessed, and clear guidance provided on obtaining appropriate support.^{3, 12}

ⁿ General symptom scores, fatigue, physical functioning, exercise performance, return to work.

^o For example: definition of recovery/improvement, diagnostic criteria of ME/CFS.

^p CBT interventions are not strictly standardised, and heterogeneity in practice may influence the interpretation of outcomes.¹⁰



3.2 Perspectives of treatments

To date, no disease-modifying therapy exists, and no pharmacological agent has yet demonstrated clinical benefit. Advances in understanding genetic factors, identification of reliable biomarkers, early disease detection and progression^q could potentially yield valuable insights for the development of future therapeutic strategies, including both pharmacological and non-pharmacological interventions.^{8, 14, 92, 102}

Reflecting a personalised medicine approach, examples of current research initiatives include multi-omic^r approaches to achieve a deeper understanding of the underlying pathophysiology^{s, 103-105}. In addition, ongoing studies, accelerated by the emergence of long COVID, have explored the effects of several repurposed drugs (e.g. m-TOR inhibitors^{106, 107})^t, and other potential drugs still in the preclinical phase.⁸

The prospect of a stand-alone solution is unlikely. Rather, effective management will need to be integrated within a broader framework of precision medicine, addressing the complex interplay of disease components and incorporating neurocognitive as well as psychological interventions.^{100, 105, 108, 109}

4 IMPACT ON QUALITY OF LIFE, OVERALL BURDEN, AND UNMET NEEDS

4.1 Quality of life

ME/CFS is associated with substantially reduced quality of life and functional capacity affecting daily activities, as well as social and occupational participation. Physical limitations are typically most pronounced, and particularly include difficulties with usual activities, pain/discomfort and impaired mobility.^{110, 111} Family members' quality of life can also be adversely affected, mainly through emotional distress and impact on work, and social relationships.¹¹¹ Evidence further suggests that long COVID can impose a comparable health-related quality-of-life burden, with impairments again most marked in physical and daily-function domains, whereas mental health domains tend to be less affected.¹¹²

4.2 Global health and socio-economic burden

4.2.1 Socio-economic burden

A recent review by Vester *et al.*, has analysed the economic burden and the distribution of costs associated with the illness, based on 16 studies published between 2003 and 2023 and conducted in several countries.¹¹ The analysis revealed that indirect costs dominated total costs across countries (up to ~94%), driven mainly by productivity losses linked to unemployment/reduced work capacity among patients (and sometimes informal caregivers). By comparison, direct costs were smaller and largely reflected outpatient care (the largest component, up to ~75% of direct costs) plus other medical and non-medical expenses.

^q including a better prediction and refined characterisation of PEM.

^r Analysis of several 'omics' results (e.g. genomics, transcriptomics, proteomics and metabolomics) to provide a comprehensive understanding of biological mechanisms.

^s <https://www.omfcanada.ngo/collaborative-research-center-montreal/>

^t <https://www.omfcanada.ngo/precision-medicine-for-myalgic-encephalomyelitis-drug-discovery-and-clinical-trials-remedial/>

4.2.2 Additional indicators of disease burden

ME/CFS imposes also a substantial burden on education. Children and adolescents with ME/CFS show markedly reduced school attendance, lower academic performance, and poorer school-related QoL, and a lower educational attainment.^{11, 113-117}

Productivity losses due to ME/CFS - Data from Belgium

This analysis is based on administrative data from the Kenniscentrum Arbeidsongeschiktheid/ Centre de connaissances en matière d'incapacité de travail at RIZIV-INAMI and includes only individuals with work absence longer than a year and an ICD 10 diagnosis code G93.3.

As of 31 December 2024, 1,189 individuals with ME/CFS were registered in the invalidity database (a database for the people on work absence longer than a year). The population shows a strong female predominance (82%), and nearly three quarters (72%) are aged 45 years or older, although a substantial share (28%) are still of younger working age (18-44 years). The sex and age distribution of ME/CFS patients in the invalidity database is illustrated in Appendix 1 Figures 1 and 2, alongside comparator diseases from previous NEED studies.

ME/CFS is characterised by very long durations of work disability. The mean duration of invalidity is 8.6 years, with women experiencing slightly longer durations than men (8.8 vs 7.5 years). Compared with other chronic conditions included in earlier NEED studies (see 8.1- Crohn's disease, psychotic disorders, melanoma and COVID 19), ME/CFS is among the diseases with the longest recorded periods of invalidity. The skewed distribution of invalidity duration, with a long tail of very prolonged absence, is shown in Appendix 1 Figure 3.

5 BELGIAN CONTEXT

5.1 Background: first initiatives and evaluation

From 2002 onwards, the Belgian authorities established tertiary-level reference centres for the management of ME/CFS. Initially, five multidisciplinary centres were set-up to deliver specialised care and coordinate treatment strategies^{u, 118}

- Reference centres were implemented with a standardised registration framework, supported by dedicated administrative resources. During this early phase, rigorous evaluation of service delivery and patient outcomes was indeed considered essential, given the paucity of evidence on care organisation. The centres were tasked with diagnosing and managing ME/CFS, particularly complex cases beyond the capacity of primary and secondary lines, delivering time-limited therapies, and providing interdisciplinary, patient-centred services. They were also responsible for educating and supporting first- and second-line care providers to facilitate future care coordination by general practitioners (GPs), as well as advising NIHDI (RIZIV/INAMI) on how to organise this care across first and second lines in the future^v. Funding was based on a lump-sum system covering the centres, rehabilitation pathways, and GPs.

In practice, patients underwent a multidisciplinary assessment involving an internist, psychiatrist, and rehabilitation specialist. Based on this evaluation, an individualised therapeutic programme

^u One centre was dedicated to children.

^v The agreement ('Convention') with NIHDI was valid until end 2008.



was proposed and communicated to the GP. Treatment could be delivered at primary or secondary care level, or within the reference centre itself.

- Following an initial evaluation of the activities and outcomes of the centres, the KCE, was commissioned, in collaboration with the Superior Health Council, to provide a scientific basis^w for patient management, and to formulate recommendations regarding the future organisation of care.¹¹⁸

The KCE report 88B indicated that the objective of delivering integrated and coordinated care across different levels of the healthcare system was not achieved^x. The impact and support provided by the reference centres to primary and secondary care were suboptimal, with few initiatives explored to foster collaborative care between these levels and the centres. Furthermore, the influence of these centres on exercise capacity, return to work, and quality of life remained unclear. In addition, the experience gained did not lead to the development of comprehensive guidelines for the diagnosis and management of ME/CFS.

Consequently, the KCE recommended a reorganisation of care centred on first-line health professionals (GPs, physiotherapists, psychologists), in collaboration with an easily accessible second-line network and reference centres, with the aim to enable early management and reduce consequently the risk of chronicity. The KCE also advocated for an evidence-based and cost-effective approach in the development of these centres. Furthermore, it emphasised the importance of collecting data on disease severity and comorbidities, and highlighted the need to build specialised expertise in ME/CFS care, including information dissemination and professional training.¹¹⁸

5.2 Current NIHDI agreement

- The NIHDI agreement with the reference centres was replaced in 2014 by a new agreement, involving only one centre in Belgium^y. The agreement continues to be in effect.¹¹⁹ This healthcare pathway includes a hospital-based multidisciplinary diagnostic centre composed of various healthcare professionals specialised in ME/CFS, and including general internists, psychiatrists, psychologists, and administrative staff.¹¹⁹ Patients are referred via their GP. The centre offers CBT when the diagnostic assessment indicates that it may improve the functioning of the referred patient. This therapy is not delivered within the centre itself but is provided by CBT therapists (ideally located near the patient's residence) and consists of 17 sessions (of which up to eight may be conducted in group format). An information session is organised to explain the holistic, multidisciplinary approach to CBT and its potential benefits. Patients are only required to pay the co-payment. GPs receive a written report detailing the diagnostic findings and the proposed treatment plan. They are invited to participate in consultations with the centre's physicians. Treatment evaluation sessions are organised by the reference centre.
- It should be noted that other centres in Belgium continue to provide services for individuals with ME/CFS outside the framework of the convention, which consequently entails out-of-pocket

^w The primary objectives of this study were to update evidence-based knowledge on diagnosis, treatment, economic evaluation of available therapies, and the challenges experienced by patients with CFS. Additionally, the study examined organisational alternatives adopted in other countries for managing CFS care.

^x The project failed to achieve its main objective: the establishment of a three-levels of care system in which most patients would receive treatment at the first or second level, as close as possible to their place of residence.

^y https://www.stradalex.com/fr/sl_src_publ_div_be_chambre/document/QRcrb_55-b085-1186-1425-2021202214905

payments by patients. In general, these centres are affiliated with hospitals, and are located in Flanders.

5.3 Strategic policy proposals

Later, in this context of limited care offer, various proposals for adapting the service offer and care structure have been suggested:

- In 2020, the Superior Health Council highlighted the need to improve knowledge and recognition of ME/CFS within society and among healthcare providers, by strengthening professional training and enhancing communication between stakeholders to facilitate early detection and management of ME/CFS. It recommended the establishment of an accessible, multidisciplinary care network, including specialised centres and support for vocational reintegration. Finally, it emphasised the importance of promoting scientific research, both biomedical and psychosocial, in order to better understand the condition and develop effective therapeutic strategies.²
- At the European level, the resolution from the European parliament, in 2020, asked The Commission and Member states to support biomedical research on ME/CFS.¹²⁰ Since then, several calls for research projects have been launched (*"Tackling high-burden for patients, under-researched medical conditions"*, *"Relationship between infections and non-communicable diseases"*).^{z aa}

6 RESEARCH QUESTIONS AND OVERALL METHODS

Methodologically, this project is anchored within the NEED framework, which is a methodology developed by KCE from 2022 onwards with the objective of developing and testing an innovative, evidence-based methodology to identify unmet health-related needs to support the transition to a more needs-driven healthcare policy and innovation system.^{121, 122} The methodology underlying this approach is more deeply detailed later in the report (see 8.1). Furthermore, patients were actively involved at various stages of the research process through a predefined patient involvement process (see 8.2).

The research questions of the present report were addressed as follows (the complete methodology is described in each chapter):

Research questions	Methodology
What are the unmet health-related needs of people with ME/CFS in Belgium?	• Quantitative approach: online survey among people who received a diagnosis of ME/CFS • Qualitative approach: in-depth interviews of a selected sample of the participants to the survey
• Is there evidence indicating that implementing specific healthcare service organisation for ME/CFS affects clinical outcomes? • Which types and organisational structures of healthcare services are recommended and what supporting evidence underpins these recommendations? • How are healthcare services for ME/CFS organised in selected foreign countries?	• Systematic review of peer-reviewed literature • Analysis of grey literature • Consultation of experts from foreign countries, governmental websites, or health agencies from selected foreign countries

^z https://www.europarl.europa.eu/doceo/document/E-10-2025-003626_EN.html

^{aa} https://www.europarl.europa.eu/doceo/document/E-10-2025-003626-ASW_EN.html#def2



◆ SECTION 1: UNMET HEALTH RELATED NEEDS OF PEOPLE WITH ME/CFS

KEY POINTS TABLE

Unmet health needs

The online survey including 749 adult patients with ME/CFS and living in Belgium, identified a marked and statistically significant deterioration in self-reported quality of life from pre-symptom baseline to the time of survey completion. These findings were complemented by semi-structured interviews conducted within a sample of 19 participants.

- Abnormal fatigue, post-exertional malaise, and sleeping problems were key physical symptoms in the sample and were perceived as highly burdensome.
- Half of the participants had been diagnosed with ME/CFS for more than 10 years. Associated reported comorbidities included most frequently fibromyalgia and irritable bowel syndrome. Long-COVID accounted for 16%.
- Impairments were evident across all EQ-5D-5L dimensions, with the greatest impact in usual activities, pain/discomfort, and mobility. Issues in these domains were common at survey completion and were related to a high perceived burden.
- Anxiety and depression were reported more frequently at the time of survey completion than before symptom onset. The perceived burden of these mental health issues was slight to moderate in 55% of respondents, and severe in 11%. Interviewees reported experiencing a cluster of interrelated psychosocial factors, including worsening day-to-day exhaustion, perceived abandonment, self-stigmatisation, reduced future prospects, impaired role functioning, identity loss, and ongoing financial insecurity.
- The majority of survey respondents reported a moderate to extreme impact of ME/CFS on their sexual life, while around half of the respondents also reported an impact on reproductive health.

Unmet healthcare needs

- Most survey respondents received their ME/CFS diagnosis during a specialist consultation or at an ME/CFS referral center and nearly half of them had to wait more than two years to receive a diagnosis.
- Reported therapeutic modalities were heterogeneous and included diverse forms of care, for example: physiotherapy (manual techniques), cognitive behavioural therapy, psychological care, as well as specific dietary regimens or prolonged antibiotic therapy. Perceived effectiveness varied considerably across participants with reports of benefit, no benefit, or worsening of symptoms. Treatment burden was also highly variable. For graded exercise therapy, a majority of patients reported more marked symptom exacerbations and substantial treatment burden.
- Interviewees emphasised the importance of developing coping strategies, and adaptive pacing to navigate daily functioning and prevent relapse and deterioration.
- More than half of survey respondents reported that care was rarely or never well organised and planned. In contrast, more than half of respondents indicated good continuity of care as they often or always encountered the same healthcare professional.
- More than half of survey respondents evaluated the level of expertise of their healthcare providers as low or rather low and indicated that they never or rarely received clear and understandable information.

- A considerable share of participants reported having foregone care in the past year. This was reported as attributed to a lack of health service offer, long waiting time, financial difficulties and care accessibility issues.

Unmet social needs

- Three quarters of survey respondents experienced a negative impact of ME/CFS on their personal relationships.
- Unmet social support needs mostly involved daily-living activities, medical, and administrative support.
- The majority of survey respondents reported a substantial impact of ME/CFS on education or work. Work participation was heavily affected; the majority of respondents stopped working or worked less.
- A large proportion of survey respondents experienced a substantial impact on their financial situation and have difficulties covering their expenses.

Unmet societal needs

- Most close caregivers who participated in the survey on behalf of someone with ME/CFS reported a substantial impact on their physical health and social life.

Equity issues

- Equity is explicitly recognised in the Belgian NEED framework as a cross-cutting dimension of unmet health-related needs, highlighting avoidable and unfair inequalities in health outcomes and access to care.
- Subgroup analyses showed greater deterioration in health-related quality of life among people aged 30-50 years, those diagnosed after more than two years, those reporting more unmet needs, poorer care experiences, limited care accessibility and financial difficulties.
- Respondents who stopped working due to ME/CFS reported a higher symptom burden, greater social impact, more care-related accessibility problems and increased financial strain.
- Interview findings pointed to a strong lack of societal and institutional recognition of ME/CFS, contributing to invisibility, inequalities in care, and insufficiently structured healthcare responses.
- The political and societal stance towards ME/CFS was perceived as stigmatising, characterised by psychologisation, simplification, underestimation and denial of the disease.
- Psychologisation of ME/CFS was frequently experienced as inequitable care, with psychological explanations and interventions offered in place of biomedical investigation.
- Patients reported an overall underestimation of symptom severity by healthcare professionals and society, leading to reduced disclosure of symptoms and social withdrawal.
- A perceived lack of scientific and political interest in ME/CFS was reported, resulting in limited progress in disease recognition, research and treatment development, despite increased attention following Long COVID.
- Access to disability support was described as particularly challenging due to complex procedures and assessment tools insufficiently capturing fluctuating, non-visible and cumulative impairments such as fatigue and post-exertional symptom exacerbation.



7 SCOPE

The specific scope of this part is the determination of unmet health-related needs in individuals living with ME/CFS. These unmet health-related needs were assessed across the health, healthcare and social domains as reported by patients or their informal carers. In addition, the variability of their needs was examined for several variables across demographic, disease-related, healthcare-related and social strata.

8 METHODS

8.1 Overall approach using the NEED framework

The approach to assess the unmet-related needs was based on the NEED (*Needs Examination, Evaluation and Dissemination*) assessment framework developed by KCE in 2022^{bb, 103, 104}

The NEED initiative was launched to engineer and test an evidence-based methodology for identifying unmet health needs, with the objective of reorienting health policy and innovation toward needs-driven prioritisation. The prevailing innovation landscape remains, indeed, primarily supply-driven: capital allocation in research and development is skewed toward domains with superior expected commercial returns, resulting in underinvestment in indications with the highest burden of disease and unmet societal demand.

Initial NEED findings were released in 2024 and presented in the setting of the Belgian Presidency of the Council of the European Union (EU).¹²¹ Deliverables included the NEED assessment framework, defining criteria and indicators to appraise unmet health needs for patients, caregivers, and society.¹²² It was initially applied in two conditions (Crohn's disease and malignant melanoma), with results posted to the public NEED database^{cc, 123}. In 2025, the NEED Assessment Framework and its core module were revised and published following European-level consensus among experts and patient representatives (whereas the previous version had been developed primarily with Belgian stakeholders).¹²⁴ In addition, the framework's suitability for rare diseases was also evaluated and a study on sickle cell disease was started. In 2026, the NEED assessment framework was used for patients living with schizophrenia.¹²⁵

8.2 Patient involvement

To enhance the quality and relevance of its research on patient-related issues, KCE seeks to actively engage patients throughout the research processes. This participatory approach is intended to inform and guide methodological decisions regarding the most appropriate strategies for evaluating aspects related to patient experiences and outcomes. The approach to determining the extent and nature of patient involvement was conducted using the guidance described in the *KCE Process Note No. 340*, which provides methodological recommendations for integrating patients into research activities.¹²⁶

The patient involvement was implemented through collaboration with patient-representative organisations, rather than through the direct inclusion of individual patients with experiential expertise. This approach was chosen to ensure a broader and comprehensive representation of patient perspectives. A process of patient involvement throughout the entire project was implemented using

bb <https://health-needs.eu/index.php/en/>

cc <https://health-needs.eu/index.php/en/news-en/need-database-now-available-online>

an embedded consultation approach (rather than targeted consultation approach^{dd}), which is a type of involvement where patients are regularly consulted throughout the research process.

The following process was followed:

- Opportunities for patient involvement were identified across the various phases of the project, including the scoping stage, study design and data collection, data analysis and interpretation, as well as the formulation and dissemination of recommendations. This involvement primarily took the form of collaborative working sessions conducted via online meetings. The stages of the project in which patient involvement was planned, along with the anticipated contributions from patient collaborators, are detailed in Table 1.
- To initiate the recruitment process, the Belgian umbrella organisations LUSS^{ee} and VPP^{ff} were contacted to disseminate a call for interest among their affiliated patient associations. As a result, four associations expressed their willingness to participate in the research project and included Millions missing Belgique^{gg}, 12ME^{hh}, CVS-contactgroepⁱⁱ and BOFiP^{jj}.
- Each representative from the participating patient associations was invited to sign a declaration of potential conflicts of interest. The terms of collaboration were formalised through a memorandum of understanding, which outlined the objectives of the partnership, the working modalities, and the confidentiality policy (all items of the documents are available in the KCE report 340- section 3.4.6.2¹²⁶).

Table 1 – The research phases, objectives and methods for patient involvement

<i>Phase</i>	<i>Objective</i>	<i>Methods for involvement</i>
Scoping	To better describe the context of the research topic, taking patient issues into account To capture the patient-related elements, specific to ME/CFS that need to be addressed in the research process	Work meeting
Design and data collection	To select the patient-relevant outcomes to be included in the survey To help recruit participants and disseminate the publicity of the data collection	Work meeting
Data analysis	To help with the interpretation of the results	Work meeting
Recommendations and dissemination	To get input about the formulation of the policy recommendations and to help for dissemination	Two work meetings (one including stakeholders)

^{dd} In a targeted consultation approach, patients are consulted on specific aspects of the research study, on an ad hoc basis.

^{ee} <https://luss.be/>

^{ff} <https://vlaamspatientenplatform.be/nl>

^{gg} <https://www.millionsmissingbelgique.com/>

^{hh} <https://12me.be/>

ⁱⁱ <https://www.cvscontactgroep.be/>

^{jj} <https://www.bofip.be/>



8.3 Patient survey

8.3.1 Participants

8.3.1.1 Recruitment strategy

As part of the recruitment strategy, the survey was announced on the KCE website, where a Quick Response (QR) code was provided to grant direct access to both the informed consent form and the online questionnaire. The link was further distributed via the KCE newsletter and the social media platforms. In addition, patient associations were invited to circulate and display information about the survey within their respective networks.

The questionnaire was launched in 2025 and accessible during a 2-month period. Throughout this period of data collection, the recruitment strategy was subject to continuous monitoring. No formal sample size calculation was performed. Instead, we set a pragmatic recruitment target 500 participants. Based on commonly cited estimates of ME/CFS prevalence, it could correspond to approximately 1% of the ME/CFS patients in Belgium.

8.3.1.2 Inclusion and exclusion criteria

Respondents had to meet the following criteria to participate in the survey:

- Being 16 years of age or older;
- Having received a diagnosis of ME/CFS;
- Living in Belgium (main residence).

In cases where the survey was completed by a proxy respondent on behalf of an individual who met the inclusion criteria but was unable to participate directly, the proxy respondent was instructed to answer the questionnaire as the patient would have, to the best of their knowledge. Participants who did not meet the inclusion criteria were automatically redirected to the end of the survey and provided with a brief explanation regarding their ineligibility.

8.3.1.3 Recruitment material

Flyers were produced in both French and Dutch (see Appendix 2) to inform potential participants and guide them to the online survey via a QR code. They were distributed in healthcare settings (i.e. medical consultation centres) directly through patient associations, which also circulated them within their own networks.

8.3.2 Quantitative data collection tool

8.3.2.1 Generic NEED questionnaire

The generic questionnaire was developed within the NEED project by KCE through a literature review and expert consultation and was subsequently validated via a Delphi process involving representatives of umbrella organisations of patient associations and sickness funds. For application to a specific health condition, the generic questionnaire must be adapted using information on symptoms, treatments, adverse effects and other condition specific features. The purpose of this tool is to capture multiple dimensions of unmet health related needs from the perspective of individuals living with a specific condition. In addition, several questions addressed exclusively to informal caregivers assess caregiving burden. The questionnaire is intended to be completed by individuals with the condition but if it is not feasible, family members or informal caregivers may respond on their behalf.

8.3.2.2 Adaptation of the generic questionnaire for ME/CFS

The adaptation of the NEED questionnaire to ME/CFS was entirely conducted through a patient involvement process as described above (Table 1). A first scoping meeting was organised with the patient associations willing to join the project. The aim of this first brainstorming session was to capture the patient-related elements, specific to ME/CFS that need to be addressed in the research process and to discuss the potential adaptations of the generic questionnaire by reformulating some questions and by adding questions that are specific to ME/CFS. The adaptations and the additional questions were, therefore, developed and structured in close collaboration with patient associations representatives. During the second meeting, the questionnaire was presented to the patient association representatives, and their feedback was taken into account to adjust the questionnaire. This iterative process of revising items and integrating patient feedback continued through multiple rounds of email correspondence until consensus was reached on the final questionnaire.

8.3.3 Languages adaptations, online publication and distribution

The questionnaire was drafted directly in Dutch and French by the research team. A comparison of the two language versions was carried out to ensure that both were identical in terms of structure and content, and during a second phase, the KCE communications team reviewed and adapted the language and the formulations.

Following proofreading by the research team, the questionnaire was transferred to the online platform LimeSurvey. A pre-test was then conducted with a sample of native Dutch and French speakers from the KCE to evaluate the comprehensibility of the survey items and the usability of the online interface. Based on the feedback received, the questionnaire was revised. Subsequently, the research team carried out a test of the online versions in both languages to verify their consistency.

8.3.4 Questionnaire structure

The final questionnaire consisted of 63 questions, organised around 10 themes summarised in Table 2^{kk}. The survey platform allowed participants to save partial responses and complete the questionnaire over multiple session.

For most questions, respondents were given a choice of answers (often based on the literature and/or patient involvement process) but were also given the opportunity to give additional responses in an open field. For several questions, respondents were asked to indicate their level of satisfaction or level of burden. In many instances, responses to specific questions led to new sub-questions (e.g. when the respondent indicated 'Yes', he/she had to specify his/her answer). At the end of the questionnaire, participants were asked whether they would be willing to participate in an individual interview. Information from respondents who indicated their willingness was retained for shortlisting, and to evaluate eligibility according to the characteristics required to define subgroups.

Table 2 – Main themes of the survey

Theme	Description
General information	Demographic information, employment status and financial information
Disease	Disease-specific information such as duration of ME/CFS, frequency and level of burden of symptoms, etc
Other illnesses	Comorbidities associated with ME/CFS and other comorbidities
General health status and health-related quality of life	Health status before and after onset of ME/CFS using the EQ-5D-5L Impact on sexual life and reproductive health

^{kk} Upon request, the questionnaire can be provided in French and Dutch.



Treatment	Frequency, perceived effectiveness and burden of side effects of different treatment options
Healthcare use and access to care	Experience with organisation of healthcare services and contact with healthcare professionals
Impact of the disease on education and occupation	Impact on education and work Time patients could not work due to ME/CFS
Financial impact	Impact on financial situation
Impact on social relationships and support needs	Impact on personal relationships and needs for social support not received
Priority needs and other needs	Priority rating of need categories and other important needs not covered in questionnaire

8.3.5 Quantitative data analysis

Descriptive statistics were calculated for all variables. For continuous variables, the mean $\pm$ standard deviation (SD) or median with interquartile range (IQR) for normally and non-normally distributed data respectively, was calculated. For categorical variables, the associated percentages were calculated. The number of missing answers varies among the questions, the effective sample sizes therefore vary across questions. The number of responses indicating “*I don’t know*”, “*not applicable*”, or “*I don’t want to answer*” were mentioned in the text and/or in the legend of the tables and figures for transparency reasons, but were not taken into account for the analysis or presentation of results, except for the question regarding other comorbidities, and self-reported effectiveness of treatment.

The impact of living with ME/CFS on quality of life was evaluated by means of EQ-5D-5L utility scores. EQ-5D-5L utility scores were derived from respondents’ EQ-5D-5L health descriptions on five dimensions of health (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression). Responses for each dimension were coded as a number ranging from 1 (no problems) to 5 (extreme problems). These numbers were combined for all dimensions resulting into a 5-digit number, reflecting the participant’s health state, which was converted into EQ-5D values using the most recent Belgian value set.¹²⁷ The EQ-5D-5L utility score was collected within the same survey for two time points: retrospectively before the start of ME/CFS symptoms and at the time of survey completion (“today”). A paired sample t-test was conducted to determine whether there was a significant difference (mean delta) between the mean utility scores before vs after the onset of the disease.

Exploratory subgroup analyses were conducted for changes in “EQ-5D-5L utility scores” and change in working status (“still working” versus “stopped working”), stratified by demographic, disease-related, healthcare-related and social variables. Change was defined as the difference between pre-symptom status and status at the time of survey completion. Accordingly, a new variable was created by subtracting the pre-symptom value from the value at survey completion. For EQ-5D-5L utility scores this means that higher negative values indicate greater decline. A separate exploratory subgroup analysis was also performed for French speaking versus Dutch speaking participants for relevant demographic, disease-related, healthcare-related and social variables.

Student’s t tests, or one-way Anova tests were performed for the continuous variable EQ-5D-5L utility score, while chi-square tests or Fisher’s exact tests (in case one cell had < 5 observations) were applied for the categorical variables. The significance level was set at $\alpha=0.05$ for all statistical tests. Since these subgroup analyses had an exploratory purpose, no correction for multiple testing was applied. All statistical analyses were performed using R version 4.3.0 (R Foundation for Statistical Computing, Vienna, Austria). The R code is available on GitHub®.

Open field responses were first analysed thematically and recoded into existing or newly defined categories when appropriate. Any recoding was done by preserving the main idea(s) behind the initial response and was verified by another native-speaking member of the research team. If the response could be interpreted in different ways, the response was classified under ‘Others’ to avoid

misinterpretation. Open field answers in the online survey have also been processed using NVIVO® software.

8.4 Qualitative interviews

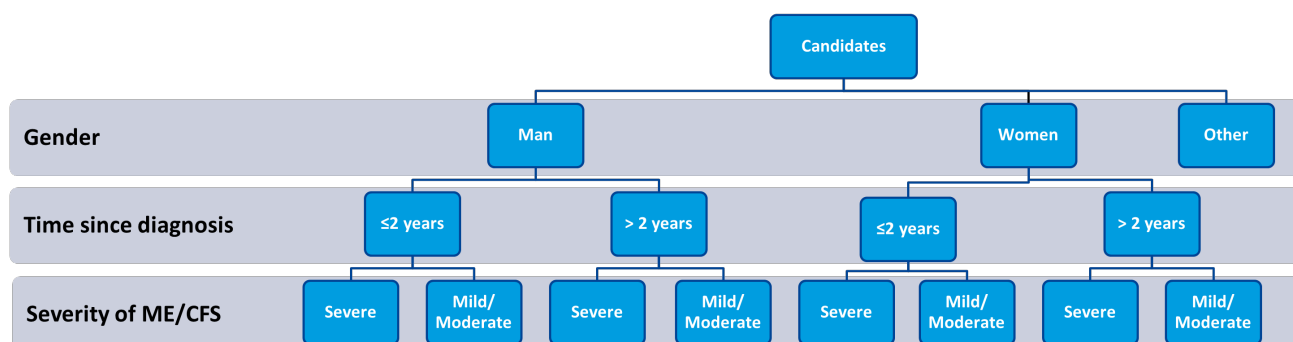
The main aim of the interviews was to get a general view of patient's experiences with ME/CFS, as well as their health, healthcare and social needs, in complement to the survey, and in correspondence with the NEED framework (see 8.1).

8.4.1 Selection of participants and recruitment

The aim was to conduct 24 individual interviews, 12 in French and 12 in Dutch. All interviewees were selected from survey respondents who indicated their willingness to participate in a subsequent in-depth interview. To ensure maximum variability among respondents, researchers prioritised recruiting participants in each language according to the following segmentation criteria as outlined in Figure 1:

- Gender
- Time since diagnosis ($\leq$ or $>$ 2 years)
- Severity of the disease (Classification was derived from responses to a mobility-related survey question^{II}. A "severe" category was assigned to participants selecting "severe/extreme," while responses of "no problems," "mild," or "moderate" were combined into the reference (non-severe) group.)

Figure 1 – Segmentation criteria for the recruitment of interview participants



For each arm of the sample segmentation, up to three volunteer participants were randomly selected: one primary participant and two substitutes. Participants for the interviews were randomly selected using PROC SURVEYSELECT in SAS® (version 9.4; SAS Institute Inc., Cary, NC) in each segment. The selected primary participant was contacted by telephone to ask whether they still agreed to take part.

- If they accepted, an appointment was scheduled.
- If they declined, the first substitute was contacted.
- After two unsuccessful phone calls (no answer), the next substitute was contacted.

^{II} The question was: "We would like to know your current health status as you experience it today. Please tick the box next to the statement that best describes your health today. Mobility today (walking about): (1) no problems; (2) slight problems; (3) moderate problems; (4) severe problems; (5) unable to walk."



Because some segments could not be fully completed, additional participants were drawn at random to fill the remaining places. We prioritised individuals who had received their diagnosis more than two years earlier, in order to avoid excessive overlap with participants who might have long COVID.

8.4.2 Qualitative data collection tools

The semi-structured interview guide used for this study was developed based on the generic version for both patients and proxies as described in KCE Report 348^{mm, 128}.

The generic interview guides covered the following topics:

- Impact on psychological and physical health;
- Symptoms;
- Daily life functioning, including sexual life and reproductive health;
- Interaction with other health problems;
- Patient journey leading up to diagnosis;
- Experience with treatment (received, refusal, administration);
- Therapeutic needs: information needs, forgone care, accessibility, care network;
- Perception of quality of care: coordination, continuity, relationship with care providers;
- Financial burden;
- Administrative burden;
- Impact of the disease on social activities, relationships, work, education and finance;
- Identification of formal and informal support networks: received and forgone support;
- Identification of potential solutions to address unmet needs in healthcare themes;
- Unmet needs expressed throughout the interview that should be addressed as a priority;
- Impact on informal caregivers' health, time, social life, support and information needs.

Disease-specific adaptations were subsequently made to the generic version to ensure its applicability and relevance in the context of ME/CFS.

8.4.3 Qualitative data collection process

One-to-one interviews were conducted on-line face-to-face via Zoom® or Teams® in French by a native French-speaking researcher (MD), or in Dutch by 2 native Dutch-speaking researchers (JC and WC). Participants had the opportunity to complete the interview in several sessions if they felt too tired to continue. In cases of extreme severity of the disease (maximum of four patients), the interview could have been conducted via chat on the Zoom® application. If the patient did not have the energy to speak, the interviewer could have asked questions verbally and the person could have responded in writing. This data-collection approach was not part of the generic NEED protocol and was introduced to accommodate the specific symptom profile of ME/CFS, at the request of the patient associations involved in the project.

The online interviews occurred in October and November 2025. Prior to starting the interviews, the interviewers clarified the research objectives, ensured confidentiality and sought permission for audio recording. Each interview lasted approximately 90 minutes, was audio- and video-recorded, and

^{mm} Patient and proxy interview guides are available upon request in French and Dutch.

transcribed using Whisper AI® or an external transcription office. Transcripts were then checked and completed *ad verbatim* by native French- and Dutch-speaking team members (LK and WC). They reviewed and completed the transcripts to ensure linguistic accuracy and completeness. All names of participants, institutions, and care providers were removed from the transcripts.

8.4.4 Qualitative data analysis

We conducted a thematic inductive analysis using QSR NVivo 13®. One researcher (WC) who was also involved in the interviewing and quality check established a basic node structure through open coding of three Dutch transcripts. Subsequently, this initial node tree was discussed with a second researcher (LK) who coded the French transcripts. When the open coding of the Dutch transcripts was finished, nodes were grouped (axial coding) to align with the main themes of the NEED framework.¹²⁸ The coding of the French transcripts started, using the same node tree, but with the possibility of creating new nodes when new themes emerged from the French data that were not represented in the initial node tree based on Dutch data only.

8.4.5 Reporting of the qualitative findings

To illustrate findings from interview data, participant testimonials were incorporated in the report using literal quotations drawn directly from the interviews in the participants' mother tongue.

8.5 Ethical approval

The methodological approach, including the research protocol, the questionnaire and recruitment material (reference: P2025/142/CCB B4062025000053), was submitted to and approved by the Ethics Committee from Hôpital Universitaire de Bruxelles-Université Libre de Bruxelles - HUB-ULB (021/406)ⁿⁿ.

9 RESULTS

9.1 Description of respondents

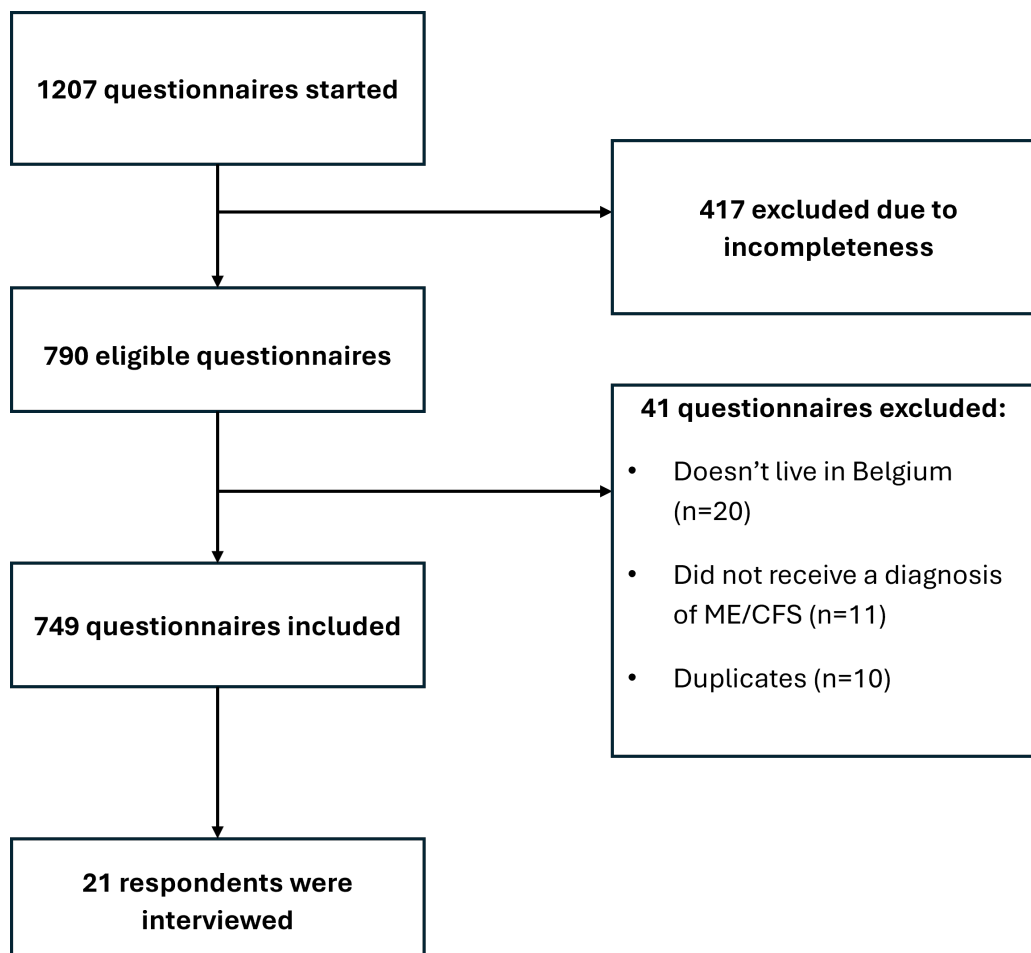
9.1.1 Survey respondents

Figure 2 shows that 1207 individuals started the survey. Of those, 790 respondents completed all questions. Of these, 41 participants did not meet inclusion criteria and were therefore excluded, yielding a final sample of 749 participants of whom 21 also participated in the interviews.

ⁿⁿ A copy of the informed consent form for participation in the interviews is available upon request.



Figure 2 – Flow diagram of number of respondents participating in the online survey and interviews



9.1.1.1 Demographic information

Almost all respondents (97.2%) completed the online questionnaire on their own behalf, while a smaller proportion (2.8%) responded on behalf of another adult (family member or informal caregiver). Most participants were women (89.9%), while 9.6% were men and 0.5% identified as another gender (Table 3). Belgium was the birth country of most respondents (95.9%) and Flanders was the most represented region of residence, accounting for 88.6% of respondents. In terms of language, 88.7% of respondents filled out the questionnaire in Dutch. The mean age of respondents was 50.6 years (SD: 0.03); 1.1% were younger than 25, and 14.3% were 65 years or older.

Most respondents completed secondary education (35.2%) or higher education (63.6%). A small proportion had no formal diploma or only primary education (1.2%). Regarding living situation, more than half of the respondents (53.9%) lived together with a partner (with or without children) and 37.9% lived without a partner (with or without children). Only a minority lived in shared accommodation/housing (1.1%) or a nursing home (0.3%).

Table 3 – Demographic characteristics of online survey respondents

Variable	n (%)
Respondent (n=749)	
<i>Individual with ME/CFS</i>	728 (97.2%)
<i>Legal representative</i>	21 (2.8%)
Gender (n=749)	
<i>Woman</i>	673 (89.9%)
<i>Man</i>	72 (9.6%)
<i>Other</i>	4 (0.5%)
Survey language (n=749)	
<i>Dutch</i>	664 (88.7%)
<i>French</i>	85 (11.3%)
Country of birth (n=749)	
<i>Belgium</i>	718 (95.9%)
<i>Other country</i>	231 (4.1%)
Region of residence (n=748)	
<i>Flanders</i>	663 (88.6%)
<i>Wallonia</i>	67 (9.0%)
<i>Brussels</i>	18 (2.4%)
Age group (in years) (n=749)	
<i>18-24</i>	8 (1.1%)
<i>25-44</i>	193 (25.8%)
<i>45-64</i>	441 (58.9%)
<i>65-74</i>	93 (12.4%)
<i>>75</i>	14 (1.9%)
Highest level of education (n=744)	
<i>No diploma or primary education*</i>	9 (1.2%)
<i>Secondary education**</i>	262 (35.2%)
<i>Higher education***</i>	473 (63.6%)
Living situation (n=747)	
<i>Alone (with or without children)</i>	283 (37.9%)
<i>With partner (with or without children)</i>	403 (53.9%)
<i>With one or two parent(s)</i>	51 (6.8%)
<i>Shared accommodation/housing</i>	8 (1.1%)
<i>Nursing home</i>	2 (0.3%)
* No diploma or primary education, ** Lower secondary education, Upper secondary education; *** Bachelor's, master's, or doctoral degree	
Note: percentages may not add up to 100% because of rounding	

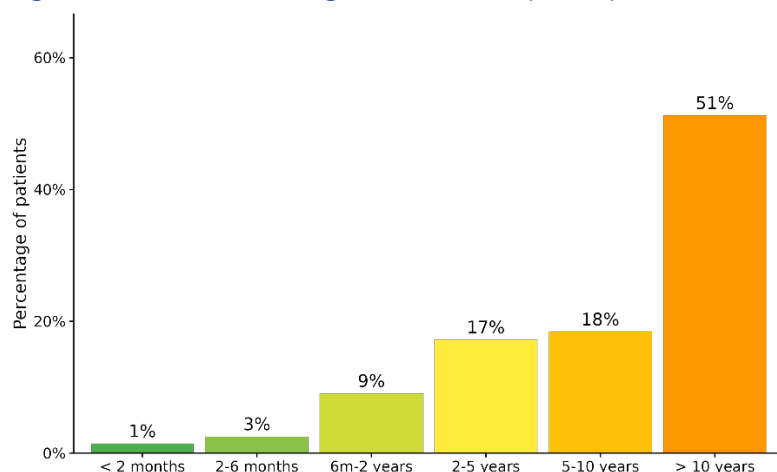


9.1.1.2 Disease-specific information

Time since diagnosis and hospitalisation

Around 50% of the respondents received their ME/CFS diagnosis more than 10 years ago. Only a very small percentage (4.0%) received their diagnosis in the past six months (Figure 3). 19.6% of the respondents have already been hospitalised due to ME/CFS, whereas the remainder have not yet been hospitalised (78.0%), or replied don't know (2.4%).

Figure 3 – Time since diagnosis ME/CFS (n=749)



Legend: percentages may not add up to 100 % because of rounding.

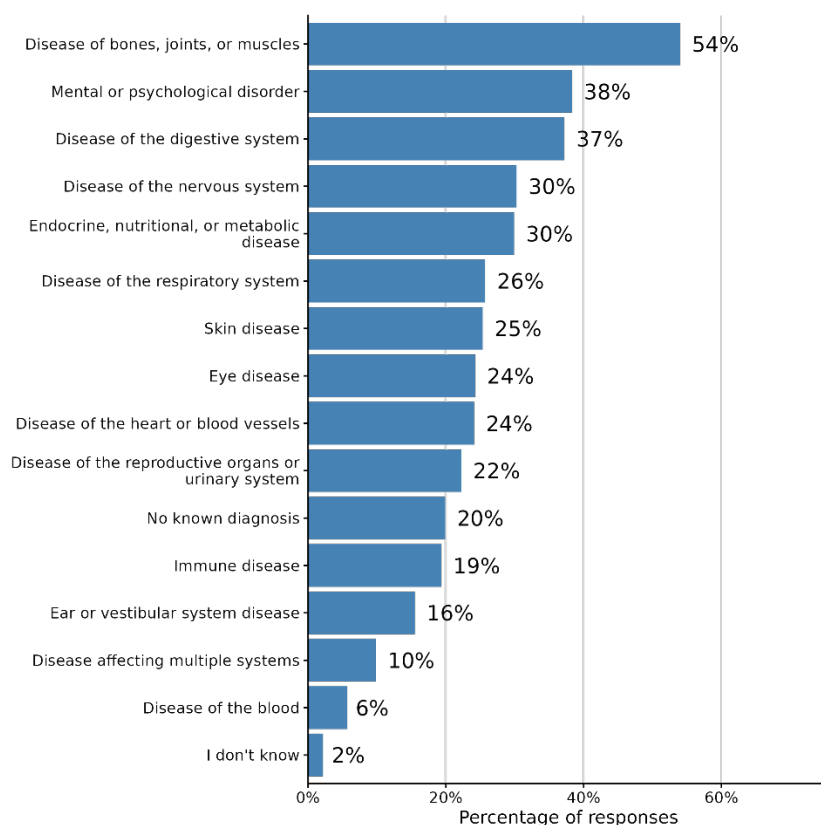
Comorbidities related to ME/CFS and other comorbidities

In addition to ME/CFS, 73.2% of the respondents also reported having one or more other health problems related to ME/CFS while 18.7% only had ME/CFS, and 8.1% replied don't know. Fibromyalgia was the most frequently reported comorbidity (63.3%), followed by irritable bowel disease (56.9%) and orthostatic hypotension (20.6%) (Table 4). Regarding other comorbidities, more than half of the patients reported bone, joints or muscle problems (54.0%), followed by mental health or psychological problems (38.0%), and digestive problems (37.0%) (Figure 4).

Table 4 – Comorbidities related to ME/CFS

Comorbidity (n=749)	n (%)
Fibromyalgia	474 (63.3%)
Irritable bowel syndrome	426 (56.9%)
Orthostatic hypotension	154 (20.6%)
Postural tachycardia syndrome	127 (17.0%)
Long COVID	123 (16.4%)
Small fiber neuropathy	96 (12.8%)
Mast cell activation syndrome	79 (10.5%)

Legend: Percentages add up to more than 100% because participants could select multiple comorbidities.

Figure 4 – Other comorbidities

Legend: N= 749; percentages add up to more than 100% because participants could select multiple comorbidities

9.1.2 Interviewed participants

We intend to interview 24 patients according to our segmentation criteria (Figure 1). The main aim of the interviews was to get a general view of patient's experiences with ME/CFS, as well as their health, healthcare and social needs, in complement to the survey, and in correspondence with the NEED framework. The interviews aimed to get a general view of patient's experiences with ME/CFS, as well as their health, healthcare and social needs, in complement to the survey, and in correspondence with the NEED framework (see 2.1).

9.1.3 Selection of participants and recruitment

The aim was to conduct 24 individual interviews, 12 in French and 12 in Dutch. All interviewees were selected from survey respondents who indicated their willingness to participate in a subsequent in-depth interview (see 8.4.1).

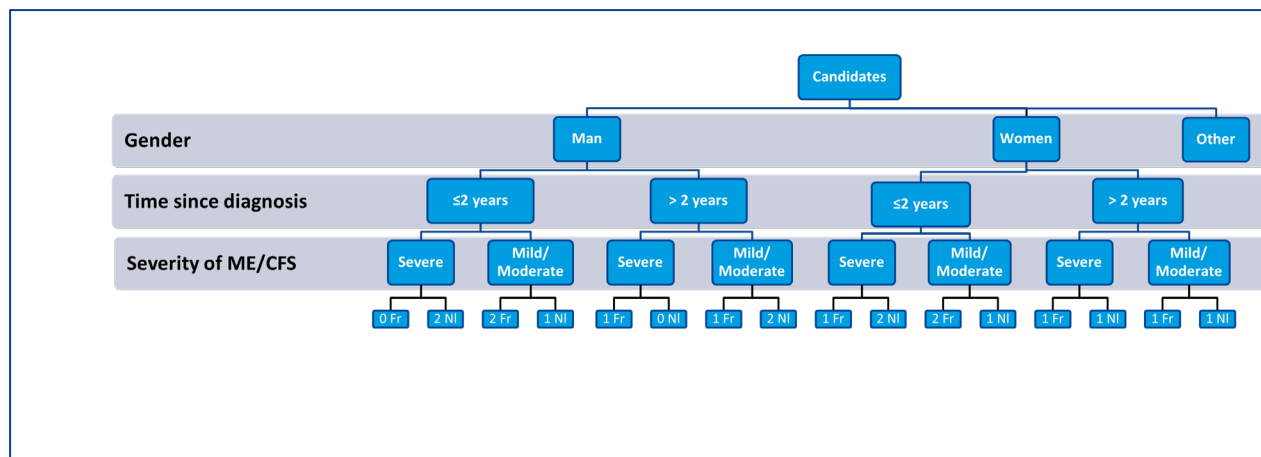
To ensure maximum variability among respondents, researchers prioritised recruiting participants in each language according to the following segmentation criteria as outlined in the section 8.4.1.

We aimed to conduct 24 individual interviews but completed 21, as we were unable to identify three participants who met the selection criteria, as described in Figure 5, in the missing cells. Although we interviewed 21 patients, two participants were excluded because it became apparent during the interviews that they were self-diagnosed, which was an exclusion criterion: one French-speaking female with 'recently diagnosed' ME/CFS and severe symptoms, and one Dutch-speaking male with 'more than two years diagnosed' ME/CFS and severe symptoms. A description of the characteristics



of the interviewees can be found in Table 5. All interviews were conducted remotely in a face-to-face videoconference format, with no participants using the chat-based response option.

Figure 5 – Final sample of interviewees: included and not reached participants



Legend: A value of 0 denotes that no participants was enrolled in that cell despite the planned sample. Fr: French-speaking individual; NI: Dutch-speaking individual.

Table 5 – Description of interviewees

ID	Language	Gender	Time since diagnosis	Severity
R1*	French	Male	≤ 2 years	Mild/moderate
R2	French	Male	> 2 years	Mild/moderate
R3	French	Male	> 2 years	Severe
R4	French	Female	≤ 2 years	Mild/moderate
R5	French	Female	≤ 2 years	Severe
R6	French	Female	> 2 years	Mild/moderate
R7	French	Female	> 2 years	Severe
R8	French	Male	≤ 2 years	Mild/moderate
R9	French	Female	≤ 2 years	Mild/moderate
R10	Dutch	Male	≤ 2 years	Mild/moderate
R11	Dutch	Male	≤ 2 years	Severe
R12	Dutch	Female	≤ 2 years	Mild/moderate
R13	Dutch	Female	≤ 2 years	Severe
R14	Dutch	Female	> 2 years	Mild/moderate
R15	Dutch	Female	> 2 years	Severe
R16	Dutch	Male	> 2 years	Mild/moderate
R17	Dutch	Female	≤ 2 years	Severe
R18	Dutch	Male	> 2 years	Mild/moderate
R19	Dutch	Male	≤ 2 years	Severe

Legend : * Patient representative

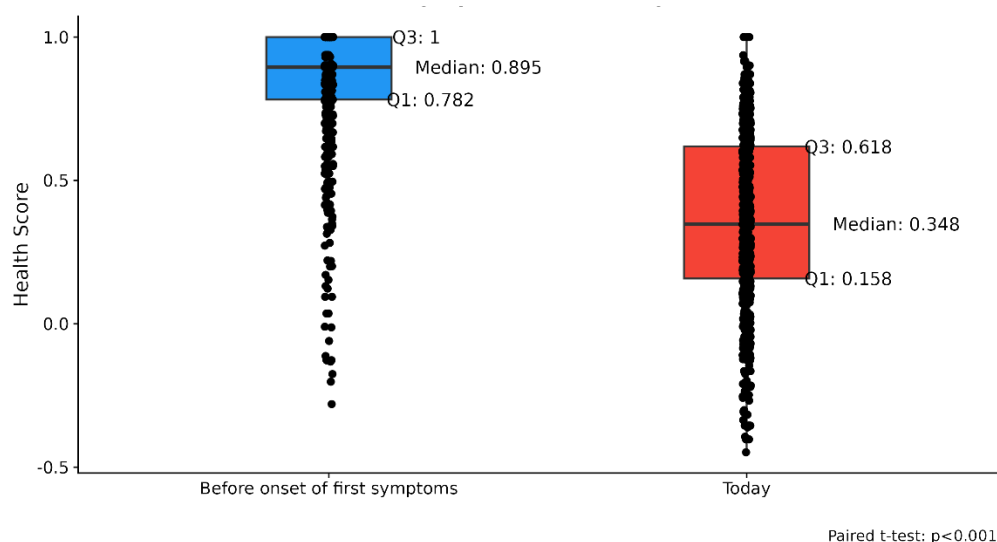
9.2 Unmet patient needs: Health needs

9.2.1 Impact on general health-related quality of life

9.2.1.1 Health-related quality of life (EQ-5D-5L utility scores)

The impact of living with ME/CFS was assessed by calculating the difference between the two EQ-5D-5L scores, as reported in the survey by each participant. One score retrospectively referred to their status before the onset of ME/CFS, and the other reflected their current situation at the time of survey completion ("today"). The EQ-5D-5L utility score decreased significantly from a mean of 0.84 (SD, 95% CI = 0.21, 0.82 to 0.85) before first symptoms to 0.36 (0.30, 0.34 to 0.38) at the moment of survey completion ($p < 0.001$). Median, Q1 and Q3 values before onset of first symptoms and today are depicted in Figure 6.

Figure 6 – Changes in EQ-5D-5L utility score before onset of first symptoms and today (n=745)



The exploratory subgroup analyses (not adjusted for multiple comparisons), conducted according to a selection of relevant health-, healthcare-, and social-related variables assessed in the survey (which are described in broad terms later in the *Results section*), for change in EQ-5D-5L utility scores (Table 6), yielded several noteworthy findings:

- When looking at **subgroup analyses for relevant demographic variables**, the decrease in utility scores significantly differs for age, with participants between 30 and 50 years old reporting the largest decrease in utility scores (mean: -0.52, reflecting a mean drop of 0.52 from pre-symptom status to the status at survey completion), compared to participants younger than 30 (-0.41) or older than 50 years old (-0.46).
- **Regarding disease-related variables**, a higher number of reported unmet needs, and a more important burden of symptoms was associated with larger decreases in utility scores ($P < 0.001$). People who were also diagnosed with long COVID (n=123) showed a significantly larger decrease in utility scores (-0.55), compared to people without long COVID (-0.45, $P: 0.007$).
- Subgroup analyses **stratified for healthcare-related variables** showed significant differences regarding the time to diagnosis with participants who had to wait more than two years for their diagnosis, reporting a larger decrease in utility scores (-0.51) compared to participants who received their diagnosis within two years (-0.43). Also, participants who found their care was not well organised, were not satisfied with the information they received, had low involvement in their treatment decisions, rated the level of experience of healthcare practitioners to be low, and



experienced accessibility issues reported significantly larger decreases in utility scores compared to participants who had better experiences with their care (see Table 6).

- Subgroup analyses **stratified for social variables** showed that participants with a high or very high impact of ME/CFS on their social relationships reported substantially higher decreases in utility scores, compared to participants experiencing less impact (Table 6). Participants having financial difficulties reported a significantly larger decrease in utility scores (-0.54), compared to participants without financial difficulties (-0.36).

No significant differences were found between subgroups ($P > 0.05$) for the other variables considered in the subgroup analysis. An overview of all descriptives, and P-values can be found in Appendix 3.

Table 6 – Difference in EQ-5D-5L utility score before onset of symptoms versus today by demographic, disease, healthcare and social variables

	Mean (SD)	Median (Q1, Q3)	Missing (n)	P-value
Demographic variables				
Age				0.026
<30 years (n=31)	-0.41 (0.38)	-0.42 (-0.62, -0.20)	1	
30-50 years (n=292)	-0.52 (0.32)	-0.51 (-0.74, -0.28)	4	
>50 years (n=426)	-0.46 (0.35)	-0.48 (-0.71, -0.22)	15	
Disease-related variables				
Presence of long COVID				0.007
Yes (n=123)	-0.55 (0.33)	-0.59 (-0.77, -0.32)	1	
No (n=509)	-0.45 (0.35)	-0.45 (-0.68, -0.22)	13	
Number of unmet needs				<0.001
0-1 (n=336)	-0.38 (0.34)	-0.36 (-0.62, -0.17)	10	
2-3 (n=267)	-0.52 (0.31)	-0.52 (-0.73, -0.31)	7	
>=4 (n=146)	-0.65 (0.32)	-0.66 (-0.87, -0.46)	3	
Burden of symptoms				<0.001
Low (n=263)	-0.34 (0.31)	-0.31 (-0.53, -0.15)	6	
Medium (n=259)	-0.51 (0.31)	-0.51 (-0.70, -0.34)	6	
High (n=227)	-0.62 (0.36)	-0.67 (-0.85, -0.36)	8	
Healthcare-related variables				
Time to diagnosis				0.010
<2 years (n=297)	-0.43 (0.34)	-0.42 (-0.70, -0.23)	4	
>2 years (n=249)	-0.51 (0.34)	-0.50 (-0.73, -0.28)	6	
Satisfaction with organisation of care and planning over time				<0.001
Yes (n=288)	-0.39 (0.35)	-0.37 (-0.62, -0.17)	1	
No (n=422)	-0.56 (0.32)	-0.56 (-0.76, -0.35)	14	
Satisfaction with information received from HCPs				<0.001
Yes (n=356)	-0.43 (0.33)	-0.43 (-0.67, -0.19)	4	
No (n=363)	-0.54 (0.35)	-0.55 (-0.76, -0.30)	11	
Satisfaction with level of involvement in treatment choice				<0.001

Yes (n=430)	-0.46 (0.34)	-0.47 (-0.70, -0.24)	4	
No (n=161)	-0.57 (0.32)	-0.59 (-0.76, -0.37)	6	
Satisfaction with level of expertise of the HCPs with ME/CFS				<0.001
Low (n=416)	-0.54 (0.34)	-0.55 (-0.76, -0.30)	12	
Medium/high (n=278)	-0.42 (0.33)	-0.43 (-0.66, -0.19)	2	
Having accessibility problems				<0.001
Yes (n=316)	-0.59 (0.32)	-0.60 (-0.80, -0.38)	9	
No (n=288)	-0.37 (0.33)	-0.36 (-0.60, -0.16)	5	
Social variables				
Impact on social relationships				<0.001
Low to medium (n=190)	-0.28 (0.31)	-0.26 (-0.48, -0.10)	2	
High (n=347)	-0.49 (0.28)	-0.51 (-0.68, -0.28)	8	
Very high (n=206)	-0.66 (0.36)	-0.68 (-0.90, -0.47)	6	
Financial situation				<0.001
Easy (n=232)	-0.36 (0.32)	-0.34 (-0.57, -0.16)	5	
Hard (n=506)	-0.54 (0.34)	-0.56 (-0.76, -0.31)	15	

We conducted exploratory subgroup analyses (not adjusted for multiple comparisons) comparing French- versus Dutch-speaking survey participants across demographic, disease-related, healthcare-related, and social variables. These findings should be interpreted as hypothesis-generating rather than definitive results and should be treated with caution since the marked imbalance in the number of French speaking (n=85) versus Dutch speaking participants (n=664). An overview of all descriptives can be found in Appendix 4. Among the main findings, French-speaking respondents were more likely to live alone, and a higher proportion of Dutch-speaking participants had been diagnosed with ME/CFS for more than five years (72.3 %), compared with French-speaking participants (49.4%). French-speaking participants experienced a higher decline in the utility score (-0.57 vs -0.47; P=0.005) and reported a higher number of unmet needs, when compared to Dutch-speaking individuals. Regarding healthcare-related variables, more French-speaking participants more frequently reported dissatisfaction with the organisation of their care (73.8%) and accessibility issues (75.4%), compared to the Dutch-speaking respondents (57.5% and 49.5% respectively). French-speaking participants also highlighted a higher impact of ME/CFS on their social relationships. For the other variables considered, no remarkable differences were observed between subgroups (see Appendix 4).

Analysis of the interview data revealed a substantial burden on patients' quality of life.

« Imaginez que vous avez quatre heures de vie en moins tous les jours, parce que vous devez vous reposer et que le reste du temps, dans la journée active, vous fonctionnez à 50 % de vos capacités. » (R8-M, ≤ 2 yrs, Mild/moderate)

“En nu eigenlijk van bed naar de zetel gaan. Ik ben afhankelijk hier binnen van een kappersstoel met wielen onder. [...] Ik ben afhankelijk van een scootmobiel. Weinig nog tot niet buiten komen. Eerder beperkt tot doktersraadplegingen, onderzoeken. En die keren dat je denkt, ik voel mij zo verschrikkelijk eenzaam. Alles komt op mij af. [...] Eigenlijk word je met alles afgestraft. Onlangs heb ik hier meer dan 22 dagen in een heel zware PEM gelegen. Geen licht kunnen verdragen, geen geluid kunnen verdragen. Mijn man niet rond mij kunnen verdragen. Geen bezoek van familieleden, dus mijn dochter ofzo. Zodat je eigenlijk zelfs zoiets hebt van, wat loop ik hier nog te doen.” (R13-F, ≤ 2 yrs, Severe)

The experiences we collected are further illustrated across the quality-of-life subdimensions.



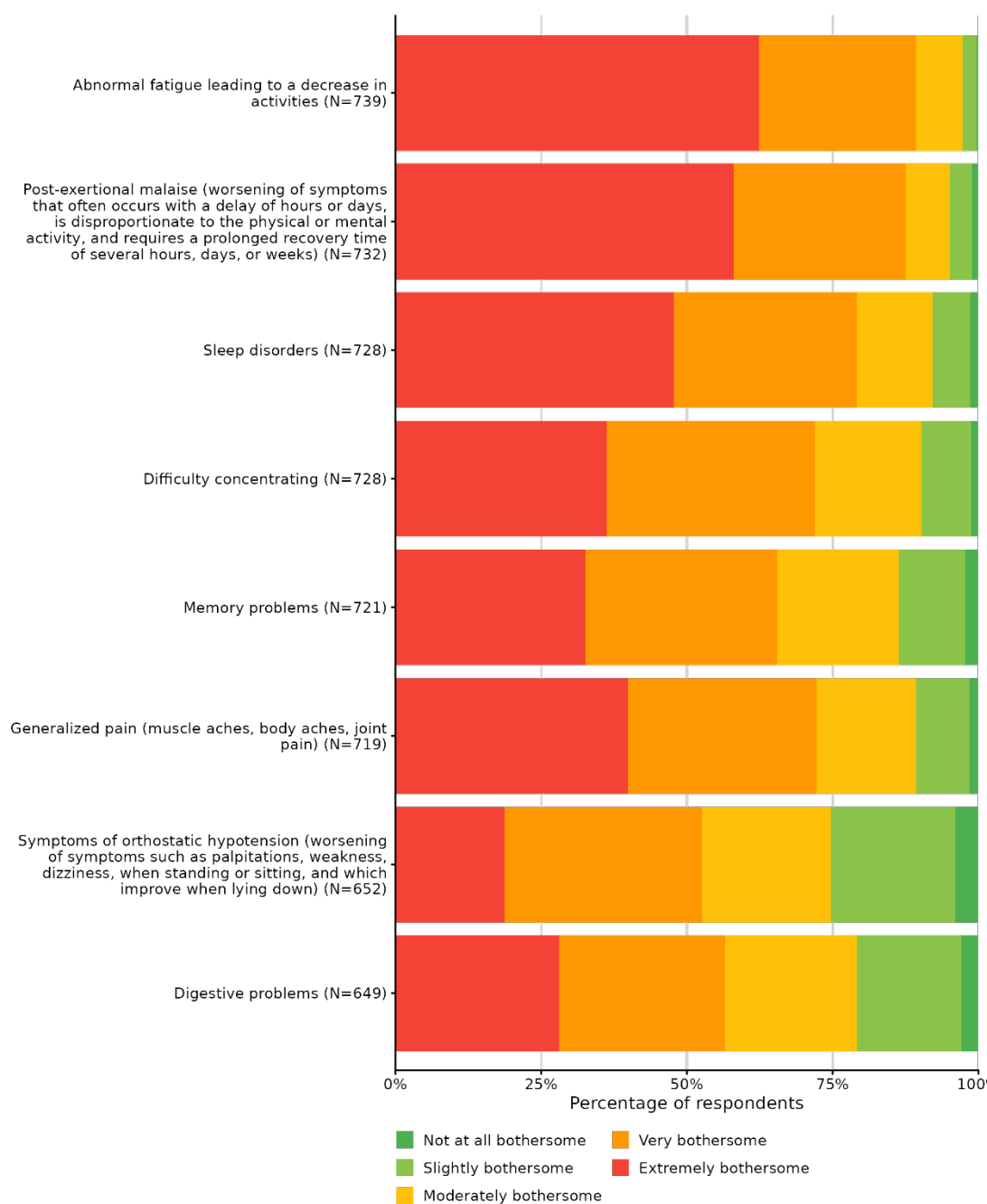
9.2.2 Impact on physical health

9.2.2.1 Experienced burden of physical symptoms related to ME/CFS

Figure 7 and Figure 9 show the frequency and perceived burden of the ten most commonly reported symptoms, selected from the 30 symptoms listed in the questionnaire.

Regarding physical symptoms (Figure 7), nearly all respondents reported abnormal fatigue (98.7%), post-exertional malaise (97.7%), and sleeping disorders (97.2%). These symptoms also carried the greatest perceived burden, with more than 75% of the respondents rating them as very or extremely bothersome.

Figure 7 – Frequency of physical symptoms and distribution of their perceived burden ordered by decreasing symptom frequency



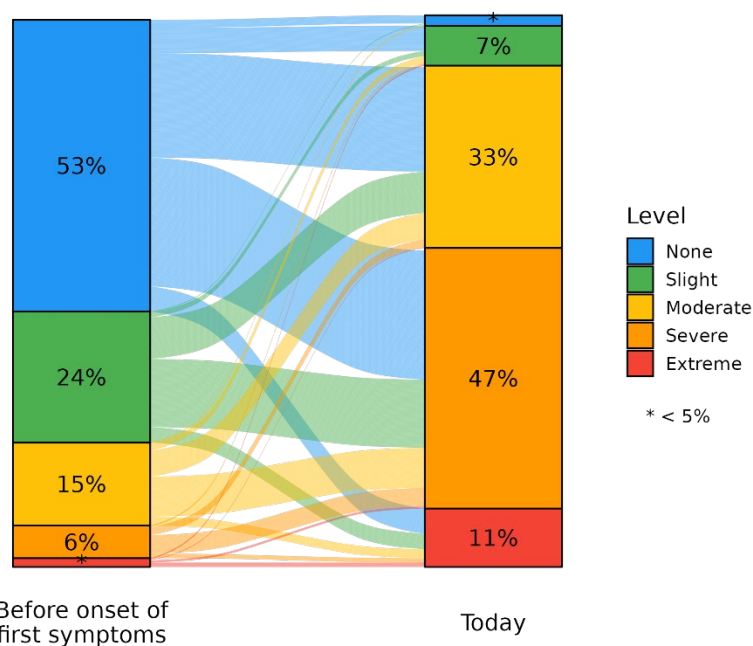
Responses “I don’t know” (n=1 for abnormal fatigue; n=2 for post-exertional malaise; n=4 for sleep disorders; n=1 for generalised pain; n=17 for symptoms of orthostatic hypotension; n=8 for digestive problems) and “I didn’t get this symptom” (n=5 for abnormal fatigue; n=11 for post-exertional malaise; n=13 for sleep disorders; n=17 for difficulty concentrating; n=24 for memory problems; n=25 for generalised pain; n=76 for symptoms of orthostatic hypotension; n=88 for digestive problems) were not displayed in the figure.

9.2.2.2 Pain/discomfort before disease vs today (EQ-5D-5L subdomain)

To assess changes in physical health over time, respondents reported their levels of pain and discomfort both before the onset of ME/CFS symptoms (left) and at the moment of survey completion (“today”) (right), using the EQ-5D-5L tool. The Sankey diagram depicted in Figure 8 visualises how respondents moved between categories of perceived pain or discomfort. Prior to symptom onset, most respondents reported no problems (53.0%) or only slight problems (24.0%). At the time of the survey, this distribution had shifted substantially: only a minority (<5.0%) still reported no problems, while 33.0% and 47.0% rated moderate and severe discomfort, respectively.

Figure 8 – Changes in the self-reported level of problems with pain/discomfort (EQ-5D-5L) before onset of first symptoms and today

Changes in self-reported level of problems with pain/discomfort of the EQ-5D-5L before onset of first symptoms and today (N=745)



Legend: Responses “I don’t know” (n=6 before onset of first symptoms and n=0 today) were not displayed in the figure

Patient interviews indicated that some symptoms are not outwardly visible yet can be deeply disabling. The analysis highlighted the ongoing burden of repeatedly documenting, justifying, and explaining debilitating symptoms.

In addition, the lack of a clear clinical picture and the variations in symptoms between patients may generate uncertainty as to whether a symptom is attributable to ME/CFS or caused by another condition, such as cancer. When a (new) symptom arose, this uncertainty contributed to health-related anxiety, with concerns that an additional condition might have been overlooked because symptoms were mistakenly attributed to ME/CFS.



« Dans cette fatigue chronique, c'est ça qui... C'est étonnant. Moi, j'ai des grosses difficultés à dormir. Je suis fatiguée. Mais je me demande si je n'ai pas un autre souci à ce niveau-là peut-être maintenant. Parce que... Et c'est ça que... J'aimerais passer un test du sommeil pour voir un peu... Pour que ce soit examiné correctement et... » (R4-F, ≤ 2 yrs, Mild/moderate)

In addition, health issues seemed framed as self-perpetuating cycles, in which one symptom triggered another, resulting in a persistent state of sickness with fluctuating complaints. For example, migraine-related pain was described as intensifying fatigue, which in turn was perceived to precipitate migraine episodes.

« Quand je fais une migraine, par exemple, là, ça va être plus douloureux, évidemment. Mon corps le paye, enfin vraiment, ça va chercher dans mon énergie. Je vais le payer par après. Donc, dès que la migraine est partie, oui, il y a un mieux, la douleur n'a plus là, mais mon corps a trinqué. Et à nouveau, de prime abord, on n'y penserait pas nécessairement de se dire la douleur est fatigante aussi, en fait. Donc, oui, c'est compliqué parce que ces symptômes font une espèce de cercle vicieux où, du coup, on ne se sent pas bien. » (R6-F >2 yrs, Mild/moderate)

Another example is the PEM occurring after an activity. A day with less fatigue or pain was perceived as an opportunity to do something extra. Although this could initially seem manageable and was not necessarily followed by immediate deterioration in physical condition relapse was described as occurring after a delay (e.g. the next day). Overall, a “good day” was described as being followed by a number of “bad days.”

“En dan is er eens een dag dat je wel energie hebt. Dan wil je je die uiteraard gaan benutten. Want dan heb je zelfs zin. En dan zeg je ik voel me nu goed. Ik heb nu energie. Ik kan terug dingen gaan doen. En dan ga je al direct dingen doen. En het heeft lang geduurd voor mij eer ik door had dat de dag daarna dat ik terug heel fel achteruitga. Het voelt niet aan dat dat komt door de activiteiten die ik gedaan heb. Het is helaas niet zo dat als je te veel doet dat ik me dan onmiddellijk terug slecht voel. Dat is vaak pas de dag erna. Of zelfs dagen erna. Dat dat zijn weerslag heeft.” (R17-F, ≤ 2 yrs, Severe)

As soon as fatigue or exacerbation of symptoms becomes apparent, one possible strategy to avoid worsening of symptoms is to withdraw from external stimuli and remain in bed.

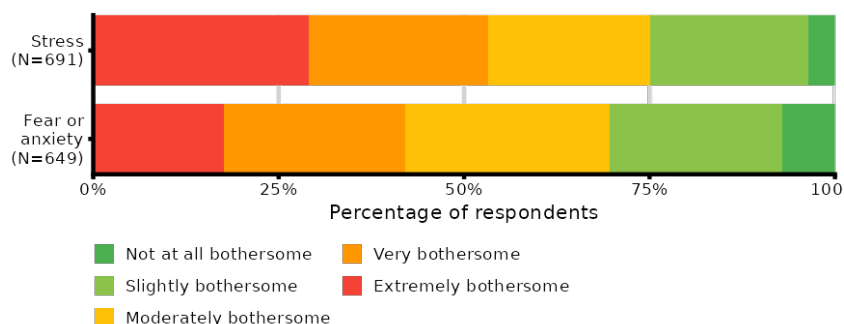
« Une journée feu rouge, c'est une journée majoritairement alitée. Donc, sinon, je suis semi-allongée. Là, c'est vraiment alitée. Et avec à nouveau un peu de télévision, mais très peu,... il faut que ce soit pas trop émotionnel, pas trop bruyant, qu'il n'y ait pas des images trop rapides. Quelque chose de... Des documentaires animaliers, par exemple. C'est très bien. Et oui, je vais être beaucoup plus isolée dans les journées rouges, (...) Et je ne vais pas dormir, mais être comme dans un état un peu de sieste quasiment toute la journée. » (R6-F, >2 yrs, Mild/moderate)

9.2.3 Impact on psychological health

9.2.3.1 Experienced burden of psychological symptoms related to ME/CFS

While most of the 10 most frequent symptoms assessed in the survey were related to physical wellbeing, stress and fear or anxiety were symptoms related to psychological health and therefore classified separately (Figure 9). Almost 75 % of respondents experienced these psychological symptoms to be moderately, very or extremely bothersome.

Figure 9 – Frequency of psychological symptoms and distribution of their perceived burden ordered by decreasing symptom frequency

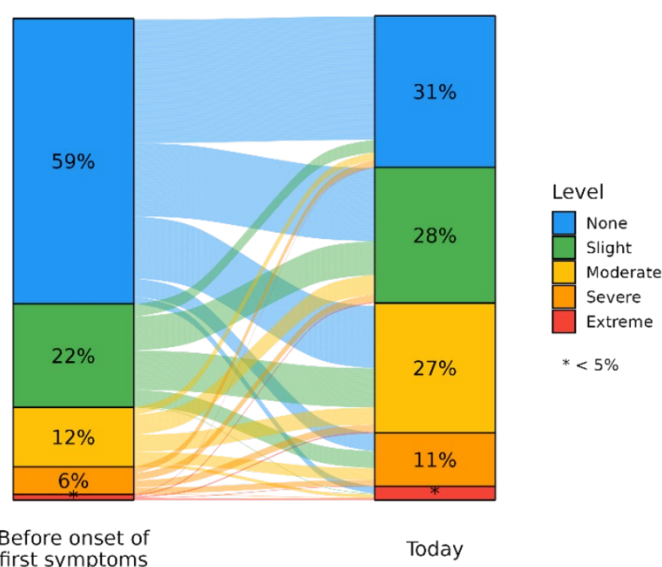


Responses “I don’t know” (n=1 for stress; n=6 for fear or anxiety) and “I didn’t get this symptom” (n=53 for stress; n=90 for fear or anxiety) were not displayed in the figure.

9.2.3.2 Anxiety/depression before disease vs today (EQ-5D-5L subdomain)

To assess changes in psychological health over time, respondents reported their levels of anxiety and depression both before the onset of ME/CFS symptoms (left) and at the moment of survey completion (“today”) (right), using the EQ-5D-5L tool. The Sankey diagram depicted in Figure 10 visualises how respondents moved between categories of anxiety/depression. Prior to symptom onset, most respondents reported no problems (59.0%) or only slight problems (21.0%). At the time of survey completion, the proportion of respondents with no problems dropped (31.0%), while reports of slight problems (28.0%) and moderate problems (27.0%) became more common. The proportion of respondents with extreme problems remained low (<5.0%).

Figure 10 – Changes in the self-reported level of problems with anxiety/depression (EQ-5D-5L) before onset of first symptoms and today (n=745)



Legend: Responses “I don’t know” (n=7 before onset of first symptoms and n=3 today) were not displayed in the figure.

A slight to moderate increase in mental health problems was also prominently reflected in the interview data. Reasons included a range of interrelated problems, including the experience of daily life as increasingly exhausting, feelings of abandonment and disappointment, heightened self-doubt and self-



stigmatisation, a diminished sense of future prospects, difficulties in fulfilling expected social roles, a perceived loss of identity, and ongoing financial insecurity.

AN ENDLESS STRUGGLE

Daily life with ME/CFS was portrayed as an endless struggle. Even routine activities, including basic tasks such as household chores or self-care (e.g. shower), were experienced as highly demanding. As a result, individuals reported needing to prioritise and limit activities per day. Energy was reserved for essential tasks, which was associated with frustration, anger, and sadness linked to physical limitations, home confinement, and a loss of participation in valued aspects of life. Overall, physical symptoms were closely intertwined with a substantial emotional burden. In this context, we have even heard that, euthanasia might even be considered, should it be accessible.

“Als ik bijvoorbeeld boodschappen moet doen, moet ik niet meer zeggen, ik ga de afwas doen of ik ga eten klaarmaken. Ik moet... Ja, nee, alles... Alles is een strijd. Ja. Ik krijg het weer moeilijk. Het is zo lastig. Ja. Het leven is zo lastig. Ja. Ik probeer altijd optimistisch en opgewekt te zijn, maar eigenlijk dissocieer ik heel veel om eigenlijk niet geconfronteerd te worden met... Nu dat we daarover praten, worden we weer mee geconfronteerd, wat je eigenlijk maar kan op een dag, hé”. (R15-F, >2 yrs, Severe)

“En dat gaat niet alleen over werken, maar ook over de leuke dingen. Zo een nachtje uitgaan en zo tot een uur of drie wakker blijven. Zo ik heb daar gewoon enorm veel last van achteraf. En die dingen... Ik moet wel gewoon zo verder. Maar ik... ik voel mij vaak wel neerslachtig of ja, gewoon verdrietig. Omdat ik er eigenlijk niet mee eens ben. Zo ik vind het heel oneerlijk dat ik op mijn leeftijd mijn verwachtingen of mijn dromen niet kan waarmaken. Gewoon omdat mijn lichaam niet mee wil.” (R19-M, ≤ 2 yrs, Severe)

“Et sinon c'était vraiment le corps. Ce n'est pas que l'envie n'était plus là, c'était le corps qui ne suivait plus du tout. Donc c'était d'autant plus frustrant parce que l'envie était là, mais on sentait vraiment un blocage physique quoi. » (R9-F, ≤ 2 yrs, Mild/moderate)

« Si je n'avais pas un fils, j'irais vers la piste de l'euthanasie très clairement ! Ça fait 5 ans et demi maintenant que je suis malade (...) mais à un moment donné une fois qu'il sera installé dans la vie et plus dépendant de moi, c'est-à-dire qu'il aura son diplôme qu'il aura un job, si mes douleurs chroniques sont toujours aussi insupportables, si ma vie en est réduite à dormir 18h sur 24, mes amis je les vois encore mais de moins en moins franchement (...)... mais si je continue à me dégrader comme je me suis dégradé ces dernières années, à ne plus pouvoir aller chercher un pain dans ma rue ou des trucs comme ça ... une fois que mon fils sera vraiment autonome, je pense que je me permettrai l'euthanasie si la douleur est toujours aussi insupportable » (R2-M, >2 yrs, Mild/moderate)

FEELING BEING LET DOWN AND DISAPPOINTED

A perceived failure of the healthcare system emerged, accompanied by a pronounced sense of being let down. Experiences of not being heard were identified, as complaints were often dismissed or classified as psychological. Uncertainty about where to seek care and be taken seriously was reported, along with feelings of embarrassment or guilt associated with the need for medical support. Feeling of hurt arose when interviewees were judged as lazy or “crazy”, particularly when such judgments were made by healthcare providers. Feelings of despair were mentioned, with life itself being questioned.

“Het ergste wat mij is overkomen in heel deze traject is dat hij [de huisarts] tegen mij zei: die ziekte die bestaat niet, dat is iets verzonnen van deze tijd en als jij daarin wilt voortgaan dan kan ik je niet helpen dan mag je vertrekken. Ik zat daar met mijn mond vol tanden en ik had zoiets van oké maar ik heb hier niet voor gekozen [...] en het feit dat een arts eigenlijk de hulp naar iemand toe weigert omdat die ervan overtuigd is dat het iets is wat niet bestaat. [...] Ik had toen ook zoiets van ja wat moet ik nu doen.” (R19-M, ≤ 2 yrs, Severe)

“Maar ja, als patiënt, [...] voel je je gewoon volledig in de steek gelaten. En ja, dat is heel lastig, want je botst tegen systemen die, wat ik begrijp, maar die dan ook nog een keer onder druk komen te staan, en dan heb ik het vooral over, ja, alles wat te maken heeft met de polarisatie in de maatschappij, wat betreft langdurige zieken. [...] Maar ja, je bent natuurlijk wel, afhankelijk van iets dat groter is dan jezelf.” (R10-M, ≤ 2 yrs, Mild/moderate)

“Ik heb vaak genoeg van mensen gefrustreerd geraken door het gebrek aan erkenning, het gebrek aan hulp, die soms lachwekkend wordt weggewuifd. En daarom vond ik het belangrijk dat we een keer kunnen reageren en nemen, om dat wel eens echt duidelijk te maken. Want het is schrijnend. Het is echt zo schrijnend.” (R18-M >2 yrs, Mild/moderate)

“Maar als je bij je huisarts niet terecht kunt en je moet ergens anders en ze hebben er zo geen gehoor naar is, dan voel ik mij soms wel heel erg verloren. [Lange stilte] [...] Ik kamp heel vaak met depressieve klachten. Omdat het leven op die manier zo uitzichtloos lijkt. [...] En wat heeft het dan eigenlijk nog voor nut?” (R19-M, ≤ 2 yrs, Severe)

This lack of understanding and perceived lack of care appeared to contribute to experiences of fear, anxiety, persistent worry, and other depressive symptoms

FACING SELF-DOUBT AND SELF-STIGMA

Following repeated experiences of lacking recognition from others – whether they were healthcare professionals, family members or friends – a process of self-doubt regarding their own experiences was evidenced, with assumptions forming around “going mad” or needing to try harder. When bodily limits were disregarded, this could lead to a downward spiral of increasing fatigue and pain.

“Naarmate dan de jaren verstreken, en op dat moment denk je, dat het ligt aan jezelf, en je je best niet genoeg doet, of je twijfelt aan jezelf, of je denkt, ben ik echt depressief. Allez, ze lieten weten dat wij eigenlijk, depressieve mensen waren, in plaats van mensen die hulp zochten met die vermoeidheid. Of een oplossing, of een diagnose, want toen had ik nog niet echt een diagnose [...]. Ze zeiden ook, je hebt veel te veel stress. Het kwam eigenlijk altijd op hetzelfde neer, dat het aan ons lag.” (R15-F, >2 yrs, Severe)

LOSING A SENSE OF FUTURE PROSPECTS

A lack of hope regarding the possibility of improvement was described, leading to grief related to the perceived loss of a future. The worsening of the condition was said to make the future highly uncertain, with increasing needs for support needs, medical care, adapted housing, and further income loss. These concerns were situated within the broader context of a healthcare system perceived as failing to provide adequate and reliable care.

“Kijk, ik woon hier in een appartement. Wij huren dat. Stel, ik word zo nog slechter. Nu doe ik het met een krukje, bij wijze van spreken. Met wielmpjes onder van de ene kamer naar de andere. Maar dat ik echt rolstoelafhankelijk ga worden. Je weet dat nooit. Ik woon in een appartement. Als ik mijn voordeur hierboven opentrek, kom ik bij de liftdeur al tegen. Ja. Ik kan mij niet keren draaien. Ik kan niet weg, gewoon weg. Evenzeer op dat gebied, misschien naar de toekomst toe, aanpassingen, huisvesting. Er zijn zoveel zorgen die je bezighouden. Angst die daar allemaal mee gepaard gaat. Wat eigenlijk niet nodig zou moeten zijn.” (R13-F, ≤ 2 yrs, Severe)

“In een keer hebben die alle plannen die er op stapel lagen gedurende de jaren tevoren [moeten opgeven]. Dus we gaan een huis kopen we pakken dat op 25, 25 jaar, onze kinderen gaan verder studeren, we hebben inkomen genoeg, we moeten er niet voor sparen dat zal wel lukken, maar ineens sta je voor voldongen feiten van verdorie, ons geld is op we hadden plannen we hadden een auto gekocht op afbetaling, huis op afbetaling, (...) niet alleen ben je dan fysiek gekraakt, financieel ook, dus mentaal gaat het er niet beter van worden.” (R18-M >2 yrs, Mild/moderate)



FACING DIFFICULTIES IN FULFILLING SOCIAL ROLES

Due to physical impairment, difficulties in fulfilling social roles, such as being a parent or an employee, were experienced. The perceived inadequacy in meeting parental expectations carried a particularly heavy emotional burden.

« Voilà, je les voyais quasi plus, je n'arrivais pas à m'occuper d'eux au niveau scolaire aussi - ce qu'une maman fait en général quand on rentre. Donc j'ai de la chance que mon mari est vraiment voilà un papa poule et qui prend le relais sur tout. Mais il y a eu beaucoup de moments où on ne faisait plus d'activités ensemble et ça, ça a eu un impact quand même très fort. » (R5-F, ≤ 2 yrs, Severe)

Alongside difficulties in questioning their own experiences and accepting their own condition, questioning by others and the dismissal of the illness as unimportant was identified. Such reactions can give rise to conflicts and tension with family, friends and employers, and contributed to a narrowing of social networks. As others become irritated, fail to understand what is going on, or no longer know what to expect, misunderstandings accumulated and social withdrawal followed. At the same time, a persistent sense of being pursued by the former life became apparent, characterised by enduring expectations linked to previously held roles. Intense pressure was experienced to function as before and to return to a former sense of self. This occurred alongside expectations of recovery, despite limited perceived prospects for improvement. This placed individuals in a state of being caught between “hammer and anvil”, alongside loneliness, isolation and a lack of social and emotional support.

“Maar het is wel een heel mentale zware kwestie. Want je oude leven achtervolgt je. En je omgeving achtervolgt je. Iedereen wil dat je terug gezond bent. Dat je normaal bent. Je wilt dat zelf ook natuurlijk. Maar dat gaat niet. Je komt niet terecht van een achtbaan van emoties. Of van frustratie. Of van ontevredenheid over je eigen inspanningen. Maar ja. Je wordt eigenlijk gedwongen om terug in het oude leven te stappen.” (R18-M >2 yrs, Mild/moderate)

“Je moet vechten tegen uzelf constant en met uw omgeving, dat is kletteren alleen niet boven in uw kopke maar ook met je omgeving dat is ruzie maken dit kan niet en dit kan wel nee, nee, oh nee punt kletterende ruzie op alle fronten, voor je eigen in je hoofd ruzie, maar ook met je omgeving ruzie want iedereen wil gewoon dat je terug normaal doet dat kost ook tijd dat mensen dat accepteren.” (R18-M >2 yrs, Mild/moderate)

« Si un mi-temps médical peut être maintenu, ça permet à la personne de ne pas avoir de rechute et de ne pas avoir cette culpabilité aussi de rechuter, cette tristesse de rechuter. Ça permet de garder un rythme et dans l'EM, ben de se dire : « Tiens, aujourd'hui, tiens, ce matin j'ai su faire ça, tant mieux ! Je ne sais pas le faire, tant pis !, je me repose parce que j'en ai besoin et voilà ! ». » (R9-F, ≤ 2 yrs, Mild/moderate)

EXPERIENCING A LOSS OF IDENTITY

Pressure on social roles was closely linked to disruptions in personal identity, contributing to experiences of self-doubt, guilt, shame, and depressive affect.

« Quand on essaye de vivre normalement ou comme les autres je veux dire... oui donc il y a le renoncement. Et alors ici depuis l'arrêt ce qui est plus compliqué à gérer c'est le fait de garder un moral, je pense qu'il y a le risque vraiment de dépression réactionnelle j'ai appris ce terme donc, ce n'est pas le problème de santé de départ mais c'est la situation causée par la maladie qui fait que on est frustré et qu'on peut tomber en dépression. » (R8-M, ≤ 2 yrs, Mild/moderate)

Being defined by illness and the associated stigma, or being reduced to a burden on others, was strongly resisted. Conversely, aspirations centered on returning to work, feeling useful again, and re-establishing social roles, which were closely tied to self-worth and identity.

« Pour moi la valeur travail est quelque chose de très important! On se définit ...et puis j'ai toujours par exemple travaillé dans le secteur public parce que je veux venir en aide à la société je veux

être... moi j'étais une locomotive d'équipe à l'époque, et là je suis je ne vais pas dire un déchet, parce que ce serait un peu méchant pour moi, mais voilà, je suis un inactif et c'est terrible! quand vous rencontrez quelqu'un la première question que la personne va vous poser c'est « tiens tu fais quoi dans la vie? » ah ben « je suis malade super » donc je n'ai pas envie de me définir en tant que tel même si c'est le cas pour le moment je suis malade. » (R2-M >2 yrs, Mild/moderate)

DEALING WITH FINANCIAL INSECURITY

Worries centred on financial insecurity following reduced working hours or forced cessation of employment, in a context where replacement income was limited. Additional risks included increasing financial dependence on a partner or the potential loss of benefits due to the absence of a medically recognised diagnosis.

“C'est vrai que je ne sais pas du tout quelle est l'opinion, ou quelle est la connaissance des médecins de la mutuelle par rapport à mon cas, et je sais qu'ils vont avoir un pouvoir important, qui peut trancher du jour au lendemain dans une direction, après il y a des recours, mais... [...] Je ne suis pas si rassuré que il y ait une vraie connaissance et une vraie compréhension de la situation. » (R8-M, ≤ 2 yrs, Mild/moderate)

“Maar er zijn geen zekerheden. En zo die financiële zorg... Het baart mij ook enorm veel zorgen. Want ik heb nu een verloning van ongeveer 1100 euro voor die 20 uur per week dat ik doe. Ik heb soms het gevoel dat ik heel veel moeite heb om zelfs dat met momenten vol te houden. En dat geeft mij een heel onzekere toekomst. Waar dat ook nog eens mee doorweegt met de mentale klachten die er dan bij komen.” (R19-M, ≤ 2 yrs, Severe)

9.2.4 Impact on autonomy

The impact of ME/CFS on individuals' autonomy was assessed at three levels including (i) problems with usual activities, (ii) problems with self-care and (iii) problems with mobility. The following sections highlight the key results for each level.

9.2.4.1 Impact on usual activities (EQ-5D-5L subdomain)

Figure 11-a depicts the change in self-reported ability to perform usual activities on the EQ-5D-5L scale. Before symptom onset, the majority (77.0%) reported no problems with performing usual activities whereas at the time of survey completion, less than 5.0% still reported no problems while the majority reported a transition to slight (5.0%), moderate (27.0%), severe (49.0%) and extreme (17.0%) problems with performing usual activities.

Interview findings indicated that levels of autonomy varied substantially according to symptom severity. In the most severe cases, autonomy was profoundly limited, resulting in complete dependence on others, most often a partner.

« Donc, c'est des journées où je ne vais pas me laver, je ne vais pas préparer à manger, je vais être dépendante pour tout, en fait. » (R6-F >2 yrs, Mild/moderate)

9.2.4.2 Impact on problems with self-care (EQ-5D-5L subdomain)

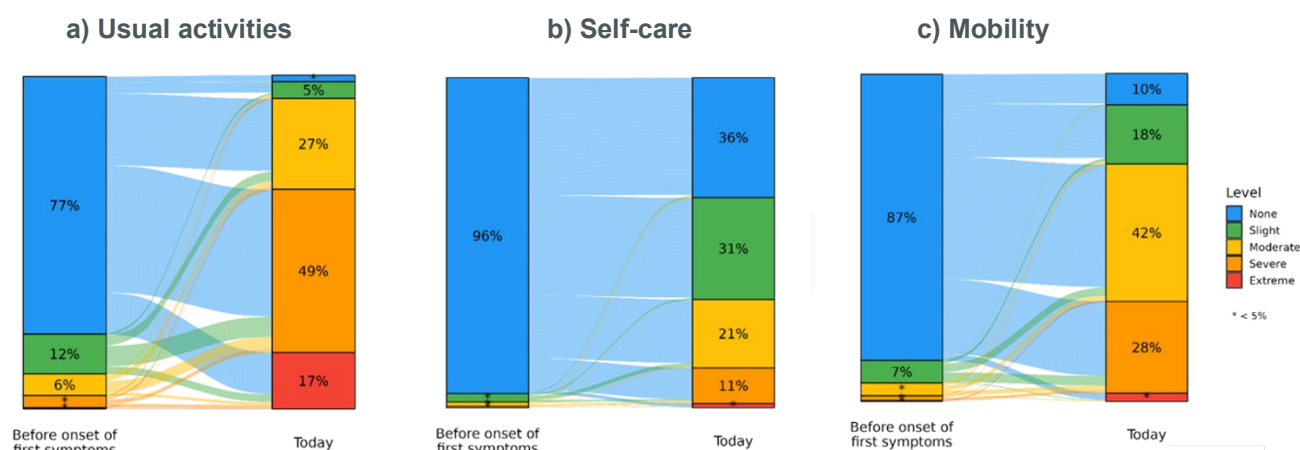
Figure 11-b depicts the change in self-reported problems with self-care on the EQ-5D-5L scale. Before symptom onset, almost everyone (96.0%) reported no problems with self-care whereas at the time of survey completion, this proportion was reduced to 36.0% and the majority reported a transition to slight (31.0%), moderate (21.0%), severe (11.0%) and extreme (<5.0%) problems with self-care.



9.2.4.3 Impact on problems with mobility (EQ-5D-5L subdomain)

Figure 11-c depicts the change in self-reported problems with mobility on the EQ-5D-5L scale. Before symptom onset, the large majority (87.0%) reported no problems with mobility whereas at the time of survey completion, this proportion was reduced to 10.0% and the majority reported a transition to slight (18.0%), moderate (42.0%), severe (28.0%) and extreme (<5.0%) problems with mobility.

Figure 11 – Changes in self-reported level of problems with a) usual activities, b) self-care, and c) mobility (EQ-5D-5L) before onset of first symptoms and today (n=745)



Legend: Responses “I don’t know” (n=3 before onset of first symptoms and n=0 today for usual activities; n=1 before onset of first symptoms and today for self-care; n=2 before onset of first symptoms and n=0 today for mobility) were not displayed in the figures.

Insights from the interview material indicated that mobility difficulties required active resolution, at times through the use of mobility aids such as scooters or wheelchairs, and at other times through increased reliance on others for assistance.

« Quelqu’un a bien voulu m’aider à faire les courses puisque j’ai investi dans une voiturette électrique, mais à mes propres deniers puisque voilà. » (R7-F, >2 yrs, Severe)

« Je sais reprendre la voiture pour des tout petits trajets, mais autrement, tout ce qui est déplacements quand je dois partir avec ma chaise roulante, ben c’est d’office avec lui ou avec le CAP mobilité puisqu’il faut pouvoir transporter la chaise roulante, la porter, etc. Donc ça je sais quand même pas faire, même si je me sens bien pour conduire » (R5-F, ≤ 2 yrs, Severe)

9.2.5 Impact on reproductive health and sexual life

9.2.5.1 Impact on reproductive health

Figure 12-a shows the impact of ME/CFS on reproductive health. The question was filled out or was considered applicable by 349 respondents (46.6%). Of these respondents, 39.0% indicated no impact of ME/CFS on their reproductive health whereas 57.0% reported at least a moderate impact of ME/CFS, and 29.0% indicated an extreme impact on their reproductive health.

One interviewee explained that, as a couple, the perceived “burden” associated with having a child was too difficult to take on.

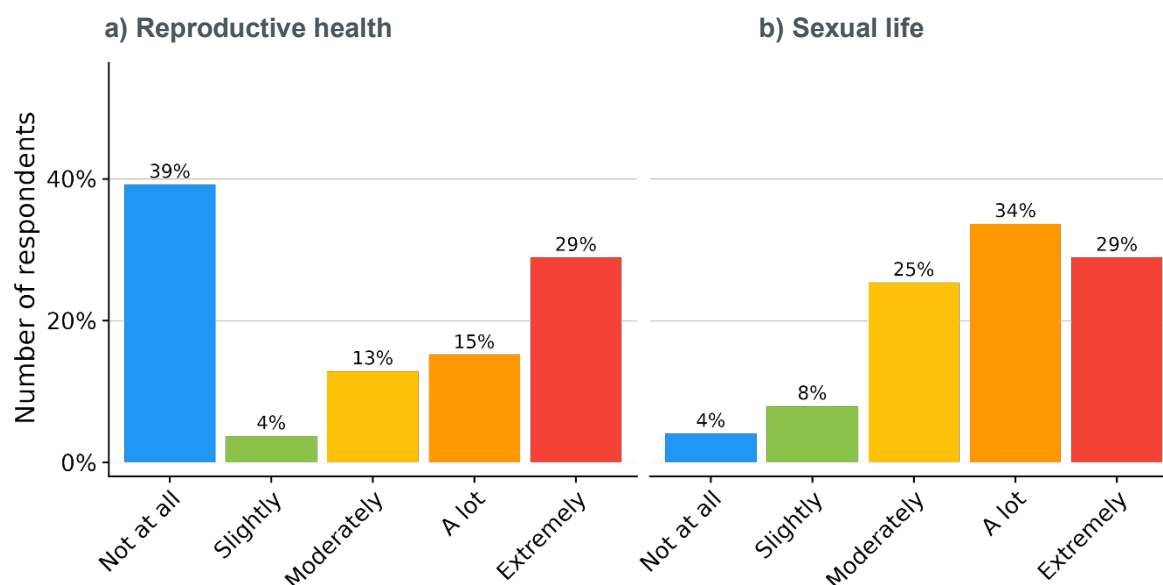
“Donc il y avait cette volonté d’avoir un enfant en commun et mon mari m’a dit, mais on n’y arrivera jamais, on n’y arrivera jamais à gérer et on n’arrivera jamais, toi avec ton état de santé, comment ? Comment mettre en place une grossesse ? Et puis comment s’occuper d’un enfant au quotidien ?

Donc ça c'est un de mes regrets de ne pas avoir pu mener à bien ce projet de parentalité. » (R5-F, ≤ 2 yrs, Severe)

9.2.5.2 Impact on sexual life

Figure 12-b depicts the impact of ME/CFS on sexual life. The question was filled out or was considered applicable by 591 respondents (78.9%). While the minority of respondents reported no impact (4.0%) or a slight impact (8.0%) on their sexual life, the majority of respondents indicated a moderate (25.0%), severe (34.0%) or extreme (29.0%) impact.

Figure 12 – Impact of ME/CFS on reproductive health (n=349) and sexual life (n=591)



Reproductive health: responses "I don't know" (n=28), "Not applicable" (n=353), "I don't want to answer" (n=15) and those who did not answer (n=4) were not displayed.

Sexual life: responses "I don't know" (n=28), "Not applicable" (n=109), "I don't want to answer" (n=17) and those who did not answer (n=4) were not displayed.

9.3 Unmet patient needs: Healthcare needs

9.3.1 Treatment effectiveness

9.3.1.1 Treatment use

There were 377 respondents who indicated having received at least one treatment for ME/CFS in the past two years^{oo}. The most commonly reported treatments included pain medication (59.9%), sleeping medication (41.9%) or other medication (40.1%), physiotherapy (manual techniques) (64.5%), psychological counselling (58.9%) and relaxation techniques (43.2%) (Table 7). Other common treatments included cognitive behavioral therapy (37.9%), alternative treatments (31.6%), pacing (30.0%) and following a specific diet (28.4%). Gradual exercise therapy was reported by 20.7% of respondents, while physical rehabilitation program in a rehabilitation center was reported by 17.5% of the respondents.

^{oo} 346 people answered they did not receive treatment in the last 2 years, 22 people answered "I don't know" and 4 people did not answer this question.



Table 7 – Use of treatments for ME/CFS (n=377)

Treatment type (n=377)	n (%)
Medication	
<i>Pain medication</i>	226 (59.9%)
<i>Sleeping medication</i>	158 (41.9%)
<i>Prolonged antibiotics</i>	18 (4.8%)
Other medication*	151 (40.1%)
Physical therapy	
<i>Physiotherapy (manual techniques)</i>	243 (64.5%)
<i>Relaxation techniques</i>	163 (43.2%)
<i>Gradual exercise therapy</i>	78 (20.7%)
<i>Physical rehabilitation program in rehabilitation center</i>	66 (17.5%)
Psychological and psychosocial therapy	
<i>Psychological counseling</i>	222 (58.9%)
<i>Cognitive behavioral therapy</i>	143 (37.9%)
<i>Acceptance and commitment therapy</i>	38 (10.1%)
Alternative treatments[§]	119 (31.6%)
Pacing	113 (30.0%)
Specific diet	107 (28.4%)

Legend: participants could select one or more treatment options from a prespecified list of treatment options. Percentages add up to more than 100% because participants could select multiple treatments; * other medications included mainly: naltrexone, ivabradine, antidepressant, antiviral drugs, magnesium, vitamins; § alternative treatments included mainly: osteopathy, fascia therapy, auriculotherapy, acupuncture, homeopathy, vagal nerve stimulation, yoga, mindfulness.

9.3.1.2 Self-reported effectiveness of treatment

Table 8 provides an overview of the self-reported effectiveness of the treatment(s) received by the 377 people that indicated to have received at least one treatment for ME/CFS during the past five years. Among the most frequently used treatment types, at least 50.0% of the respondents who reported they received this treatment type indicated that their condition somewhat or substantially improved due to pain relievers, sleeping medication, psychological counseling and relaxation techniques (55.8%). For physiotherapy (manual techniques) this was 46.8%. However, 7.3% (psychological counseling) to 18.9% (physiotherapy, manual techniques) reported somewhat or substantially worsening of their condition following these treatments.

Other frequently used treatments that were experienced as somewhat or substantially improving their condition included cognitive behavioral therapy (44.5%), alternative treatments (51.8%), pacing (66.7%) and following a specific diet (57.7%). However, 6.7% (specific diet) to 18.7% (cognitive behavioral therapy) reported somewhat or substantially worsening of their condition following these treatments.

Table 8 – Self-reported effectiveness of treatments for ME/CFS (n=377)

Treatment type (n=377)	Effect on the symptoms					
	Substantially worsened	Somewhat worsened	Unchanged	Somewhat improved	Substantially improved	I don't know
Medication						
<i>Pain relievers</i>	13 (5.8%)	9 (4.0%)	74 (32.7%)	102 (45.1%)	23 (10.2%)	5 (2.2%)
<i>Sleeping medication</i>	7 (4.4%)	9 (5.7%)	53 (33.5%)	67 (42.4%)	21 (13.3%)	1 (0.6%)
<i>Prolonged antibiotics</i>	3 (16.7%)	1 (5.6%)	4 (22.2%)	5 (27.8%)	3 (16.7%)	2 (11.1%)
Physical therapy						
<i>Physiotherapy (manual techniques)</i>	18 (7.4%)	28 (11.5%)	80 (32.8%)	97 (39.8%)	17 (7.0%)	3 (1.2%)
<i>Relaxation techniques</i>	7 (4.3%)	8 (4.9%)	55 (33.7%)	77 (47.2%)	14 (8.6%)	2 (1.2%)
<i>Graded exercise therapy</i>	30 (38.0%)	15 (19.0%)	8 (10.1%)	21 (26.6%)	4 (5.1%)	1 (1.3%)
<i>Physical rehabilitation program in rehabilitation center</i>	14 (20.9%)	7 (10.4%)	12 (17.9%)	27 (40.3%)	6 (9.0%)	1 (1.5%)
Psychological and psychosocial therapy						
<i>Psychological counseling</i>	7 (3.2%)	9 (4.1%)	79 (35.6%)	100 (45.0%)	24 (10.8%)	3 (1.4%)
<i>Cognitive behavioral therapy</i>	15 (10.4%)	12 (8.3%)	50 (34.7%)	59 (41.0%)	5 (3.5%)	3 (2.1%)
<i>Acceptance and commitment therapy</i>	3 (7.9%)	1 (2.6%)	14 (36.8%)	14 (36.8%)	5 (13.2%)	1 (2.6%)
Alternative treatments	5 (4.5%)	3 (2.7%)	38 (34.5%)	41 (37.3%)	16 (14.5%)	7 (6.4%)
Pacing	2 (1.8%)	9 (7.9%)	21 (18.4%)	63 (55.3%)	13 (11.4%)	6 (5.3%)
Specific diet	4 (3.8%)	3 (2.9%)	35 (33.7%)	45 (43.3%)	15 (14.4%)	2 (1.9%)

Legend: participants could select one or more treatment options from a prespecified list of treatment options. Percentages add up to more than 100% because participants could select multiple treatments.



Survey findings indicated that the perceived effectiveness of symptomatic treatments varied considerably between individuals.

Interview data showed that attempts to alleviate ME/CFS symptoms may involve prolonged processes of trial and error. In the absence of curative treatments, care was limited to symptoms management. This can result in an extensive list of medication including antidepressive drugs, nutritional supplements, or physical therapies or energetic alternative therapies.

« J'ai une liste de médicaments, vous n'êtes même pas prête ! parce qu'il y a la fatigue chronique et il y a les conséquences de la fatigue chronique (...) au niveau antidouleur, j'en suis à des antidouleurs de classe 3 donc quand dalflgan ne fonctionne pas, (...) parfois je prends quand même de l'ibuprofène si j'ai mangé ou des trucs comme ça, mais je passe au dafalgan codéine ou au tradonal quand ça ne marche pas, j'en suis à l'oxycodone qui est deux fois plus puissant que la morphine. Ça par exemple il y a des troubles du sommeil qui sont parfois entraînés indirectement par les amphétamines. Ou bien je prends aussi du provigile qu'on donne pour ceux qui souffrent normalement de narcolepsie ou militaires qui partent en guerre, et donc ça a été prescrit off the label comme on dit par le spécialiste de [Hôpital A], tout comme les amphètes ça a été prescrit off the label quelque part. Il y a la mélatonine, il y a malheureusement des aides drugs donc du zolpidem, du piracétam qui aide à faire circuler le sang dans le cerveau, l'oxycontin qui est en fait un ginkgo biloba mais ultra concentré, il faut une ordonnance, ça coûte assez cher... Je prends plein de compléments alimentaires aussi si il y a la N-acétylcystéine, il y a le précurseur la dopamine la L-thyroxine, (...) c'était aussi ma neuropsychologue qui m'avait dit peut-être que les amphet chez vous ça ne fonctionne pas parce que vous n'avez pas assez de L-thyroxine, donc je prends ça ... je prends de la taurine dans les excitants, (...) de Ashwagandha c'est aussi pour le mental, la L-carnitine ça je pense que c'est plus anti-inflammatoire... j'en oublie j'en ai encore d'autres (...) des oméga 3 des machins comme ça, (...) moi j'ai envie de guérir donc j'essaie tout en fait ! j'ai même, le pire, c'est vraiment parce que c'est mon ami d'enfance qui m'a conseillé parce que c'est vraiment c'est un peu ridicule, mais c'est pour vous montrer aux extrémités auxquelles on va, j'ai été à des soins de médecine traditionnelle philippine, 170 boules à Bruxelles pour trois soins ! C'était surtout des endroits où on appuie sur le dos ou sur le corps des massages spéciaux (...) moi il n'y a strictement aucune amélioration ...mais c'est pour vous dire j'ai tout essayé même les soins énergétiques et ce genre de trucs là quoi on essaye tout ce qu'on peut quand on veut guérir ! (...) Le docteur X m'avait dit régime vraiment sans graisse etc il faut essayer et finalement ça a été qu'une hypothèse mais même j'ai essayé 15 jours vraiment sans graisse de manière hard mais ça n'a pas du tout joué sur ma fatigue ou quoi que ce soit mais par contre ça a quand même joué sur mon système gastrique. Donc j'essaie de maintenant de prendre que des fromages light presque plus de charcuterie ou en tout cas des charcuteries vraiment light par exemple du jambon cuit sans la couenne c'est que 3% de matière grasse on peut... voilà j'essaie de... au niveau alimentation. » (R2-M, >2 yrs, Mild/moderate)

“Ja de de de twee dokters die of de derde ik weet het al niet meer die schreef mij zelfs antidepressiva voor terwijl ik helemaal niet depressief was maar hij dacht ja het is een onderliggende depressie en dat ook wel zou kunnen helpen of of wat was zijn redenering ehm ja dat een neven effect van antidepressiva is ook dat je meer energie krijgt dus ik heb mijn tijd aan antidepressiva genomen maar dat heeft totaal ook niet geholpen” (R17-F, ≤ 2 yrs, Severe)

Interview data further indicated that a wide range of symptoms were classified under the diagnostic label of ME/CFS and tended to be addressed through largely uniform treatment approaches. This lack of differentiation suggested that potentially heterogeneous underlying mechanisms were managed in similar ways, despite indications that greater diversification according to presumed aetiology might be desirable to optimise treatment strategies.

“Van de veelheid aan mensen met gelijkaardige symptomen die daarin worden ondergebracht, zonder abstractie makend van wat is de oorzaak van die diagnose. En ik kan dat best vergelijken met eigenlijk een parallel voorbeeld. Als iemand hoofdpijn heeft omdat er een tumor in zijn hersenen zit. Of iemand heeft hoofdpijn omdat hij met een stomp voorwerp op zijn hoofd geslagen

is en een trauma heeft opgelopen. Ja, beide zijn hoofdpijn. Maar je gaat beiden niet op dezelfde manier behandelen. Terwijl ik het gevoel heb, iemand die trauma heeft, die een kind verloren heeft of wat dan ook en daardoor vermoeidheidsklachten heeft, wordt op dezelfde manier, zagezegd, behandeld voor die vermoeidheidsklachten als iemand die na een infectie postvirale symptomen ontwikkelt.” (R10-M, ≤ 2 yrs, Mild/moderate)

From the interviews, we heard a strong willingness among patients to find a cure and to adhere to treatments with demonstrated effectiveness. At the same time, we heard that patients often lack the energy required to continuously search for accurate, up-to-date information and new scientific developments. Within this context, a clear need for a centralised and easily accessible knowledge database or information platform became evident.

In response to their suffering and in the absence of effective treatments, patients adopt a wide range of coping strategies to manage daily life and prevent the worsening of symptoms. These coping strategies aim to conserve energy, avoid symptom exacerbation, and improve quality of life. These are often developed spontaneously, based on patients' own experiences, or acquired through education on approaches such as 'pacing'.

We identified practical, organisational coping/pacing strategies:

- Routines, ways of organising daily life

“Ik heb een redelijk vast patroon dat ik wel aanhoud omdat ik dan toch wel wat regelmaat heb maar het is eigenlijk s' morgens opstaan, douchen is niet meer elke dag is wanneer ik iets meer energie heb want dat vraagt ook heel veel energie, dat is ontbijten dan ga ik meestal stretch oefeningen en meditatie oefeningen doen om wat in evenwicht te brengen, en ademhalingsoefeningen dan ga ik rusten en dan als ik dan een huishoudelijke taak doe dan is dat ook in stukken dat is de was ophangen, ja soms in ene keer soms moet dat in twee keer, dat is ook al zittend, dat is niet meer al rechtstaand. Achter iedere activiteit waar ik echt wel energie heb gebruikt moet ik eigenlijk rusten.” (R12-F, ≤ 2 yrs, Mild/moderate)

- Plan activities – Save energy before an activity – Rest after an activity – Dosing energy

« Parce que donc, dans ces cas-là, il faut que les jours avant, je ne fasse rien, et les jours après, je ne fasse rien non plus. [...] Du coup, ça, ça permet de justement pas faire de malaises post-efforts qui sont vraiment extrêmement difficiles à vivre. Oui. Du coup, c'est tout. Il y a très peu de spontanéité, en fait. C'est vraiment une vie plutôt calculée, et encore avec les aléas de parfois je calcule, et puis le jour même, c'est quand même pas possible. » (R6-F >2 yrs, Mild/moderate)

- Alternate physical and mental activities:

“Dat is een techniek die ik heb geleerd: je moet altijd tegenovergestelde activiteiten doen hersenactiviteit, fysieke activiteit die staan los van elkaar. Ze zijn allebei vermoeiend maar het fietsen seffens maakt mijn hoofd leeg mijn hoofd heb ik dan niet nodig. Nu wel. Ik ben niet aan het bewegen dat kan seffens nog wel omdat dat gewoon tegenovergestelde activiteiten zijn voor mijn lichaam die fysiek is ik ben seffens wel moe maar dat is niet teveel. Mijn hersenen seffens kan ik aan de kant schuiven je moet gewoon trappen, muziek luisteren en dat kost mij geen focus [...], maar dat is een hele aanpak van hoe je je dag indeelt en fysiek.” (R18-M >2 yrs, Mild/moderate)

- Avoid stimuli and rush (e.g. noise, stress)

“Telefoongesprekken afgeschaft. Zelfs met familieleden. Het kost mij een hoop energie, dan lig ik daar maar. Dat is energie waarmee ik bijvoorbeeld kan eten, andere dingen doen. Ook omdat je het onbegrip van de mens hebt. Dat je het niet meer waard vindt om je energie daar allemaal aan te geven, maar aan de dingen die echt nodig zijn om te kunnen overleven.” (R13-F, ≤ 2 yrs, Severe)



“Je merkt wat je wel kan doen en wat je niet kan doen. Het een vraagt heel veel energie, het ander vraagt minder energie. Ook gemerkt dat je stressfactoren moet vermijden. Je moet je leven zo proberen in elkaar te steken dat je heel weinig stress hebt. Dat je je geen zorgen moet maken.” (R16-M >2 yrs, Mild/moderate)

- Cutting activities into pieces to make them feasible – pacing
« Au début ma pelouse je la tondais en une fois et puis en deux fois et puis en trois fois et puis à un moment donné je me suis rendu compte que même un tiers de pelouse, c'était un peu j'étais complètement trempé, mon coeur battait, j'étais pas bien, j'avais mal à la tête il fallait que j'aille m'allonger un jour ou deux et donc c'est pour ça que j'ai dû payer un homme à tout faire ...même pour tondre la pelouse quoi ! donc il y a des efforts que je ne fractionne même plus voilà, il y a des efforts qu'on ne peut pas fractionner. » (R2-M >2 yrs, Mild/moderate)
- Find comfort in lying down - Take naps during the day or after an activity or household chore
« Hier j'étais à l'anniversaire de ma filleule, c'est vrai que je vais prendre cet exemple là, mais les parents savent que j'ai un soucis de santé après un certain temps un petit temps de discussion etc je sentais que je forçais un peu je leur ai dit « est-ce que je peux m'étendre un moment ? » ils m'ont amené dans une chambre où j'ai pu m'étendre 40 minutes et puis je suis revenu dans la fête et ça fait une grosse grosse différence. » (R8-M, ≤ 2 yrs, Mild/moderate)

Setting boundaries and communicating limitations can help to implement these coping strategies, while also enabling the identification of priorities and decision-making.

“Je moet de rest van de buitenwereld gewoon weg keihard nee zeggen en dat wil je niet, dat is niet fijn, maar je moet het doen want anders kun je gewoon weg de energie die je wilt geven aan de juiste mensen, die ben je dan al kwijt nog voor je begonnen bent. Dus je moet eigenlijk tegen iedereen erbuiten zeggen [...] ik ben moe ik ga naar huis. Als ze dat niet snappen, dan moet je kunnen zeggen: als jij dat niet kunt snappen wat ik zeg of niet kunt appreciëren wat ik zeg dan hoeft het voor mij niet meer. Ik ga me niet elke keer gaan uitleggen.” (R18-M >2 yrs, Mild/moderate)

“Je moet aanvaarden leren aanvaren je moet kiezen dingen te doen maar je moet dat kiezen op basis van wat echt moet en wat je niet anders kunt en waar je wil En daar moet je een balans in vinden van als je alleen maar dingen gaat doen die je moet doen dan ga je er niet gelukkig van worden dus je moet ook dingen doen die je wilt doen en daarom moet je dan ook de gevolgen van accepteren je weet van ik ga seffens een fietstochtje maken dat weet ik nu al het is buiten mooi weer ik ga met de fiets elektrisch trouwens anders komt het niet vooruit maar ik weet dan ook wel van dan zal het zal gedaan moeten zijn want die inspanningen van voormiddag betaal ik cash in de namiddag en als ik dat niet doe dan ga ik daar morgen de gevolgen van dragen dus kiezen, alles kiezen. Het is moeilijk dat je moet kiezen.” (R18-M >2 yrs, Mild/moderate)

Other examples of practical coping strategies were mentioned:

- Adapt leisure activities to limit displacements or holidays to be able to rest
« Ou pour partir en vacances, cette année, j'ai renoncé à partir en vacances parce que l'énergie était trop basse pour la première fois j'ai renoncé, je vais dire. Mais l'année avant, j'étais... Les deux, trois années avant, j'étais parti chez un ami très proche, qui peut m'accueillir en France [...] qui est un peu au courant. Et donc, là, il y a moyen de me reposer chez lui, en fait, quelque part. Il a des enfants qui s'amuse avec mes enfants aussi, mais c'est très particulier parce que là aussi, pour partir en vacances, sinon, dans des endroits classiques, c'est... J'ai pas encore trouvé vraiment d'autres solutions, quoi. » (R8-M, ≤ 2 yrs, Mild/moderate)
« Pour les restaurants. On n'en fait plus, mais on prend à emporter. » (R1-M, ≤ 2 yrs, Mild/moderate)*

- Adapt lifestyle and diet (e.g. no more alcohol consumption, warm meal at noon rather than in the evening)

“En je leert natuurlijk wat je wel en niet mag doen. Alcohol drinken is een boosdoener hé. Oké. Het vraagt zoveel energie om te verwerken.” (R16-M >2 yrs, Mild/moderate)

- Saving energy in small things like using less heavy and/or noisy objects, e.g. plastic plates and cutlery.

« Ben voilà, les jours où je ne vais pas bien, je vais prendre une tasse moins lourde parce que ça me demande moins d'efforts et que du coup, ça va être plus facile pour moi et ça va épargner de l'énergie. C'est assez dingue de se dire qu'il y a des malades qui vont utiliser que des assiettes et des couverts en plastique parce que, à nouveau, et ce sera plus léger et ça ne fera pas de bruit. Enfin, voilà, ça peut vraiment aller loin. » (R6-F >2 yrs, Mild/moderate)

- To cope with brain fog, the use of technology (e.g. AI, subtitles, audio's of books) could be helpful.

« Il était vraiment dans un gros brouillard cérébral, c'était compliqué de lire. Alors que si... Écouter une vidéo, ça peut être compliqué de se concentrer aussi. Donc ce qu'il fait, c'est qu'il met des vidéos avec sous-titres. Comme ça, il a à la fois le son, l'image le texte qui s'affiche. Et c'est plus facile de rester concentré, de comprendre. » (R1-M, ≤ 2 yrs, Mild/moderate)*

Emotional burden of the symptoms could also be alleviated by using some coping strategies, e.g.:

- Changing attitudes, perceptions and expectations

“Door te accepteren, is wel mijn levenskwaliteit verbeterd. Vroeger zag ik mezelf als... Ja, ik ben gewoon gehandicapt. Nu focus ik mij op dingen die ik wel nog kan.” (R17-F, ≤ 2 yrs, Severe)

- Accepting incapacities - Listening to the physical limitations due to the disease - Accepting own limits

« Mais il y a deux ans maintenant, deux ans et demi, je m'étais vite rendu compte que ma vie professionnelle était derrière moi. Je me suis dit, tu as le choix. Ou tu te laisses glisser. Et dans deux ans, tu es dans le cercueil. Parce que ton état, enfin, tu vois un peu le bazar. Ton état va se dégrader, machin. Oui, oui. Ou alors, tu acceptes de changer de vie et d'être heureux avec ce que tu as ! Finalement, c'est cette acceptation de la situation qui me permet de tenir.... Il y a des coups de blues,mais qui me permet de tenir le coup. Et d'avoir des chouettes moments parce que j'ai fait quelque chose qui me plaisait. » (R3-M, >2 yrs, Severe)

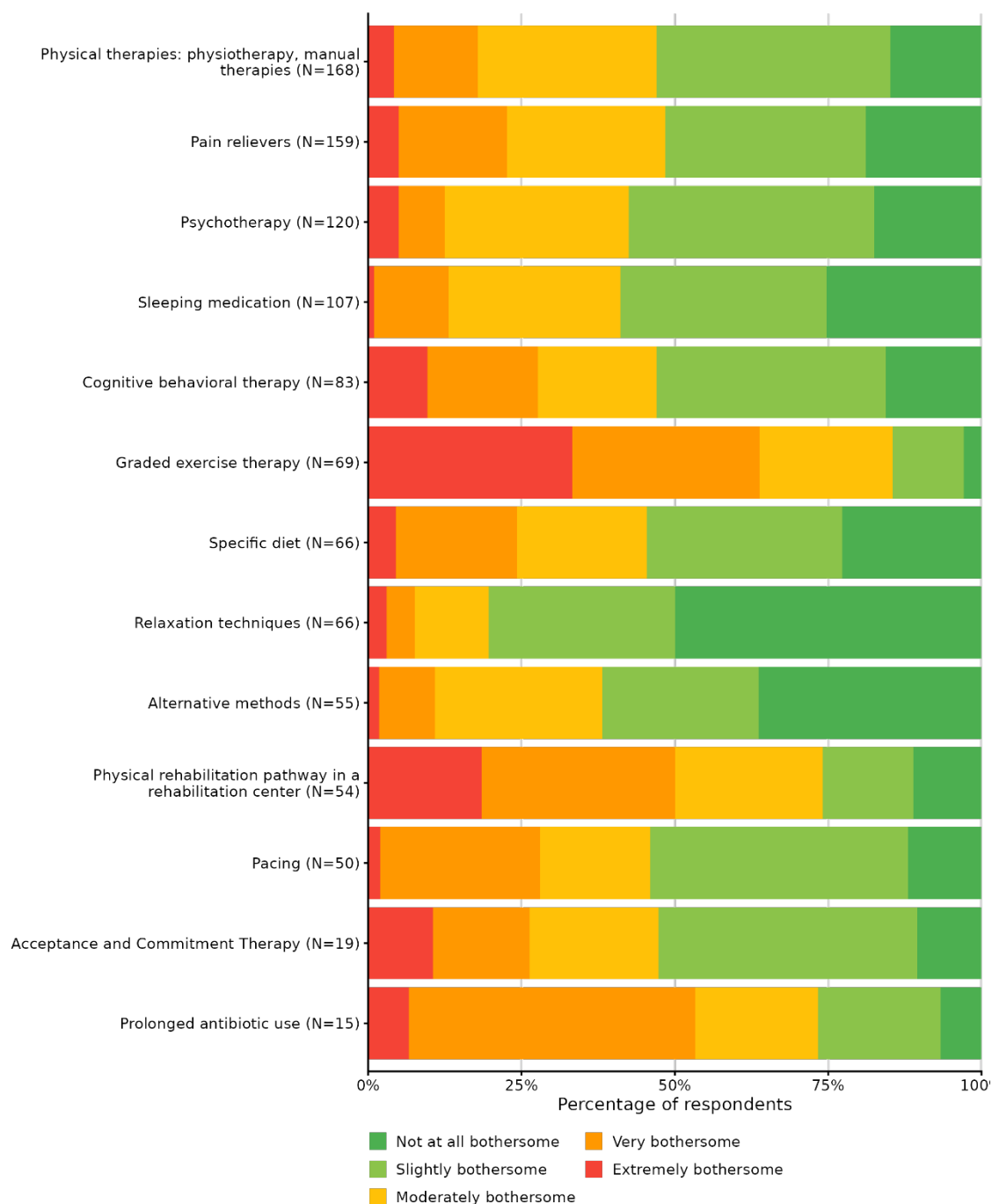
9.3.2 Burden of treatment

9.3.2.1 Self-reported burden of treatment side effects

Figure 13 illustrates the burden of side effects associated with the different treatments the respondents received over the past five years. For most treatments, more than 50.0% of respondents indicated the treatment as slightly or not burdensome, whereas the remainder indicated the treatment to be moderately, very or extremely burdensome. For relaxation therapy more than 75.0% reported the treatment to be not or slightly bothersome whereas in contrast, around 75.0% of the respondents indicated to experience a physical rehabilitation program in a rehabilitation center and prolonged use of antibiotics to be moderately, very or extremely burdensome. Regarding graded exercise therapy, more than 75.0% of respondents experienced this treatment as moderately, very or extremely burdensome.



Figure 13 – Self-reported level of burden of treatment side effects



Legend: Some patients replied 'I didn't have any side effect related to this treatment' (n=6 for pacing; n=2 for graded exercise therapy; n=26 for physical therapies; n=35 for relaxation techniques, n=13 for cognitive behavioral therapy; n=6 for physical rehabilitation pathway; n=2 for acceptance and commitment therapy; n=21 for psychotherapy; n=15 for specific diet; n=20 for alternative methods; n=29 for sleeping pills; n=30 for pain relievers; and n=1 for prolonged antibiotic use).

9.3.2.2 Experienced burden of treatment

To the question to what extent the received treatment for ME/CFS was experienced as burdensome, apart from the side effects (for example due to transport or other practical issues) most participants experienced treatment as not (22.4%), slightly (26.8%) or moderately (29.3%) burdensome while the other participants experienced treatment as very (14.9%) or extremely (6.6%) burdensome (n total for this question=362).

From the interviews we heard that graded exercise therapy can be particularly burdensome because of the side-effects.

« J'ai suivi un programme en 16 séances multidisciplinaires qui était assez bien organisé, mais c'est le médecin (...) que j'ai rencontré qui m'a mal conseillé. C'était un peu un médecin sportif qui était dans l'optique du reconditionnement à l'effort. Il m'a dit « Il faudrait augmenter votre activité physique de 2% par semaine. » Ce qui, en fait, est un très mauvais conseil. Voilà. Et je lui ai dit, parce que j'avais déjà lu que c'était un peu controversé, le graded exercise therapy. (...) parce que les médecins ne connaissaient pas l'avis du Conseil supérieur de la santé de 2020. C'est quelque chose qui est un peu crédible en Belgique et qui dit aussi que c'est un peu... que c'est pas vraiment conseillé, et que c'est un peu controversé. Mais même quand je lui ai parlé de ça, il avait l'air tellement sûr de lui ! Moi, je me suis dit, bon, ben, j'essaye, quoi ! Il avait l'air d'avoir vu des patients un peu similaires que ça avait aidé de refaire plus d'exercices. Et en fait, je pense que j'ai perdu suite à ça. J'étais un peu trop dans l'approche physique, dans les exercices, etc. Et puis je me suis craché. » (R8-M, ≤ 2 yrs, Mild/moderate)

A negative impact of cognitive behavioral therapy was raised, i.e. a 'blaming the victim effect' because it is perceived as if the responsibility of getting better is on the patients' shoulders.

“Maar die cognitieve gedragstherapie was zo stuitend. Omdat je voelt van... Ja, maar... Omdat de verantwoordelijkheid een beetje voor je ziekte in de schoenen geschoven wordt van een patiënt. Terwijl... Dat kwam zo raar over en zo hard binnen eigenlijk. Dat ik dacht van... Ja, maar... Wat is dit hier eigenlijk?” (R10-M, ≤ 2 yrs, Mild/moderate)

An indirect consequence of pacing, as highlighted in the interviews, was increased social isolation, as the need to renounce certain activities, including routine tasks such as grocery shopping, and limited social participation.

« On demandait s'il y avait des effets secondaires du pacing, et moi j'ai pensé aux effets secondaires médicamenteux et donc je me suis dit : mais non, il n'y a pas d'effets secondaires, enfin il n'y a rien qui arrive à mon corps, mais en fait, ce qui arrive, c'est un isolement social terrible. Parce que ce n'est plus possible d'enchaîner les activités, de faire les choses, de voir des gens. Et donc il y a un isolement social extrêmement important. Moi, je ne fais jamais les courses, par exemple, c'est toujours mon mari qui les fait. Ça peut sembler bête, mais c'est un peu comme ce qu'on a vécu pendant la pandémie, quoi. Il y a vraiment un isolement social qui se met en place. Il y a plus de spontanéité, il y a plus de : « Ah ben, OK, on va pouvoir planifier les choses ». » (R5-F, ≤ 2 yrs, Severe)

9.3.3 Quality of care

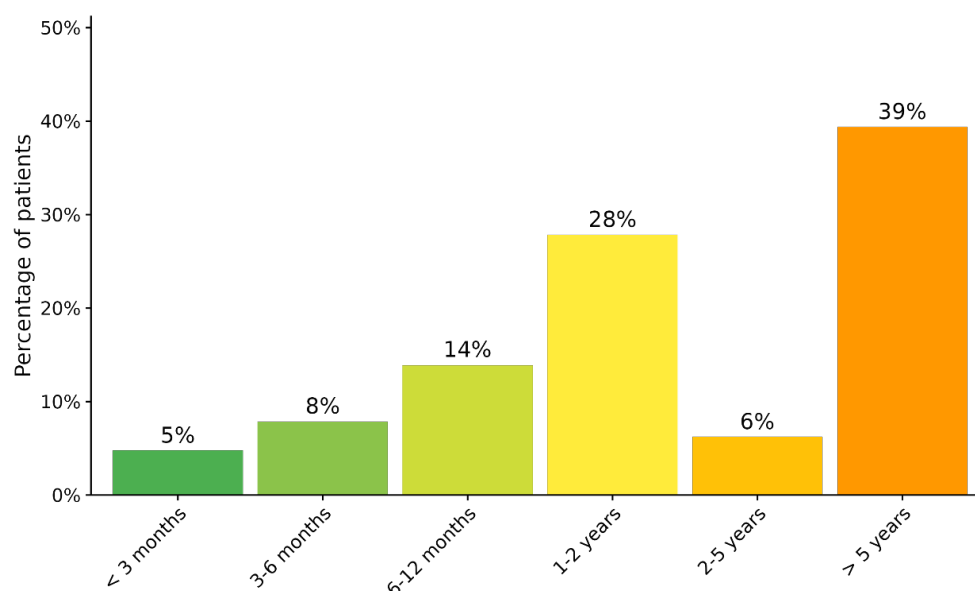
9.3.3.1 Diagnosis of ME/CFS

For this item, time to diagnosis was reported by 546 participants while 203 were unable to provide an estimate. Figure 14 shows that almost three quarters of the respondents had to wait more than one year to receive a diagnosis of ME/CFS, 39.0% received their diagnosis more than five years after their first medical consultation. Only 27.0% of the respondents were diagnosed within one year after their first medical consultation, with 5.0% having received a diagnosis within three months. Most respondents received their diagnosis during a specialist consultation (52.7%) or at an ME/CFS referral center (34.1%). A smaller proportion of the respondents received their diagnosis during a GP



consultation (7.5%), during a hospital stay (2.2%) or didn't know where they got their diagnosis (0.5%). 2.9% of the respondents answered they got their diagnosis in another place, including physicians consulted in foreign countries, and 11 people did not provide an answer.

Figure 14 – Time between first medical consultation and diagnosis of ME/CFS (n=546)



Legend: Responses "I don't know" (n=203) were not displayed in the figure.

Findings from the qualitative data revealed a long process leading to diagnosis. This diagnostic trajectory appeared to follow a broadly similar pattern, beginning with the gradual onset of multiple, non-specific symptoms that were difficult to clearly attribute (e.g. reduced energy levels, persistent fatigue, influenza-like symptoms, muscle pain, and sleep disturbances). In some cases, interviewees mentioned a preceding infection (e.g. Covid-19 or Lyme disease) or to a triggering event such as an accident with major injuries. In response to these symptoms, individuals initially sought medical support through their GP.

Before a diagnosis of ME/CFS was established, symptoms were frequently attributed to a range of other conditions, such as burnout, depression, postural orthostatic tachycardia syndrome (POTS), or Alzheimer's disease. Also co-occurring or related conditions such as dysautonomia, fibromyalgia or irritable bowel disease were reported by patients.

During interviews, sequences of returning to work followed by being put on sick leave with intervals of varying durations were described. After a period of rest ranging from a few days to a few weeks, a clear clinical picture was not established, and complaints were attributed to stress. When interviewees did not receive satisfactory answers, they tended to return repeatedly to the same healthcare provider or consult multiple specialists, either with or without a referral from their GP.

"Mijn energie was vaak zoek maar toch bleef ik doorgaan tot op een bepaald moment dat ik gewoon niet meer kon. Het was echt, ik had geen energie meer. Ik had, ja ik kon niet meer en ik ben dan toch naar de huisarts geweest en toen zeiden ze het is een burn-out, maar ik voelde dat het dat niet was. Ik had er allemaal wel zin in, maar ik had gewoon de energie niet en ik heb dan eigenlijk ja heel lang dat ik niet wist naar welke dokter dat ik moest gaan. De ene dokter zei van je moet naar die dokter gaan en toen ik bij de bij de neurologe was zei die ja maar ik kan jou niet helpen maar eigenlijk weet ik het niet. Ik wil jou wel doorsturen maar ik weet eigenlijk echt niet naar wie. Dus ja het stopte weer en ik weer naar de huisarts want die vermoeidheid bleef duren."
(R12-F, ≤ 2 yrs, Mild/moderate)

Over time, periods of sickness absence tended to become increasingly prolonged and more frequent. Extensive diagnostic investigations were commonly undertaken, including a wide range of clinical tests that failed to yield a definitive diagnosis (for example, assessments conducted in sleep clinics). This process often continued until contact was made with a healthcare professional who, frequently through a diagnosis of exclusion, ultimately established the diagnosis of ME/CFS. This occurred irrespective of whether individuals had already reached this conclusion themselves through personal information-seeking.

« Et en fait, c'est une émission à la télévision qui m'a permis de me dire « mais ça ressemble quand même très fort à ce que je vis ». Donc, c'était sur le journal de la santé. Il recevait le professeur chez qui je suis allée pour recevoir le diagnostic et une dame qui témoignait et ça ressemblait quand même vraiment très très fort à ce que à ma vie, en fait, à mon quotidien, à mes symptômes, à mon ressenti. Donc, j'en ai parlé d'abord aux différents médecins que j'ai vus et en fait, en Belgique, c'est soit pas du tout connu, (...) vous savez, c'est comme le burn-out, en fait, c'est pas une vraie maladie, quoi, on s'en remet avec du repos. » (R6-F >2 yrs, Mild/moderate)

During this process of searching for the right diagnosis, patients faced long waiting times, and had their symptoms dismissed as psychological or insignificant, against a backdrop of anxiety caused by a lack of a clear diagnosis, the discomfort of the symptoms, struggles with family members who do not understand what is going on, and the search for a balance between work and the burdens of the illness.

The result of this long search for a diagnosis, which often lasts several years, is a deterioration in both their mental and physical health and a shrinking of their social network.

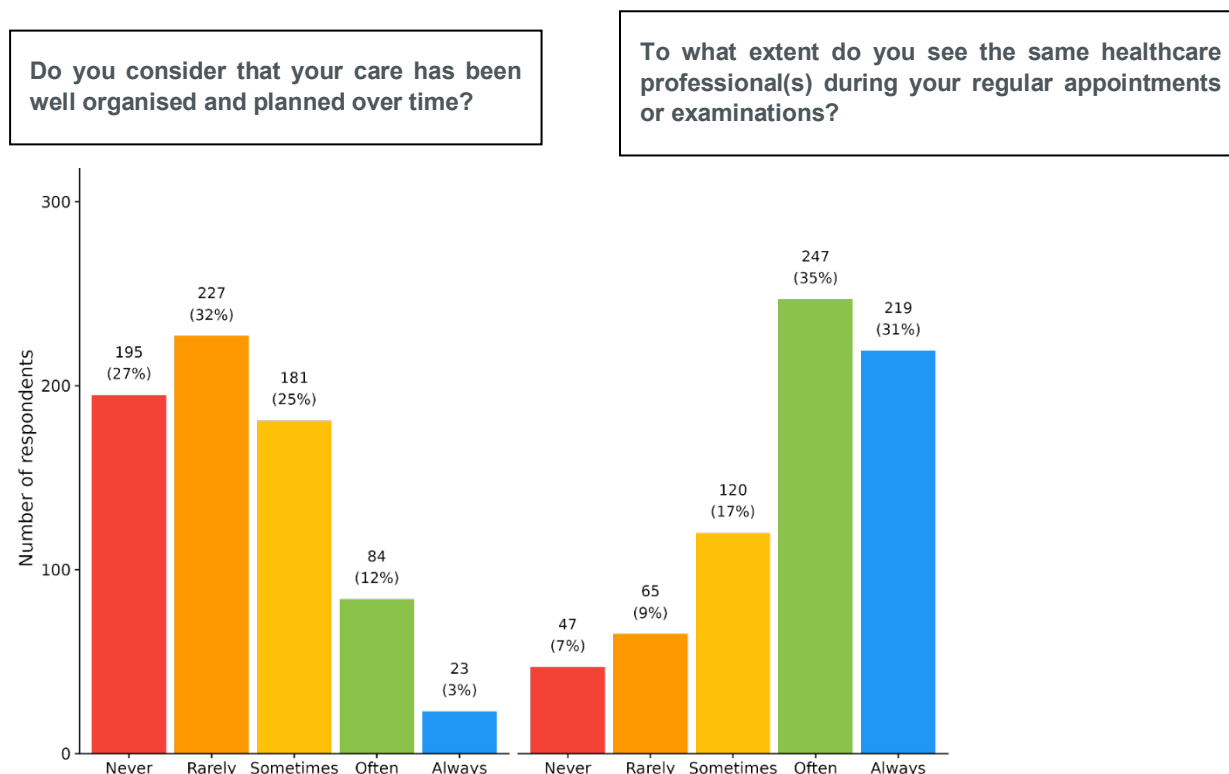
9.3.3.2 Coordination and continuity of care

Figure 15 shows that more than half of the respondents indicated that they rarely (32.0%) or never (27.0%) experienced their care to be well organised and planned. Only a minority of respondents reported that they often (12.0%) or always (3.0%) experienced their care to be well organised and planned. In addition, most respondents indicated that they often (35.0%) or always (31.0%) see the same healthcare professionals during their regular consultations or examinations. Only a small proportion reported that this was rarely (9.0%) or never (7.0%) the case.

Regarding the question on care coordination (n=745), most respondents indicated that they have no healthcare professional responsible for the coordination of their complex care (66.6%) while the remainder indicated that they have a coordinating professional (20.0%) or didn't know (6.6%). For 7.8% of respondents this question was not applicable since they didn't need complex care requiring coordination.



Figure 15 – Experience with organisation and planning of care (n=710) and having the same healthcare providers during their care trajectory (n=698)



Legend: Responses "I don't know" (n=35 for organisation of care and n=47 for continuity of healthcare providers) were not displayed in the figures.

In response to the lack of organisation and continuity of care, it appears from the interviews that patients may develop a kind of do-it-yourself (DIY) approach for managing their healthcare. This included independently searching for information, consulting a wide range of specialists based on specific symptoms, actively asking for referrals, examinations, or specific medication. In this context, management of care was then assumed by patients themselves, as the healthcare system was perceived to insufficiently provide medical expertise and coordination.

« C'est au patient de faire son propre parcours, de constituer sa propre clinique, en fait, en demandant des rendez-vous. » (R1-M, ≤ 2 yrs, Mild/moderate)*

« J'ai dû quand même beaucoup moi-même trouver des solutions, que ce soit pour avoir finalement le diagnostic confirmé ou pour trouver les endroits où. Et je pense qu'il doit y avoir plein de gens qui ne sont pas diagnostiqués parce que, c'est un peu comme les primes à la rénovation. Il faut avoir fait des études supérieures pour s'y retrouver quelque part. Et pour savoir où chercher, etc. » (R8-M, ≤ 2 yrs, Mild/moderate)

Patients we heard who developed additional health problems alongside ME/CFS and who required further care such as surgery, experienced difficulties in explaining their specific needs related to ME/CFS to healthcare providers. Patients struggle to obtain understanding and appropriate accommodation (e.g. access to a quiet room to avoid overstimulation). In other words, we heard experiences where healthcare services in general are not adapted to the specific challenges faced by patients with ME/CFS.

« Ici, j'ai dû être opérée, on a dû me faire une hystérectomie parce que j'avais un fibrome, voilà qui était mal placé, etc. Mais j'ai dû aller voir l'anesthésiste. Ben, ça a été super compliqué en fait

de noter sur le formulaire, de lui expliquer. Ils ont été vraiment géniaux et j'ai eu la chance d'être prise en charge dans une maternité, dans le service maternité grossesses à haut risque. Ils ont été hyper aux petits soins avec moi. Mais voilà, c'était compliqué de lui dire : « Ben voilà, j'ai peur qu'après l'anesthésie je ne sache plus parler, je ne sache plus bouger, je ne puisse pas m'exprimer, je ne peux pas ci, je ne peux pas là » (R5-F, ≤ 2 yrs, Severe)

9.3.3.3 Level of expertise healthcare providers

The survey results showed a negative pattern concerning how respondents rate the level of expertise of the healthcare providers they consulted for ME/CFS (N=745). The majority of respondents evaluated the overall level of expertise as very low (32.1%), rather low (23.8%) or average (20.5%). The other respondents rated the level of expertise as rather high (10.1%), very high (6.7%), or they didn't know (6.8%).

In the interviews, the following problems related to a lack of technical skills and knowledge regarding the diagnosis and treatment of ME/CFS among physicians were described. This deficit manifested in multiple ways including uncertainty about appropriate referral pathways, the continuation of ineffective treatments, and an inability to provide clear answers or relevant information. In addition, they hold prejudices about the illness and those affected by it, alongside doubts regarding the diagnosis or the existence of the illness.

A reluctance to seek further information or to engage in learning about ME/CFS, either proactively or reactively, was also apparent. Consequently, they misdiagnosed and had the tendency to fit individuals into more familiar medical categories rather than recognising the specific features of ME/CFS.

« Donc, je suis allée voir plusieurs internistes qui me disent... d'abord qu'ils remettent systématiquement en cause le diagnostic que j'ai reçu. Et donc, on est à chaque fois reparti pour toute une batterie d'examens pour finalement arriver à, « ben, si c'est certainement bien de ça que vous souffrez. Mais je ne connais pas votre pathologie. Donc, je ne peux rien vous proposer ». Et donc, voilà. Donc, à l'heure actuelle, je n'ai aucun médecin, en fait. Là, j'ai rendez-vous en avril avec une nouvelle personne qui, je pense, s'y connaît un peu. Mais sinon, c'est vraiment un peu du ping-pong comme ça où on nous renvoie d'une personne à l'autre. » (R6-F >2 yrs, Mild/moderate)

Interview accounts suggested that this lack of expertise was attributed to the limited or insufficient coverage of ME/CFS within medical training curricula, and to its insufficient recognition as an official diagnosis.

« Dr X m'a dit : « C'est une pathologie qui n'est pas expliquée en faculté de médecine. En médecine on ne parle pas de cette pathologie-là. Donc il y a un manque de reconnaissance de cette pathologie. Donc forcément, comme les médecins ne sont pas formés, ça génère un délai pour trouver le diagnostic avec des démarches. » (R9-F, ≤ 2 yrs, Mild/moderate)

“Er staat nergens in een boekje, ah, iemand met ME/ CVS dan moet je dat en dat en dat doen. Dus ze zeggen, het is iets psychologisch, of men gaat naar iets trekken dat wel in een boekje staat. Terwijl dat... Ja, dat is ook maar een impressie dat ik heb, maar... Het lijkt wel alsof men weet dat het dat niet is, maar dat men liever een bepaalde procedure volgt. Ook al is het niet de juiste. Dan te zeggen, ja, sorry, maar ik kan u niet helpen. Dat heb ik nooit gehoord. Dus men gaat u slaappillen voorschrijven, of naar een psycholoog sturen, of antidepressiva voorschrijven, en van die zaken.” (R17-F, ≤ 2 yrs, Severe)

A continuous search for help from physicians with accurate expertise was also addressed.

« Quand on a plusieurs refus de personnes que finalement ça n'intéresse pas qui préfèrent traiter les tumeurs cérébrales ou l'Alzheimer ou d'autres pathologies et bien vous vous dites je vais m'adresser à qui? Je vais m'adresser à qui? J'en ai marre qu'on me dise que c'est dans ma tête qu'il faut que je bouge mon cul. » (R3-M, >2 yrs, Severe)



This lack of expertise can lead to several consequences:

First, examples of patient blaming were given, including accusations of laziness and unwillingness to cooperate, despite patients being physically unable to comply with what was asked to them. Traumatic experiences were reported as a result.

“Dus ja, ik was al zagezegd lui. Ze noemden me lui. En ik niet gemotiveerd. Maar dat lukte mij gewoon niet. Ik werd kapot tegen dat ik daar was. Zelfs de rit naar daar, je klaarmaken, de rit naar daar, dat zie je nu, dat was eigenlijk al te veel. Ik moest dan ook nog oefeningen doen, en de loopband, en fietsen. Allee, dat lukte mij totaal niet. Maar dat lag dus aan mij. Ik was lui. Ik was niet gemotiveerd. Allee, zwaar. Eigenlijk is dat wel ook traumatisch geweest.” (R15-F, >2 yrs, Severe)

« Quand je m'y suis présentée pour moi, pour moi-même, donc pas pour ma fille et que j'ai expliqué un petit peu tout ce parcours il m'a dit « vous savez des gens comme vous en fait moi je les vois dans les magasins ils rigolent en faisant leurs courses, je m'excuse mais je vous comprends pas ! » oui et c'est vrai que j'allais pour la prolongation de mon certificat médical pour mon employeur... qu'est-ce que vous voulez dire il dit ben oui je vais vous mettre la prolongation mais moi voilà vous seriez bien ma première patiente au sein de la maison médicale ici ! Oui donc en gros il n'apporte aucune crédibilité à ce que je lui avance... j'ai eu vraiment... pour moi j'ai été maltraitée j'ai été hyper maltraitée. » (R4-F, ≤ 2 yrs, Mild/moderate)

Second, limited expertise may also contribute to physicians' reluctance to prescribe or extend sickness leave, refer their patient to specialists, arrange further investigations or provide requested treatments, including off-label prescriptions.

« Il y a, par exemple, deux molécules qui sont assez prescrites chez les malades dans d'autres pays qui sont des molécules qui sont utilisées, enfin, des traitements qui sont utilisés pour de toutes autres pathologies et qui donnent des bons résultats pour chacun, à peu près, un quart des malades. Et je pense qu'à nouveau, c'est quelque chose qu'on pourrait essayer ici, mais la plupart des médecins sont frileux à l'idée de prescrire vu qu'ils ne connaissent pas la pathologie et qu'ils ne nous connaissent pas en général suffisamment. Ben, la dernière fois, ça a été ça. Pour moi, ça a été un refus de... « Oui, donc, vous avez bien cette pathologie, mais non, je ne suis pas assez expérimentée pour essayer ces molécules avec vous. » (R6-F >2 yrs, Mild/moderate)

It was claimed that Belgium was lagging behind its neighboring countries in terms of medical expertise, research, and development regarding ME/CFS. As a result, some interviewees sought care abroad, even when it was not reimbursed and therefore came at a higher cost. It was also suggested to consider international collaboration.

“En gelukkig in het buitenland, onze buurlanden, die wel investeren. Maar het is dan eigenlijk wraakroepend van een land als België dat altijd prat gaat op zijn biomedische sector en farmaceutische sector, dat er hier nougat bollen wordt gedaan.” (R10-M, ≤ 2 yrs, Mild/moderate)

“Toen zijn wij verder gaan zoeken. Toen zijn wij in Nederland terechtgekomen bij [naam organisatie]. En die zijn in ieder geval in Nederland volledig erkend. En ook door de hoogste rechter. Dus de uitkerende instantie moet de diagnoses van deze stichting accepteren als zijnde waarheid. En zij diagnosticeren op basis van de ICC en nog wat dingen. Door allerlei... Testen enzovoort. Uiteindelijk ben ik daar dus beland in maart 2025. Oké. En zij hebben dus uiteindelijk na allerlei intake en ook allerlei vragenlijsten, onderzoeken enzovoort... hebben zij de diagnose gesteld.” (R11-M, ≤ 2 yrs, Severe)

Nevertheless, the interviewees noted that there were significant differences: some doctors were very supportive, while others were quite discouraging and even humiliating.

“Ça dépend lesquels. On a rencontré des médecins qui sont vraiment formidables. Notre généraliste, (...) quand on la voit, c'est un plaisir parce qu'on sait qu'elle connaît [mon fils], qu'elle

comprend, qu'elle se souvient de lui, qu'elle sait très bien par quoi il passe. Le généraliste qu'on avait avant, il commençait par faire un grand sourire et faire « Alors, comment ça va la forme ? » Déjà ça, cette réaction au début, ça nous mettait mal à l'aise. Il y a des médecins qui sont tout simplement humains, qui se renseignent ou qui au moins admettent qu'ils ne savent pas mais qui reconnaissent la maladie et qui comprennent.» (R1-M, ≤ 2 yrs, Mild/moderate)*

“Dan moet je je stress verminderen, je moet, ja, want ik heb dan met de deur geslaan, ik weet, ik ga dat nooit vergeten, ik was zo kwaad op haar. Ik zeg, allee, op een bepaald moment zei ze, of was er een evaluatie, ik zeg, ja, ik zeg, het is iets beter gegaan, maar ik heb het gevoel dat het nu weer achteruit gaat. Dat was een eerlijk gevoel op dat moment, en haar antwoord was, ja, het is niet moeilijk, je ziektebriefje loopt af zeker. En ik dacht, ja, oké, ik stop ermee. Dat was echt traumatisch. Het was echt, ja. Maar ik hoorde dat dat, zelfs nu, zoveel jaar later, we zijn 10 jaar later, dat (...) ze nog geen meter veranderd is naar empathie toe. Oké. Dus dat, daar heb ik echt niets aan gehad, dat heeft belachelijk veel geld gekost, en ik heb daar echt niets aan gehad.” (R15-F, >2 yrs, Severe)

The goodwill of physicians who tried to help despite limited expertise and tools was appreciated.

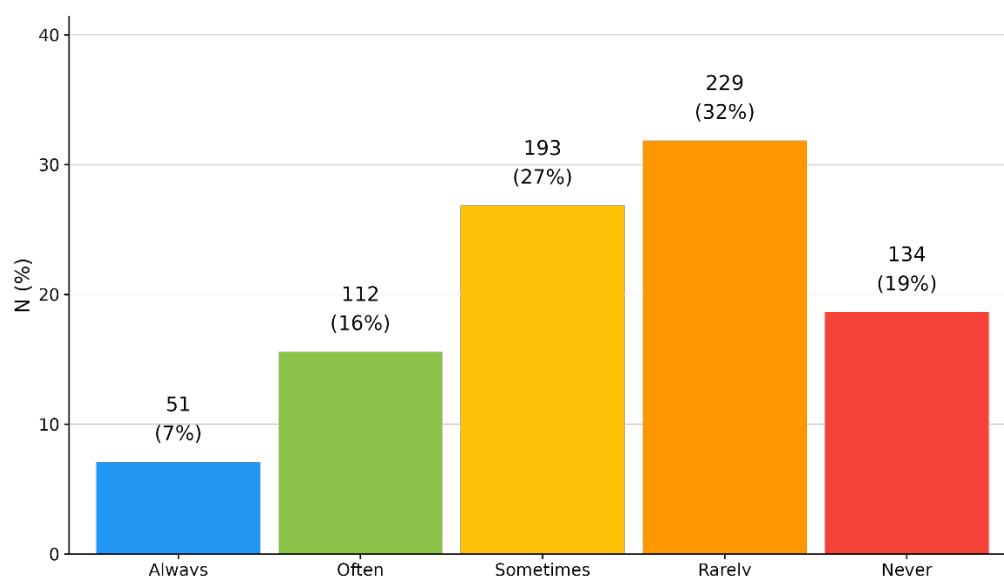
“Ik denk dat er heel veel zorgverleners zijn die het goed menen met de patiënten, maar die gewoon ook te weinig tools hebben om een wel overwogen op wetenschappelijkheid evidentie gebaseerde beslissing te nemen.” (R10-M, ≤ 2 yrs, Mild/moderate)

“De samenwerking naar andere landen ook naar onderzoek dat er daar misschien echt wel er is daar veel te weinig onderzoek naartoe en in Duitsland of in Nederland staan ze ook wel een stukje verder dan hier.” (R12-F, ≤ 2 yrs, Mild/moderate)

9.3.3.4 Information received

Figure 16 shows respondents' satisfaction with the quality and quantity of information they received from their healthcare professionals. Overall, survey results provided a generally negative pattern since most respondents reported that they never (19%), rarely (32%) or sometimes (27%) received sufficiently clear and understandable information from their healthcare professionals. Only around one fifth of the respondents had positive experiences and reported that they received often (16%) or always (7%) clear and understandable information from their healthcare professionals.

Figure 16 – Satisfaction with information received from healthcare providers (n=719)



Legend: Responses “I don’t know” (n=26) were not displayed in the figure.



We heard from the interviews that obtaining adequate information about the disease and available support service was difficult, particularly when healthcare providers themselves are insufficiently informed.

« Je pense que j'aurais aimé (...) qu'on m'explique les choses, qu'on m'explique... parce que j'ai eu un diagnostic, et c'est tout. On ne m'a pas parlé de malaise post-effort... Enfin si, on m'a dit, ce que vous faites, c'est des malaises post-effort. D'accord. Mais c'est tout. Je ne savais pas ce que c'était exactement, ce qui les déclenchait, ce que je pouvais faire par rapport à ça. Donc j'aurais aimé (...) qu'on m'explique en tout cas, ce que c'est réellement, et puis qu'on m'enseigne (...) le dosage d'activité, donc qu'on appelle le pacing, Oui, je pense que j'aurais aimé (...) qu'on me dise qu'il y avait des ressources par rapport à ça, parce qu'il y a des associations qui ont fait des guides par rapport à ce dosage énergétique. Et pour tout, en fait, je n'ai eu aucune explication, ni par rapport au degré de sévérité de la maladie, ni le fait que faire des malaises post-effort aggravait la sévérité de la maladie à long terme. Donc j'ai eu un diagnostic, mais je n'ai eu aucune explication. Et je pense que déjà, les explications m'auraient permis de délimiter un petit peu la maladie, de lui donner des contours, de voir un petit peu ce que c'était, ce que je pouvais faire ou pas. » (R6-F, >2 yrs, Mild/moderate)

We heard that treatments are sometimes initiated without sufficient information being provided on potential side-effects. Patients rely on multiple sources of information, including scientific literature and peers found online, social networks or patient associations. However, the information available on social media is not always verified by official bodies, which can lead to misinformation.

To satisfy their information needs, interviewees mentioned:

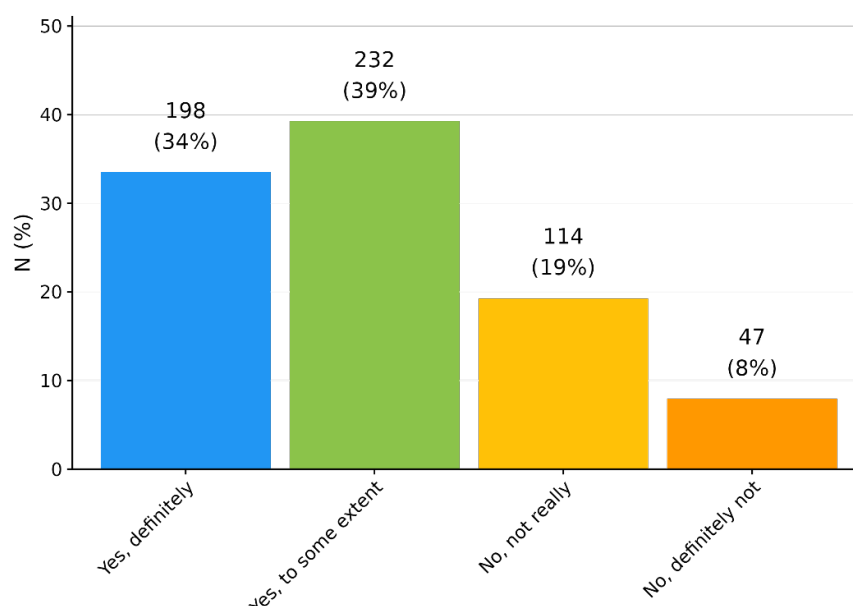
- The centralisation of accurate, scientific-evidence based information in a national expert network, including access to an up-to-date database on the current state of medical knowledge about the causes and progression of the disease, diagnostic criteria and effective (off-label) treatments, or validated online information.

« [Les médecins] disent 'ne cherchez pas sur Internet des informations. Ça entretient une certaine forme de stress etc. Donc si vous êtes d'accord avec le diagnostic, simplement faites le pacing etc. Mais c'est quand même un domaine où il y a de la recherche, de la nouveauté et des choses qui changent et on les voit passer chez les patients et donc quelque part ce serait utile d'avoir un peu l'avis des professionnels de la santé sur ces études-là. (...) Il y a des moments où on va sur Internet. Donc, c'est pour ça que ce serait mieux d'avoir de temps en temps une évaluation des dernières études ou des tendances. S'il y a quelque chose qui devient vraiment à la mode dans les groupes de patients, etc., pouvoir avoir un commentaire officiel. » (R8-M, ≤ 2 yrs, Mild/moderate)

- The provision of easily accessible information for patients about the disease and its management with preference for auditive or printed formats. In complement, guidance on appropriate healthcare providers to consult as well as information on available services and entitlements to reimbursements.
- The sensibilisation of the general public about ME/CFS and its consequences.
- International collaboration in research and development about ME/CFS.

9.3.3.5 Involvement in treatment decisions

For the survey respondents for whom this question was applicable (n=591), almost three quarters indicated that the level of involvement in their treatment decisions was definitely (34.0%) or to some extent (39.0%) in line with their preferences (Figure 17). For the remaining 27.0% this was not the case.

Figure 17 – Level of involvement in treatment decisions (n= 591)

Legend: Responses “I don’t know” (n=18), “No but I don’t want to be involved” (n=0), “Not applicable: no decisions made regarding treatment” (n=136) and missing (n=4) were not displayed in the figure.

Interview data suggest that these figures require careful interpretation. Participants highlighted that patients are often compelled to initiate diagnostic or therapeutic steps themselves, largely due to the limited awareness of ME/CFS among healthcare providers. Nevertheless, these patient-led initiatives were not systematically implemented because doctors do not endorse the proposed procedures. When patients manage to find a healthcare provider who shares their vision, their sense of involvement in their care increases.

In addition, consultations were described as too short to adequately address the full range of patients’ complaints.

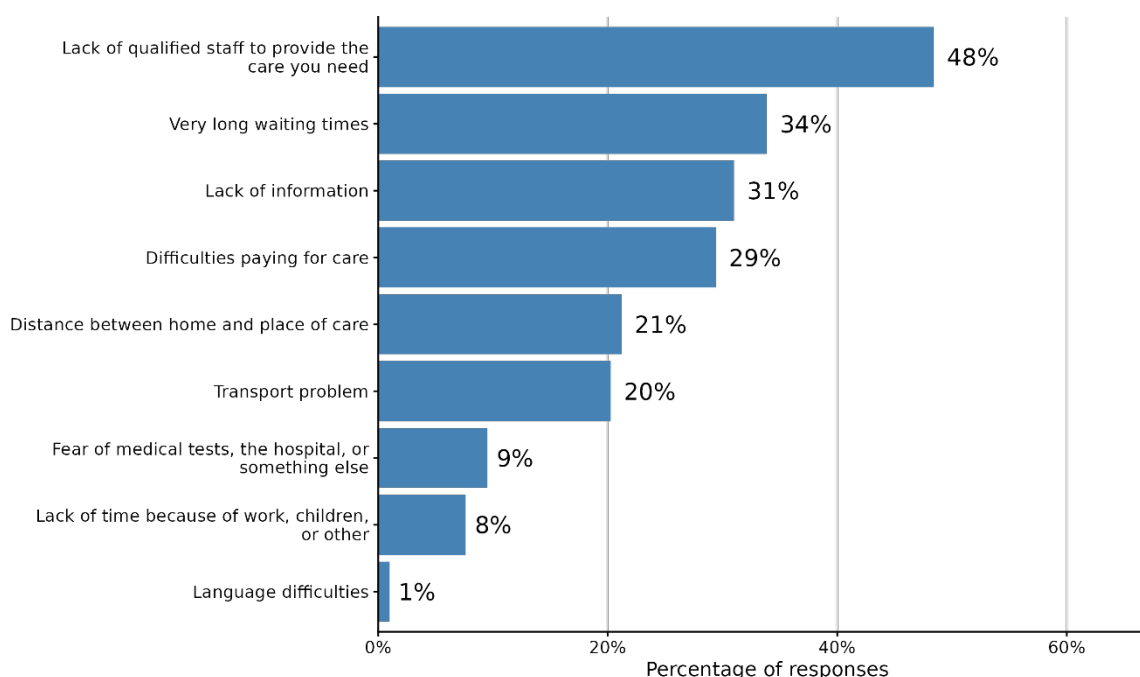
« Donc voilà, ce n’est pas toujours évident. Et puis comme ces rendez-vous sont tellement précieux, c’est un quart d’heure une fois par an ou tous les six mois, mais il y a tellement de choses à mettre dedans que voilà. » (R5-F, ≤ 2 yrs, Severe)

9.3.3.6 Forgone care

Figure 18 provides an overview of the respondents’ experiences with the causes of forgone care in the past 12 months. Among the respondents who completed this question (n=745), 316 people (42.4%) indicated they had forgone care in the past 12 months, whereas 288 people (38.7%) experienced no forgone care, and 18.8% indicated they didn’t know. The most frequently reported causes of forgone care included a lack of qualified staff available at the moment they needed care (48%), very long waiting times to access care (34.0%), and a lack of information (31.0%). Financial difficulties with paying care (29.0%) and logistical barriers like distance between home and care facilities (21.0%) and transport problems (20.0%) were also frequently reported.



Figure 18 – Causes of forgone care identified by survey participants (n=316)



These problems were also mentioned in the interview data. Patients encountered difficulties in finding a physician with adequate medical expertise regarding ME/CFS and appropriate treatments within an acceptable timeframe. Moreover, simply finding a GP or a physician who was empathetic and supportive, who did not downplay or dismiss the symptoms, who showed an interest, and who was willing to work towards finding solutions (e.g. initiate symptomatic treatment or refer to a specialist) was reported as difficult. These difficulties and negative experiences may lead to anxiety about how an unfamiliar physician might respond to the symptoms or diagnosis.

As noted previously, when information about a potential treatment was obtained through fellow patients or other sources, difficulties were reported in finding a physician willing to prescribe an off-label treatment with which they were unfamiliar. The same was said to apply for examinations: once patients were aware of a particular examination, they still had to find a prescribing physician, a hospital, or a healthcare provider who was willing and able to conduct the examination and interpret the results.

“Ik heb hier een jaar gezocht, niemand gevonden. Dus dat is echt een issue in België. Dus door het feit dat er ook geen huisartsen worden opgeleid, dat er ook niet meer nagedacht wordt. Je bent als patiënt dokter van je eigen. En je bent ook geen baas over je eigen lichaam. Dus daar heb je dus ook een constatacie waarachter het komt als je een keer aan de patiëntzijde zegt dat je niet zomaar alles kunt proberen, maar dat je eigenlijk eerst een jaar moet zoeken achter een huisarts die alsjeblieft op beide knietjes kussen u wil helpen om dat voor te schrijven. Ook al is dat in Nederland wel relatief courant en wordt er wel aangegeven dat er 60% beterschap ervaart. Toch werkt het dat je geen partner hebt in het zorgsysteem om u... Dus ik heb dat dan... Dat was dan een andere specialist in [Stad] die dat wel onderzocht en voorschreef. Maar die mens heeft “wachtrijen van een jaar en een half.” (R10-M, ≤ 2 yrs, Mild/moderate)

« Si on sait qu'on a besoin de faire un examen parce qu'on a vu dans la littérature, dans les conseils qui sont donnés aussi en France, là où l'autorité de santé donne des listes d'examens à faire, si on va voir un médecin en Belgique en disant qu'on pense que ce serait nécessaire de faire cet examen, déjà, il faut un médecin qui soit convaincu que c'est nécessaire. Et puis si on a une prescription, après, il faut un hôpital qui fasse cet examen, qui ait de la disponibilité pour le faire et qui ait derrière un spécialiste capable d'interpréter. Ces étapes, c'est un souci. » (R1-M, ≤ 2 yrs, Mild/moderate)*

Searching for and switching between healthcare providers was perceived as burdensome, yet participants felt they had no alternative if they were to find solutions to their health problems. This is why participants continued to seek answers, despite the considerable effort required.

« En fait, je dis : « Si on m'avait entendue et si quelqu'un avait pris en charge le dossier et me donner des rendez-vous dans le même hôpital dans des délais raisonnables, je n'aurais pas cherché moi-même, je ne me serais peut-être pas autant diversifiée en fait ». C'est vrai que ça paraissait un peu particulier. Quelqu'un qui allait comme ça dans des endroits différents, mais c'est parce que je voulais une réponse. » (R9-F, ≤ 2 yrs, Mild/moderate)

From the interviews, we heard that waiting times can extend to several months for consultations with specialists and psychologists, and even for GPs, delays of more than one week were not uncommon. We heard that these long waiting times lead to considerable frustration, as patients experienced no progress in their search for a diagnosis or appropriate treatment. During this period, they remain in a state of persistent fatigue and pain, accompanied by anxiety about their condition and fear of further deterioration. Waiting time was therefore perceived by patients as wasted time.

« Voilà de pouvoir le programmer suffisamment tôt de pouvoir y accéder physiquement parfois de ne pas attendre de dire « oh vous avez besoin de ça décembre 2026 j'ai pas de place avant ». Parce que très souvent pendant ce temps-là c'est rien qui avance. » (R3-M, >2 yrs, Severe)

“En die zeiden van nou, dan moet er eerst een intake plaatsvinden. Nee, moet ik even teruggaan wanneer dat was. Dat was in december 2024. Ja, ik kijk even van het papiertje af. En die zeggen nou, in de loop van 2025... Krijg je Dan een intake. En dan in de loop van 2026 komen allerlei disciplines aan de orde. En dan als je geluk hebt in 2026 of 2027 komt er een diagnose. Oké. [...] Toen zijn wij verder gaan zoeken. Toen zijn wij in Nederland terechtgekomen.” (R11-M, ≤ 2 yrs, Severe) « Donc voilà, ce n'est pas toujours évident. Et puis comme ces rendez-vous sont tellement précieux, c'est un quart d'heure une fois par an ou tous les six mois, mais il y a tellement de choses à mettre dedans que voilà. » (R5-F, ≤ 2 yrs, Severe)

We heard also that another key issue related to accessibility is the presence of only one reference center for ME/CFS in Belgium. This results in long travel distances for many patients, and for French speaking patients an additional language barrier. We were told that access to appropriate care is even more challenging for minors than for adults, as they may be refused by adult-oriented care services.

« La seule chose que j'avais trouvée en Belgique, c'était un centre de référence à Leuven. Mais tout était en néerlandais. Je ne parle pas du tout en néerlandais. Et du coup, voilà . » (R6-F >2 yrs, Mild/moderate)

« Il y a une clinique qui pourrait ressembler un petit peu à ce qu'il y a à Rome, c'est-à-dire que normalement, ça réunit des spécialistes qui s'y connaissent en covid long, qui peuvent faire des examens. Mais je les avais contactés, ils m'avaient dit « Moins de 18 ans, on ne prend pas ». (MR1, ≤ 2 yrs, Mild/moderate)*

Moreover we heard that patients who reported in the survey having forgone care due to a lack of information may in fact be experiencing difficulties in being referred to the appropriate medical specialty.

Financial constraints are described in 9.4.4. These constraints were reported as a reason to skip, postpone, or discontinue treatments due to their cost, and to forgone support services.

Finally, as noted previously, living with ME/CFS is characterised by considerable uncertainty regarding day-to-day functioning. We heard that this unpredictability can make it difficult to attend scheduled consultations. As a result, appointments sometimes needed to be rescheduled because patients were too ill to attend, resulting in delays in accessing necessary care. Consultations were also described as requiring too much energy or causing overstimulation (for example due to the transport to the physician or clinic, the distance between the car park and the consultation room, or the noise in the waiting room). We heard a clear need for greater flexibility in care provision.



“Dus dan moest hij natuurlijk iedere keer naar Knokke voor dat infuus. En dat was iedere keer een namiddag, want dat loopt drie, vier uur. Zeer vermoeiend. Maar ik heb dat wel drie jaar gedaan, tot covid. Dat was normaal gezien de bedoeling wekelijks. Maar voor mij was dat totaal niet haalbaar. Want ik moest een week recupereren van dat bezoekje aan het ziekenhuis. Want we zaten eigenlijk allemaal samen. Ook met oncologie. Dus dat was een drukte van je welste. En ja, die prikkels, dat was gigantisch. Dus ik deed dat om de twee weken.” (R15-F, >2 yrs, Severe)

Interviewees suggested options such as online or tele consultations, as well as home visits to facilitate timely access to healthcare and prevent PEM. In addition, the need for a management that considers increased sensitivity to sensory stimuli was highlighted.

9.3.3.7 *Fraudulent practices*

Malicious healthcare professionals who promised effective treatment while charging substantial financial fees were mentioned by the interviewees. Patients can be so desperate to obtain relief that they were prepared to pay large sums of money until they were financially ruined. Abuse can also be more subtle, such as requests for payment for medical reports or the continuation of costly treatments despite a lack of demonstrated effectiveness.

“Meldonium oké, dat neem ik nu eigenlijk al bijna een jaar maar eigenlijk heeft dat weinig effect en dan stel je jezelf ook de vraag is dat nu iemand die profiteert van mij want uiteindelijk wil ik wel beter worden (...) maar hij vraagt ook veel te veel en ja dat is ook niet oké en dan zit je ook in de strijd maar wat moet je doen? Is dat oké wat hij doet? Hij beweert dat wel bij hoog en bij laag, maar houdt hij mij hier niet gewoon aan het lijntje” (R12-F, ≤ 2 yrs, Mild/moderate)

“On se retrouve face à des gens qui ont parfois peu de scrupules en disant par exemple: « Mais pour l'AVIQ, vous avez besoin d'un rapport. Donc je vais vous faire le rapport. Le rapport est facturé 50 € de l'heure. J'aurais besoin d'une heure et demie pour faire mon rapport. Donc ça va faire 75 €. Et là, on est là, on ne sait pas quoi, ça vient de nous arriver, on ne sait pas ce qui arrive et donc ça, je pense aussi que s'il y a une professionnalisation, il y aura moins de places pour des personnes malintentionnées qui viennent chercher de l'argent auprès des personnes qui se retrouvent dans des situations inconnues, en fait”. (R5-F, ≤ 2 yrs, Severe)

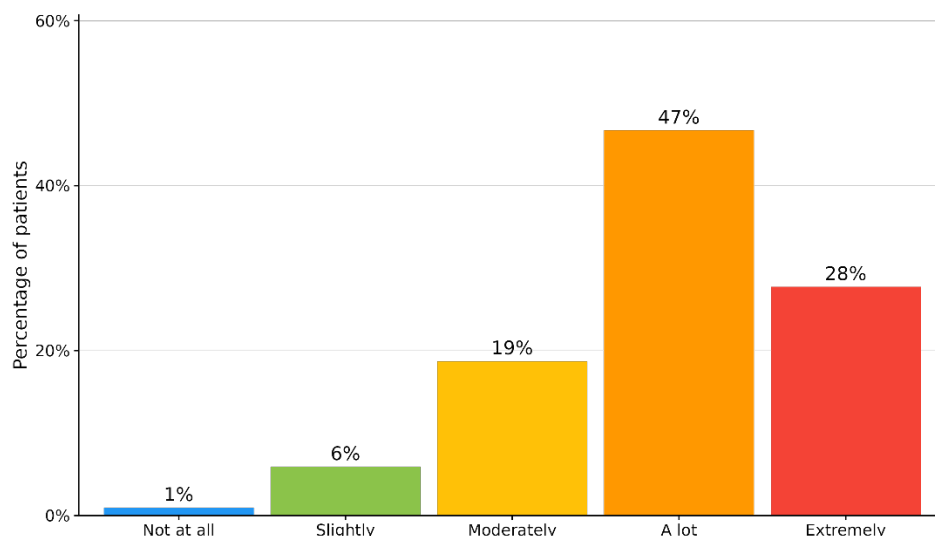
“Dus er is... Maar ik kan me wel voorstellen dat er mensen zijn die... ten einde raad zijn, en die dan... Ja, er loopt een behandeling... Ik heb daar nog eens voor gehad, zelfs iemand in België, en die vroeg ook al direct 1000 dollar, de euro bedoel ik, of nee, 5000 euro, het was ook 5000 euro. Maar dat het lijkt dat men gewoon gaat door blijven doen, totdat je gewoon financieel geruïneerd bent. En ja, ik ben er niet in gegaan, maar er zijn... Ik denk wel dat er veel mensen zijn die wel op die manier, financieel geruïneerd... Die eigenlijk het weinig spaargeld nog wel volledig opmaken aan behandelingen. En ja, ik heb toch het gevoel dat die mensen niet eerlijk zijn.” (R17-F, ≤ 2 yrs, Severe)

9.4 **Unmet patient needs: social needs**

9.4.1 *Impact on social life*

9.4.1.1 *Impact on personal relationships*

Figure 19 shows that personal relationships with family, partner, friends or neighbors were negatively affected by ME/CFS among most survey respondents. The impact was mostly rated as extreme (28.0%) or a lot (47.0%), indicating that three quarters of the sample perceived a clear negative impact. Only 7.0% reported no or slight impact, while 19% reported a moderate impact.

Figure 19 – Impact of ME/CFS on personal relationships (n=743)

Legend: Responses “I don’t know” (n=2) were not displayed in the figure.

Interview data indicated that social relationships deteriorated or broke down due to conflicts arising from the physical limitations experienced by individuals with ME/CFS. Patients felt they were not understood by family and friends. They found it exhausting to keep explaining themselves.

“Ja. Het wordt niet gehoord. En dat is nog zo'n probleem. Je naasten, je partner, je kinderen, familieleden, laat staan de buitenwereld. Dat ook ten minste na een diagnose ook je partner, je familieleden, dus je naasten, mee geïnformeerd worden. Welke impact heeft dat? Wat erbij komt kijken? Ja, dat brengt allemaal zoveel frustraties mee. Je hebt duizend keren moeten uitleggen. Ik kan dit niet. Wil je alsjeblieft stil zijn? Wil je dit? Ruzies. Verdriet. Pijn. Ja.” (R13-F, ≤ 2 yrs, Severe)

IMPACT ON FAMILY AND PARTNER

From the interviews, we heard that these difficulties may arise prior to the establishment of a diagnosis, when it is particularly challenging to understand and interpret the patient's limitations. Conflicts with partners were reported and, in some cases led to separation. We also heard that the substantial burden that ME/CFS places on partners, combined with uncertainty about the future contributed to concerns about losing their partner.

« Je pense que ça a joué un rôle dans la séparation que j'ai eue avec la maman des enfants parce qu'il y a eu il y avait... On n'avait pas de diagnostic et il y avait une frustration chez elle du fait que j'avais des limitations, que je faisais moins pour les tâches les tâches ménagères ou communes, même si j'essayais de continuer et que j'ai d'abord supprimé mes hobbies avant de réduire ma part. Oui. Mais donc, il y avait... Et alors, ça a une conséquence quand même aussi. » (R8-M, ≤ 2 yrs, Mild/moderate)

The family members can become informal caregivers.

« Et alors tout ce qui est gestion de la maison, enfin c'est lui [mon mari] qui prend charge quoi. Donc, c'est mon aidant proche. Et alors, quand on n'a pas l'infirmière, mais il devient aussi mon soignant pour me déshabiller, tout ça. Voilà, quand je suis trop fatiguée, et ma fille aînée aussi, qui a seize ans, elle m'aide aussi beaucoup. » (R5-F, ≤ 2 yrs, Severe)



IMPACT ON AFFECTIVE RELATIONSHIPS

Those who had lost their partner explained how difficult it was to enter a new relationship. Once their limitations became apparent, potential partners tended to withdraw. Those who had found a new partner felt uncertain about the stability of the relationship and how to cope if the partner were to end it.

“Moi- j'ai fait une croix là-dessus parce que j'ai essayé d'avoir - donc je suis divorcé puis j'ai eu des petites amies mais avec la fatigue actuelle voilà je ne recherche plus l'amour parce que ça me demande beaucoup trop de une nouvelle relation ça demande beaucoup beaucoup d'énergie il faut s'investir et c'est impossible pour moi. » (R2-M >2 yrs, Mild/moderate)

« Maintenant, je suis séparé depuis début 2023. J'ai essayé de, comme tout le monde, de renouer une relation. Une fois que je dis que je suis en invalidité et que j'explique un peu, c'est fini. Ah, non, t'es chouette, mais non, moi, je ne veux pas être un invalide, je ne veux pas de, je n'ai pas envie d'avoir des problèmes en plus des miens. (...) J'ai même été sur un site pour handicapés, à un moment donné. Je vois que le constat est le même. Les personnes n'ont pas envie de se mettre avec quelqu'un qui a un handicap physique ou intellectuel. Elles préfèrent se mettre avec une personne valide. Qui a une vie standard. Qui travaille, qui fait un peu de sport, qui aime le jardinage, qui aime la rando. Qui a une vie standard. » (R3-M, >2 yrs, Severe)

IMPACT ON SOCIAL RELATIONSHIPS

Interview data highlighted the disruptive impact of the condition on social relationships. Participants described a progressive reconfiguration of their social networks, with some relationships fading over time, due to reduced contact or a lack of understanding. Maintaining social interactions was reported to require significant effort and constant adjustment, leading some interviewees to deliberately distance themselves from certain social circles. While some of them reported receiving sustained understanding and consideration from close relatives or friends, others emphasised the isolating nature of the condition. Reduced energy levels, increased time spent alone due to symptom severity, and difficulties engaging with others' concerns contribute to emotional withdrawal and social isolation.

« Je vois de moins en moins mes amis et c'est normal la dernière on m'a toujours j'ai toujours fêté nouvel an avec des amis et la dernière fois, c'était l'année 2023-2024, j'ai dû partir c'était très mal poli mais c'était vraiment une raison de santé à 23h30 parce que je sentais de nouveau ces vertiges, ces maux de tête qui m'indiquaient que j'allais faire un énorme crash ... cette petite demi-heure qui aurait été nécessaire pour dire bonne année bonne santé pour être poli... j'allais la payer pendant deux ou trois jours au lit ! et depuis on ne m'a plus invité pour la nouvelle année le soir c'est très très dur. » (R2-M, >2 yrs, Mild/moderate)

“C'est pas évident à accepter au début, mais oui, ça fait un tri. Il y a des personnes qui, d'elles-mêmes, vont de moins en moins prendre de nouvelles et donc, juste disparaître de notre vie. Il y a des personnes où il faut expliquer un peu plus (...) Les personnes importantes ont compris, prennent en considération mon état, ce que je peux faire ou pas faire, sont précautionneuses. (...) Mais je pense que ça peut être très isolant pour d'autres personnes. (...) Déjà, c'est une maladie qui isole, rien à faire, ne serait-ce que de devoir passer du temps tout seul dans sa chambre, et je pense que ça peut être très isolant comme maladie. » (R6-F >2 yrs, Mild/moderate)

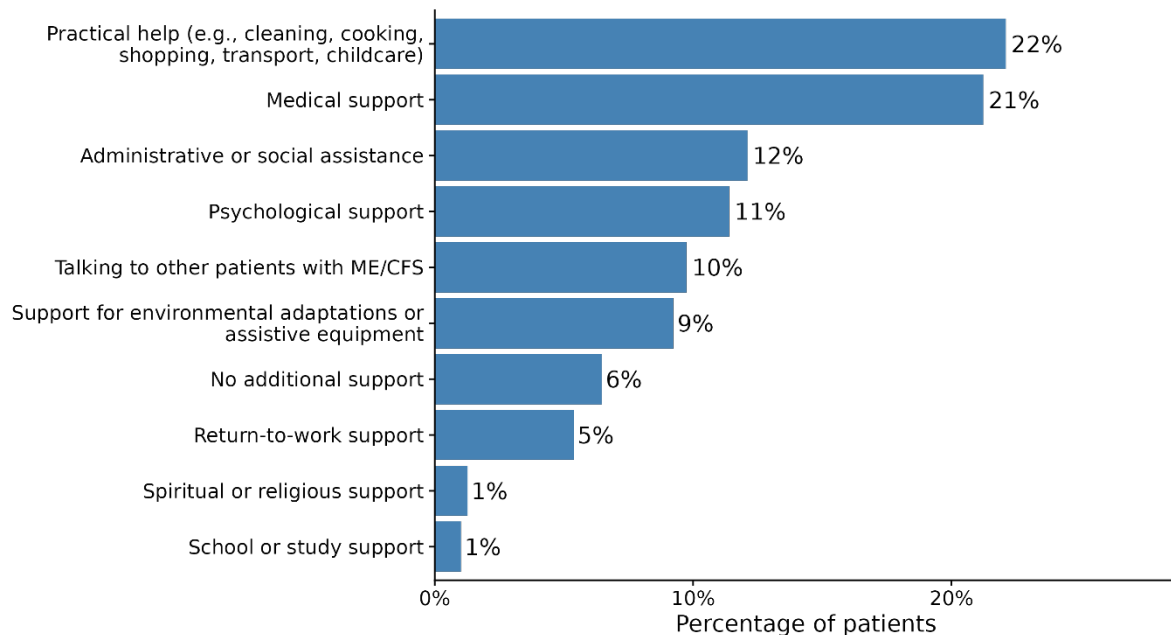
« Au niveau social, c'est un peu compliqué aussi. Parce que ça demande toute une gymnastique, euh... Alors, euh... Pour vous situer, je bossais à temps plein. Les week-ends, euh... Il y avait... Que ce soit vendredi, samedi, dimanche, une vie extrêmement bien remplie. On arrivait au boulot le lundi, (...) Voilà. J'avais vraiment une bonne vie, enfin... Maintenant, je suis loin du compte. Il y a des personnes, que je pouvais côtoyer, mais avec lesquelles j'ai pris de la distance parce que j'avais un peu de mal, j'avoue, à écouter les problèmes des autres. » (R4-F, ≤ 2 yrs, Mild/moderate)

9.4.1.2 Impact on social support needs

The most frequently reported unmet social support needs included practical support, such as assistance with cleaning, cooking, shopping, transport, or childcare (22.0%) and medical support (21.0%).

In addition, a lack of administrative or social support (12.0%), psychological support (11.0%), and opportunities to talk with other patients diagnosed with ME/CFS (10.0%) were reported. Other less frequently reported unmet social support needs are shown in Figure 20.

Figure 20 – Needs for social support that was not received



Legend: Responses “I don’t know” (n=60) were not displayed in the figure, participants could select one or more options

We learned from the interviews that accessing supportive services posed additional challenges. These services were described as costly and not always adequately adapted to ME/CFS. The importance of continuity of care (i.e. avoiding frequent changes in staff) and awareness of sensory sensitivities was emphasised. In the absence of such adaptations, support provision could become both emotionally and physically exhausting. Excessive noise, constant talking, and disruptive behaviour were reported as particularly difficult to tolerate, while repeatedly having to request quieter conditions was described as an additional burden.

“Dan moet ik nog twee, drie keer zelf mijn plan trekken. En dan komen ze terug. Ja, maar dat was ook belachelijk veel geld. Maandelijks. En altijd nieuwe mensen, dat is ook zeer vermoeiend. Ja, sommigen zijn zodanig uitbundig dat ik kapot word als ze weg waren. Ja. Ik zeg even van, kan je alsjeblieft een beetje stiller zijn, of minder babbelen, maar ja, die mensen snappen dat niet, hé. Dus eigenlijk daarin, ja, dat is ook nog iets. Dat ze ook eigenlijk met hun familiehulp, bijvoorbeeld, dan die mensen ook misschien een keer al eens maar een mini-opleiding krijgen van, er zijn prikkelgevoelige mensen. Hou daar ook rekening mee, dat ze niet... Allee, dat dat zeer vermoeiend is. Iemand die constant rondloopt en babbelt en met de deuren en de kasten slaat, dat is...je kan dat wel keer zeggen maar je kan dat niet iedere week herhalen, snap je?” (R15-F, >2 yrs, Severe)

There is a need for flexible, collaborative and practical support that complements patients’ capacities, rather than assistance that replaces them entirely. They valued doing tasks together, particularly when



physical limitations prevented these tasks from being performed independently. This type of support helped preserve autonomy while responding to physical constraints.

« C'est vrai que je peux aller voir au service social de ma mutuelle. Ou aux A.L.E. Mais moi, du point de vue physique, j'ai plus besoin de quelqu'un qui va venir et qui je vais dire, tiens, aujourd'hui, on fait la vaisselle. Tiens, la prochaine fois que tu viens, j'aimerais bien qu'on charge tous les cartons qui se trouvent dans la cuisine. Parce que ça aussi, une fois que j'ai chargé la voiture, moi, je ne sais pas aller au parc à container. Et qu'on aille ensemble parc à container... Aujourd'hui, j'aimerais bien que tu prennes les poussières. Parce que ça, je ne sais plus faire. Ou nettoyer le sol. J'ai plus besoin de quelqu'un avec qui je vais faire des choses que de quelqu'un qui les fera à ma place. » (R3-M, >2 yrs, Severe)

The interview data indicated a need for administrative and social assistance, especially for non-routine tasks that were perceived as particularly energy-intensive. Activities such as managing bills, searching for services or products, or replacing malfunctioning technology required significant effort, despite not being technically difficult. Although existing personal assistance schemes were perceived as potentially valuable, access to these services was described as difficult.

“Maar de niet-standaard dingen, die het meest energie kosten... Ad-hoc zaken. Of je hebt een probleem met iets. Of met een instantie. Of... Zeg maar met een factuur. Of stel, ik heb iets nodig. Een dienst of een product. en... Ik moet gaan opzoeken. (...) Stel, mijn telefoon is stuk. En ik heb een nieuwe telefoon of een laptop nodig. Om dan uit te zoeken van... Ja, wat is nu een goeie? Of... En bij welke winkels hebben ze goeie aanbiedingen? Ja, daar kruipt toch wel wat tijd in. Maar voor mij is dat wel... Ja... Kost dat wel... Allee, is dat wel vermoeiend? Ik moet daar echt tijd voor maken. Zoiets zou tof zijn. Dat ik aan iemand kan vragen... Zoek mij dat voor mij eens uit. En ga dat voor mij kopen. En stel dat in. Zet mijn contacten over. Stel mijn cloud in. Van die zaken. Die zijn op zich niet moeilijk. Maar dat ik moet iemand ook toegang geven tot al mijn accounts. En van zodra er iets moet gedaan worden, dat die dat allemaal voor mij doet. Maar ook voor niet IT-zaken. Dus een knecht gewoon. Ja. Een knecht. Dat zou helpen. Natuurlijk, dat is... Ik denk dat dat moeilijk realiseerbaar is. Maar wel onderdelen daarvan.” (R17-F, ≤ 2 yrs, Severe)

The need for access to a social worker to support in obtaining financial, material, and logistical support was also described during the interviews. In terms of psychological support, interview findings emphasised that it should be conceived as ongoing support, as for any other chronic condition, rather than as a time-limited therapeutic intervention.

“Le côté psychologique, c'est quand même compliqué... C'est quand même compliqué de se retrouver du jour au lendemain en ne sachant plus travailler, plus voir sa famille, plus voir ses amis, plus de loisirs. C'est vraiment très compliqué. Donc, je pense que ce serait intéressant que ce soit là aussi. Que ce ne soit pas vu à nouveau comme l'accompagnement psychologique, qui peut guérir la maladie, mais comme l'accompagnement psychologique pour accompagner, comme toute autre maladie chronique, je pense, qui est un vrai besoin. » (R6-F, >2 yrs, Mild/moderate)

The role of patient associations and online fora was also addressed. While, some interviewees expressed interest and willingness to engage with these resources, their involvement was often limited because of their low energy levels. Others reported no interest in such engagement. Such involvement can require substantial energy and some interviewees felt too fatigued to read extensive online information or experienced difficulties related to screen use and concentration.

Nevertheless, we heard that contact with fellow sufferers represented an important source of information and support. At the same time, a potential downside was noted, as exposure to others who are more severely affected can have a discouraging impacts.

“J'ai eu la chance d'avoir ce groupe de patients en fait, qui m'a vraiment aidé à ce moment-là, parce qu'on a différents patients avec différentes histoires, à différents moments de notre parcours et donc qui ont pu expliquer comment ça allait se passer, qui ont pu expliquer les

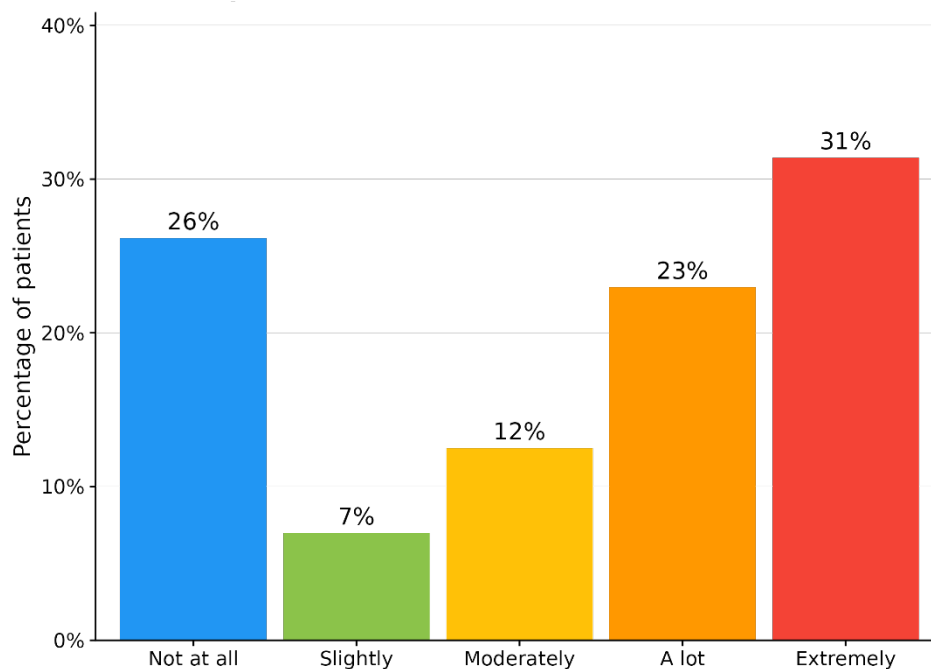
démarches administratives au niveau des reconnaissances de handicap, au niveau de la mutuelle, enfin tout en fait. Et ça, je pense que j'aurais aimé qu'il y ait une structure qui puisse accompagner là-dedans en fait." (R5-F, ≤ 2 yrs, Severe)

"Et je pense qu'en fait, les groupements de malades sont... C'est un peu à double tranchant. Mais c'est un lieu avec des personnes qui sont... dans la même situation et donc... Donc, c'est vraiment un lieu où on a l'impression que les personnes comprennent de quoi on parle, contrairement à d'autres lieux même qui sont liés aux soins de santé, parfois. Mais en même temps, ça peut être déprimant aussi parce qu'on croise d'autres personnes qui relâchent un peu tout leur... Ben oui, leur situation, leur désespoir, parfois, etc. Donc, on peut un peu absorber émotionnellement aussi cet aspect-là." (R8-M, ≤ 2 yrs, Mild/moderate)

9.4.2 Impact on education

Of the survey respondents for whom the question regarding the impact of ME/CFS on education was applicable (n=344), a large proportion of respondents reported a moderate (12%), high (23%) or extreme (31%) impact of ME/CFS on their education, regarding presence, performance, success or study choice. In contrast, around one third of the sample reported that their education was not (26%) or slightly (7%) impacted (Figure 21).

Figure 21 – Impact of ME/CFS on education (n=344)



Legend: Responses "I don't know" (n=9) or "not applicable" (n=392) were not displayed in the figure.

Interview data revealed that fatigue and concentration difficulties significantly affected school attendance, potentially leading to a complete disruption of the normal schooling trajectory.

« Donc, en fait, avant, [mon fils] était scolarisé à l'école X. Ils ont été assez ambigus dans leur attitude. D'un côté, ils comprenaient, ils voulaient nous aider. D'un autre côté, ils étaient pris dans leur carcan, leurs règlements, ils ne pouvaient pas faire tout ce qui serait nécessaire. Donc, on avait certains aménagements où [mon fils] n'allait pas en classe, quasiment plus du tout. Les examens étaient coupés en plusieurs parties. Quand c'était un examen de maths, au lieu de faire trois heures d'affilée, c'était coupé sur deux ou trois jours. Mais au bout d'un moment, c'est devenu impossible à gérer. [On est passé dans l'] enseignement à Distance, qui propose des cours en



ligne. (...) Avant, il était à l'école X jusqu'en quatrième secondaire (...) Il avait commencé à étaler cette année-là sur deux ans à l'école X. Il avait fait la première de ces deux années. Et en passant dans [l'autre système], (...) il a dû recommencer le même niveau en étalant à nouveau sur deux ans. Il a fait ça l'année dernière et là il fait la deuxième année de ce niveau. Donc ça veut dire que c'est la troisième année où il est dans le même niveau scolaire. Et même en étalant cette année sur deux ans (...), on n'est pas sûr qu'il réussisse - à faire tous les devoirs pour la fin de l'année." (R1-M, >2 yrs, Mild/moderate)*

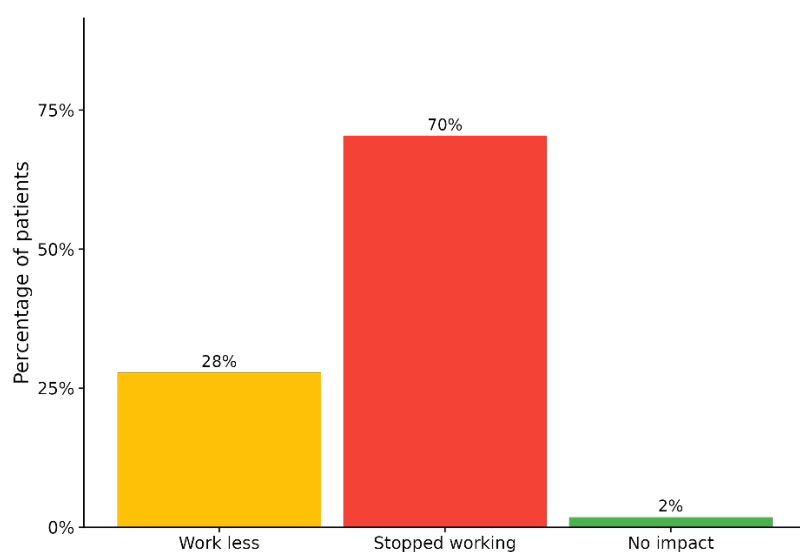
9.4.3 Impact on work

The sample (n=749) consisted of 625 respondents (83.4%) who have been in paid employment since the onset of the first ME/CFS symptoms whereas 112 respondents (15.0%) were not and 12 respondents (1.6%) didn't know.

9.4.3.1 Global impact and reduction in working time

Figure 22 shows the impact respondents experience from ME/CFS on their work. Apart from 2.0% who experienced no impact, almost all respondents started working less (28.0%) or stopped working (70.0%) because of ME/CFS. Figure 23 shows the time respondents have not been working because of ME/CFS. Around 83.0% of the respondents have not been working for more than one year, and 22.0% have not been working for more than 10 years.

Figure 22 – Global impact of ME/CFS on work (n=550)



Legend: Responses "I don't know" (n=3) were not displayed in the figure.

The exploratory subgroup analyses (not adjusted for multiple comparisons) for impact of ME/CFS on work (including the variables "working less or no impact of ME/CFS" versus the variable "stopped working due to ME/CFS"), yielded several noteworthy findings (Table 9). These should be interpreted as hypothesis-generating rather than definitive results.

- Regarding **demographic variables**, significant difference was observed across age categories, with a greater proportion of participants aged >50 years stopping work (75.7%) than in the <30 years (62.5%) and 30–50 years (63.6%) groups.
- Among **disease-related variables**, a difference was observed for burden, with the highest proportion of participants having stopped working in the group reporting the highest symptom burden (78.2%).

- Among **healthcare-related variables**, a difference was found for time to diagnosis and accessibility issues. A higher proportion of participants who stopped working was observed among the participants who had to wait longer than two years for a diagnosis of ME/CFS (75.7%), compared to participants who received their diagnosis within two years (65.4%). A similar result was seen for participants experiencing issues with accessibility of care (76.8%), compared to participants experiencing no accessibility problems (63.2%).
- Regarding **social variables**, a significant difference was observed for the perceived impact of ME/CFS on social relationships, with the highest proportion of participants who stopped working found in the subgroup reporting a very high impact on social relationships (82.9%). In addition, participants reporting a difficult financial situation were more likely to have stopped working (74.4%) than those in a more comfortable financial situation (60.2%).

For the other variables, no significant differences were found ($P > 0.05$). An overview of all descriptives and P-values can be found in Appendix 5.

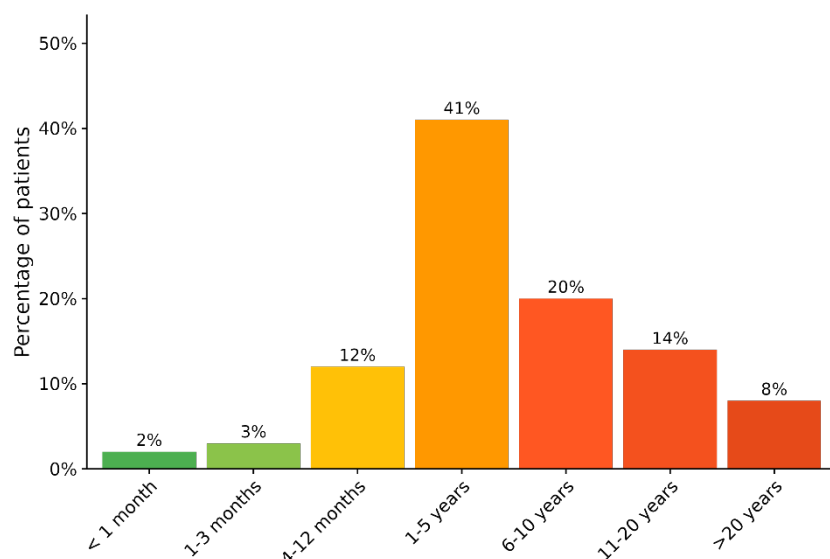
Table 9 – Difference in working status before onset of symptoms versus today by demographic, disease, healthcare and social variables

	Still working (n=163)*	Stopped working (n=387)	Missing(n) **	P-value
Demographic variables				
Age				0.006
<30 years (n=31)	(6) 37.5%	(10) 62.5%	15	
30-50 years (n=292)	(82) 36.4%	(143) 63.6%	67	
>50 years (n=426)	(75) 24.3%	(234) 75.7%	117	
Disease-related variables				
Burden of symptoms				0.013
Low (n=263)	(68) 35.8%	(122) 64.2%	73	
Medium (n=259)	(57) 30.6%	(129) 69.4%	73	
High (n=227)	(38) 21.8%	(136) 78.2%	53	
Healthcare-related variables				
Time to diagnosis				0.027
<2 years (n=297)	(73) 34.6%	(138) 65.4%	86	
>2 years (n=249)	(43) 24.3%	(134) 75.7%	72	
Having accessibility problems				0.002
Yes (n=316)	(52) 23.2%	(172) 76.8%	92	
No (n=288)	(81) 36.8%	(139) 63.2%	68	
Social variables				
Impact on social relationships				<0.001
Low to medium (n=190)	(63) 44.7%	(78) 55.3%	49	
High (n=347)	(74) 28.4%	(187) 71.6%	86	
Very high (n=206)	(25) 17.1%	(121) 82.9%	60	
Financial situation				<0.001
Easy (n=232)	(66) 39.8%	(100) 60.2%	66	
Hard (n=506)	(97) 25.6%	(282) 74.4%	127	

Legend: *Still working includes responses of “no impact of ME/CFS on work” and “working less because of ME/CFS”; ** Including answers “I don’t know”



Figure 23 – Time not working because of ME/CFS (n=596)



Legend: Responses "I don't know" (n= 26) were not displayed in the figure.

Interviews indicated that ME/CFS had a substantial impact on patients' professional lives. Some interviewees reported reducing their working hours, while others experienced downward occupational mobility, resulting in acceptance of lower-paid jobs. Others described having to leave paid employment altogether or discontinued self-employed activities, often following repeated periods of sickness absence.

"Het is wel zo dat vier jaar nadat de ja de symptomen begonnen en het erg erg ben ik ja ben ik mijn werk verloren kwijtgeraakt ja dat klinkt negatief, maar dat had eigenlijk veel eerder moeten gebeuren. ik had eigenlijk zelf ontslag moeten nemen of euch want ik ben denk ik twee jaar, nee het is later pas. Het heeft uiteindelijk zes jaar geduurd. Het is in 2008 begonnen. Ik heb in 2013 ben ik dan, ben ik dan langdurig ziek geweest. Officieel dan, want ik was dat voordien ook al. (...) Drie volle jaren, eigenlijk voltijds. En daarna deeltijds. Dus met stukken van twee maanden dat ik niet ging werken. Ziek was en ging ik terug. Dan terug ziek. Dan heb ik een tijdje met halftijds ziekteverlof gewerkt." (R17-F, ≤ 2 yrs, Severe)

"Alors, j'ai d'abord dû réduire mon temps de travail. C'est la première fois que j'ai consulté pour cette maladie, c'est en 2013. Et j'étais à temps partiel en diminuant un peu progressivement. Je vais dire avec parfois des périodes aussi de congés sabbatiques. Donc, jusqu'à mon arrêt. Mon arrêt, c'était en janvier 2024. Donc, ça fait un peu plus qu'un an et demi. J'ai pensé que j'allais pouvoir, après m'être reposé peut-être deux mois, que j'allais pouvoir reprendre, mais ça n'allait pas du tout. J'ai diminué mon temps de travail en utilisant des astuces administratives comme le crédit temps. Crédit temps pour s'occuper d'un enfant de moins de huit ans. C'est des choses qui existaient encore. Je ne sais pas où ça en est maintenant. Mais je n'ai jamais utilisé la raison médicale, ou le temps partiel médical, jusqu'à l'arrêt de travail complet dans lequel je suis maintenant, qui est le lui pour raison médicale." (R8-M, ≤ 2 yrs, Mild/moderate)

"Ik krijg nu nog wel een gedeeltelijke uitkering. Want het is ook gewoon niet mogelijk om met maar 20 uur per week te moeten rondkomen. En soms dan zou ik meer willen gaan werken of zo. Maar het is nu soms... Met die 20 uur per week... Al niet helemaal haalbaar. Of soms heb ik zoveel last van mijn klachten dat het... functioneren... het werken en zo allemaal nogal beïnvloed wordt. En dat maakt het mentaal soms gewoon heel erg moeilijk om... Om te dragen. Van... Van zover het gaat. Want elke keer ook als ik mij terug moet gaan verantwoorden bij de mutualiteit of zo. Van... Dat de situatie eigenlijk niet gebeterd is". (R19-M, ≤ 2 yrs, Severe)

9.4.3.2 Impact on employment status

633 respondents were in paid employment prior to ME/CFS onset. Table 10 shows that, at the time of survey completion, only 140 respondents (22.3%) were still in paid employment, whereas the majority of respondents indicated their employment status changed from paid employment to disability (52.9%). Other people responded that they retired in the meantime (17.4%), are seeking a job (4.1%) or stay at home without an income (3.3%). The responses from 5 participants were missing.

Table 10 – Employment status at survey completion among those who had a paid employment before ME/CFS (n=628)

Primary employment status at survey completion (n=628)	n (%)
Paid employment (employee or self-employed)	140 (22.3%)
Disability	332 (52.9%)
Job seeker, unemployed, living wage	26 (4.1%)
Stay-at-home, no income	21 (3.3%)
Retired	109 (17.4%)

Legend: Note that from the interviews we learned that (early) retirement can be granted (instead of the 'invalidity' status) to people who are long-term ill in Belgium, but this depends largely on their employment status (employee, self-employed, civil servant, etc.), age, and length of career.

9.4.3.3 Impact on work productivity

Among survey respondents who are still working and answered the question on work productivity (n=163), 4.3% indicated to experience no impact on their work productivity (0 on a scale from 0 to 10), whereas 9.8% indicated an extreme impact on their work productivity (10 on a scale from 0 to 10). Of this sample, 33 people (20.2%) scored the impact of ME/CFS on their work productivity to be 5 out of 10 or lower, while 130 people (79.8%) scored the impact to be 6 out of 10 or higher.

Interview data revealed that symptoms of ME/CFS significantly limited individuals' ability to continue working as usual, resulting in a reduction in working time. In addition, some individuals continued to work despite being physically unable to function effectively.

“De reden ook dat ik langer bleef was dat meestal naar het einde van de dag ging het wel meestal iets beter. En dan kon ik wel toch iets presteren. Maar eigenlijk ook de... Van morgens tot in de namiddag... Kon ik gewoon niets doen. Dus ik las dan mijn e-mails. En antwoordde daarop zo... Echte heel simpele dingen. Deed ik dan. En... Het echte werk was later. Terwijl ik eigenlijk al naar huis moest. Ja, omdat ik met halftijds ziek genoeg was. Dus normaal moest ik maar tot twaalf aan. Max één uur 's middags werken. En... Ja, het bezorgde ook conflicten op het werk. Waar men niet flexibel was. Ik vroeg voor dan twee uur later. Of ik twee uur later mocht toekomen. En dan ook twee uur later zou blijven. Omdat ik dan toch beter kan presteren. Maar... Daar deed men altijd heel moeilijk over.” (R17-F, ≤ 2 yrs, Severe)

“Het was waanzin op een bepaald moment dat ik het zelfs tijdens de pauze overwoog. Ik ga hier om de hoek gewoon in een hotel gaan om even één uur een kamer te huren. Voor gewoon één uur te kunnen slapen. Ik ging soms naar de toiletten. Om gewoon proberen... Ja, dat lukte wel meestal niet. Maar om even te kunnen slapen. Ik was zo moe. Ja, het is gewoon waanzin om... Omdat ik in die toestand op een bureau zat en toch probeerde te werken.” (R17-F, ≤ 2 yrs, Severe)



9.4.3.4 Employers' reactions

From the interviews we know that employers' reactions varied between empathic-supportive, granting their employee the flexibility and autonomy they needed to be able to work despite their limited energy levels and other health problems, to harsh-unsupportive, leaving no room for (creative) solutions adapted to their employees' needs. The degree to which ME/CFS patients can continue functioning, be it at a lower intensity, largely depends on the severity of the illness and the willingness of the employer. In the best cases ME/CFS patients were able to continue working on a part-time basis with flexible energy-based working hours and/or telework.

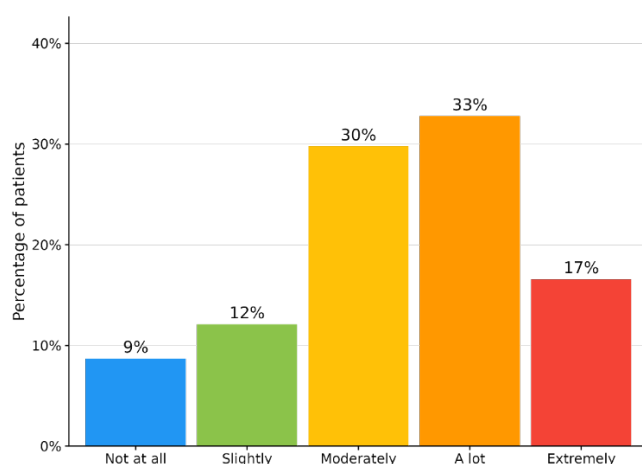
"Ja. Ik heb ook, mijn werkgever is heel lief geweest, heel lang. Dat ik een beetje kan spelen. Veel thuiswerk. Als het 's morgens niet gaat, een uur later beginnen, en uur langer werken, of twee uur vandaag en vier morgen. Maar nu is het bedrijf overgenomen en nu is dat niet meer. Het is een beetje veel moeilijker." (R14-F, >2 yrs, Mild/moderate)

"Maar [naam werkgever] heeft(...) toen heel snel laten merken van kijk, het is alles of niks. Dus "wij gaan een werknemer niet meer flexibele uren gaan geven. Straks wil elke werknemer dat. En daar gaan we niet mee beginnen". Ik kreeg dan toen ook de Vlaamse ondersteuningspremie. (...) En dan werd er gezegd van, kijk, je krijgt eigenlijk een schone vergoeding voor toch een aanpassing te doen. En eigenlijk werd er toen op die moment gezegd van, nee, doen we niet". (R19-M, ≤ 2 yrs, Severe)

9.4.4 Financial consequences

Figure 24 depicts that more than three quarters of the survey sample experience an impact of ME/CFS on their financial situation to a moderate (30%), large (33%) or extreme extent (17%). Only a small group indicated to not (9%) or slightly (12%) experience any financial impact. In addition, of the 738 respondents who answered to another question on financial impact, 31.4 % answered that their family income easily covers their expenses while 68.5% indicated that it is hard to cover their expenses with their family income.

Figure 24 – Impact of ME/CFS on financial situation (n=735)



Legend: Responses "I don't know" (n=10) were not displayed in the figure.

According to the interviews, the financial impact of the illness was due to:

- The loss of income from work (due to reduced working hours or incapacity): the extent of financial loss varied according to the severity of the condition. In most favourable situations, patients were able to rely on alternative sources of income, such as rental revenue, personal savings, or allowances or replacement income based on a previous high wage. In contrast, other participants

described a transition from dual-income households to single-earner family arrangements, resulting in increased financial dependency on their partner.

- The lack of access to financial support

“En als ik dan bij de maatschappelijk werker van de mutualiteit langs ga. Dan is het eigenlijk ook gewoon zo van. Ja, jij kunt nu zelf nog wassen. Je kunt af en toe nog zelf naar de winkel. Ja, dan kun je hier eigenlijk niets komen zoeken. Wij kunnen je niet helpen. Met je verloning kun je naar de voedselbank. En voor de rest kunnen we eigenlijk niets doen. Dan zorgt dat ook wel ervoor. Dat ik met een hele erge druk zit. Stel dat mijn relatie nu niet zou werken of zo. Dan zou ik gewoon niet weten hoe ik de rest moet aanpakken. En dat zorgt wel zo. Ik kamp heel vaak met depressieve klachten. Omdat het leven op die manier zo uitzichtloos lijkt.” (R19-M, ≤ 2 yrs, Severe)

- The lack of reimbursement for certain treatments, particularly following discharge from hospital, as costs are no longer covered by hospital insurance. Examples included treatments (e.g. experimental, alternative or off-label), psychotherapy, costs associated with treatment sought abroad, or treatments only being reimbursed for adults but not for younger patients.

“Et alors le problème, c'est que comme la maladie n'est pas reconnue, les médicaments qu'on donne sont toujours des médicaments « off-label » qui ne sont utilisés pas dans l'indication. Et donc pour le moment, la [assurance] est en train de me refuser tous les traitements en disant : vous ne l'utilisez pas dans l'indication reconnue. Okay. Et il y a aussi tout ce qui est compléments alimentaires et là, il n'y a aucune intervention et ça coûte relativement cher, quoi.” (R5-F, ≤ 2 yrs, Severe)

“Maar dan krijg je dat dat een psycholoog is die 130 euro per uur vraagt. Dat is voor iemand in mijn situatie gewoon niet realistisch dat ik mij daar kan veroorloven. Ik wil heel graag mijn levenskwaliteit verbeteren. En ik zou graag beter willen worden als dat zelfs mogelijk is. Maar ik heb gewoon de middelen niet om dat te bekostigen en dan zeggen van ik wil harder gaan werken. Om dat dan toch te kunnen bekostigen. Dat is gewoon niet realistisch. Want ik kan het nu met momenten zelfs niet volledig dragen. En dan is het soms wel zo echt uitzichtloos.” (R19-M, ≤ 2 yrs, Severe)

“Dans les traitements qu'il a, il y a de l'ivabradine pour la tachycardie posturale. Peut-être que pour les adultes, c'est pris en charge, mais en tout cas, pour les moins de 18 ans, ce n'est pas censé être donné aux moins de 18 ans donc c'est nous qui payons tout.” (MR1-, ≤ 2 yrs, Mild/moderate)*

- When a treatment shifted from hospital care to ambulatory care, the medication was no longer covered by hospital insurance and became at the patients' expense.

“Oké ja dus 1 keer per week ga ik maar vroeger kreeg je dat infuus in het ziekenhuis en van mijn werk hadden we een hospitalisatieverzekering die dat allemaal dekte waardoor ik ook mijn kine kon inbrengen eigenlijk elke doktersafpraak kon ik eigenlijk echt gewoon alles inbrengen he maar nu is er een wijziging in het ziekenhuis en sinds nieuwjaar is dat via thuisverpleging dat ik dus waardoor dat ik alles zelf betaal he je betaalt niet je thuisverpleging maar je betaalt wel je medicijnen dat je moet gaan halen je bakster is niet gratis he bij de apotheek dus dat moet je allemaal betalen die vitamines die erin gaan betaal je apart moet dus nu ook mijn kine zelf betalen stel dat je mutualiteit niet betaalt moet ik zelf betalen, terwijl dat vroeger ja alles gratis was.” (R14-F, >2 yrs, Mild/moderate)

“Het is financieel zeer duur, want het wordt bijna niet terugbetaald. Omdat het experimenteel is. Enkel... Ja, je ziekenhuis, allez, je ding [hospitalisatieverzekering] komt wel tussen. Omdat dat de opname is. Maar de behandeling zelf... Dus die producten die hij heeft... Worden nauwelijks terugbetaald.” (R15-F, >2 yrs, Severe)



9.5 Other unmet patient needs

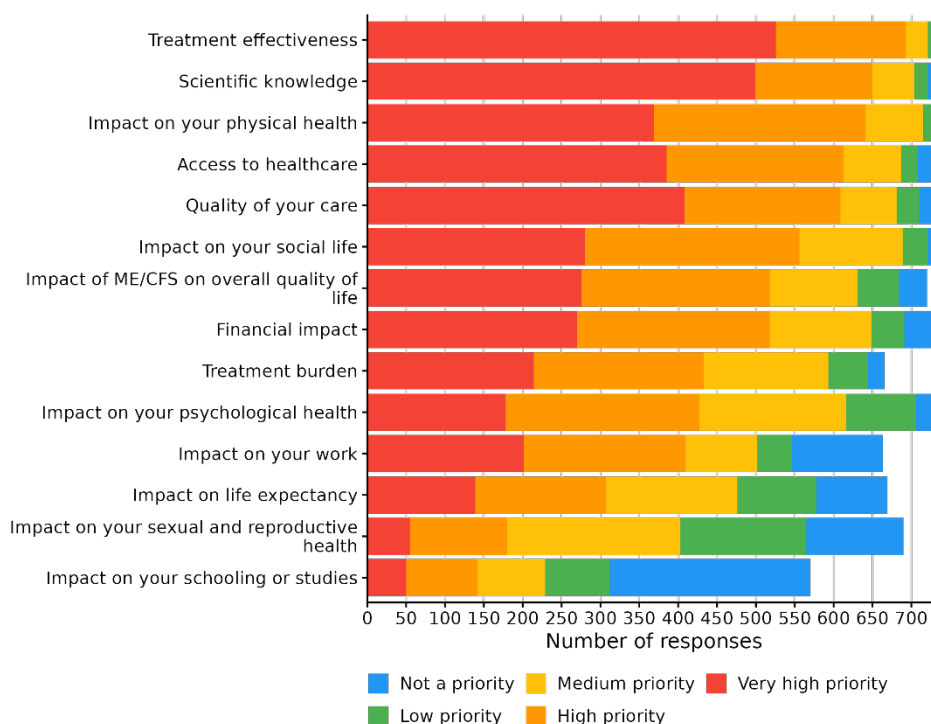
In addition to the structured survey items, respondents were invited to elaborate on unmet needs through an open-ended question at the end of the survey. No additional needs were identified beyond those captured by the qualitative analysis in response to this question. An analysis of the responses is provided in Appendix 6.

9.6 Prioritisation of patient needs

Survey respondents (n = 744) were asked to provide a priority ranking to a list of 14 needs categories, offering a comprehensive overview of perceived importance across domains of living with ME/CFS (Figure 25). The highest proportion of “very high priority” and “high priority” responses was observed for treatment effectiveness, scientific knowledge, physical health impact, health care access, and quality of care. These categories stood out as primary needs, with 50% or more of respondents rating these categories as a “very high priority” and only a minority of respondents rating these as a “low” or “no priority”.

Categories such as impact on social life, impact on overall quality of life, financial impact, treatment burden, impact on psychological health and impact on work were also rated by more than 50 % of respondents as being “very high priority” or “high priority” categories. Notably, “not a priority” and “low priority” options were most frequently selected for impact on life expectancy, sexual and reproductive life and educational impact.

Figure 25 – Priority rating of need categories in people with ME/CFS (n=744)



Legend: For this question, participants could select more than one option. Responses “I don’t know” (n=13 for treatment effectiveness; n=19 for scientific knowledge; n=11 for impact on physical health; n=24 for access to healthcare; n=21 for quality of care; n=14 for impact on social life; n=29 for impact on overall quality of life; n=23 for financial impact; n=83 for treatment burden; n=14 for impact on psychological health; n=86 for impact on work; n=80 for impact on life expectancy; n=59 for impact on sexual and reproductive health; n=179 for impact on schooling or studies) were not displayed in the figure.

9.7 Unmet societal needs

In this study, we have only limited data from the survey regarding the impact of ME/CFS on the health and social lives of informal caregivers. Other dimensions were not included in the survey.

9.7.1 Unmet health needs : health burden on informal caregivers

Of the people who completed the survey on behalf of a person with ME/CFS (n = 21), 19 considered themselves as a “close caregiver”, providing regular non-professional assistance to this person due to their ME/CFS. Apart from one person indicating no impact and one person indicating a slight impact on their physical health, all close caregivers reported a moderate (36.8%), high (26.3%), or extreme (26.3%) impact on their physical health because of their role as close caregivers. Overall, almost two thirds of close caregivers reported a substantial impact on their physical health because of their role as close caregivers.

9.7.2 Unmet social needs : burden on informal caregiver’s social life

Of the people who considered themselves as a “close caregiver” (n=19), apart from one person indicating no impact, and one person indicating a slight impact on their social life, all close caregivers reported a moderate (26.3%), high (36.8%), or extreme (26.3%) impact on their social life because of their role as close caregivers.

9.8 Transversal dimension: Equity

Within the Belgian context, the NEED framework explicitly recognises equity as a transverse dimension in assessing unmet health-related needs. It emphasises that unfair and avoidable differences in health outcomes, care accessibility, and social participation may exist by amongst others socio-economic status, ethnicity, and migration status.¹²²

Earlier analyses, in the section 9.2.1.1, had already highlighted differences in the subgroup analyses of the survey findings on deterioration in EQ-5D-5L utility scores according to age, time to diagnosis, the number of unmet needs, experiences of care and access to care, and financial difficulties. In addition, people who stopped working reported a higher symptom burden, greater disruption to social relationships, more problems accessing their care, and greater financial difficulties than those who continued working (9.4.3).

From the interviews, it appears that ME/CFS is characterised by an pronounced need for societal and institutional recognition: patients felt disregarded, experiencing inequalities in access to care and a lack of structured responses from the healthcare system. This was interpreted as reflecting the invisibility of the condition and its impact, which, in turn, leads to inequalities in care provision and rights.

In the context of their uncertainty regarding their future, interviewees referred to the political and societal attitudes towards the chronically ill in general versus towards ME/CFS patients, in particular. Interviewees characterised the latter as marked with stigmatisation, psychologisation, simplification, underestimation, denial and the lack of scientific interest in the condition.

- Political-societal stance towards the chronically ill in general

Patients spoke out against the stereotype that people with chronic illnesses are lazy and unwilling to work. Concern was expressed about prevailing discourses that place pressure on people with long-term conditions to return to employment as quickly as possible. Societal pressure was felt to be active and productive on top of their own struggle with longing to restore their professional role and identity on the one hand and the physical reality of not being able to work on the other hand.

“Het is constant vechten om gehoord en om gezien te worden en de politieke toestand nu doet er ook geen hulp geen goed bij we horen van overal van langdurige zieken ze gaan moeten



beginnen werken met zijn profiteurs en we gaan het geld en we gaan dit en dan denk je van help maar je staat daar machteloos over.” (R12-F, ≤ 2 yrs, Mild/moderate)

- Stigmatisation

In the qualitative interviews, stigmatising reactions from healthcare professionals, family and friends as well as society at large were reported. These reactions included being perceived as crazy, an exaggerator, a pretender, a lazy freeloader, or simply as unwilling to work.

“Ce serait d’avoir cette reconnaissance, LA reconnaissance de dire et ça, ça veut dire tellement de choses en fait. Ça veut dire que mais « oui tu n’es pas folle, mais oui tu ne fais pas semblant, mais oui tu ne le fais pas exprès, mais... » et « on a quelque part une considération », c’est pas l’oubliée du bidule.” (R4-F, ≤ 2 yrs, Mild/moderate)

“Ik moet ook verder met die diagnose die ik heb gekregen en waar dat de diagnose is dat kan mij eigenlijk niet schelen maar ik weet wel dat ik effectief last heb van klachten en problemen en het feit dat dat dan ook op dat moment niet erkend wordt en dat er gewoon wordt gezegd van je je bent een luierik en een profiteur want jij wilt eigenlijk gewoon niet gaan werken.” (R19-M, ≤ 2 yrs, Severe)

The interview data indicated that participants were often steered towards psychological explanations, such as depression, burnout, or stress, or told that the problem was “all in their head”. They reported being treated with medication or psychological therapy. However, they also described an absence of somatic investigation or treatment. This approach was perceived as inequitable when compared with the care provided for other conditions, for which biomedical diagnostic and treatment pathways are more consistently pursued.

“C’est vraiment quelque chose qui serait à questionner, je trouve, que ce soit très régulièrement la première réponse qu’on peut avoir. On nous apporte le... « C’est dans votre tête, mais c’est le stress mais vous êtes fatigué. Est-ce que vous n’êtes pas... Est-ce que vous êtes sûre de ne pas être dépressive ? » C’est tellement fréquent pour tellement de malades que c’est quand même quelque chose à questionner. Je pense que ce serait la première réponse apportée avant que nous, on doive batailler pour dire « mais non, il y a autre chose. » S’il vous plaît, est-ce qu’on peut faire au moins des tests ?” (R6-F >2 yrs, Mild/moderate)

“Ze lieten, weten dat wij eigenlijk, depressieve mensen waren, in plaats van mensen die hulp zochten met die vermoeidheid. Of een oplossing, of een diagnose, (...) Ze zeiden ook, ja, je hebt veel te veel stress. Het kwam eigenlijk altijd op hetzelfde neer, dat het aan ons lag.” (R15-F, >2 yrs, Severe)

“Mensen krijgen geen antwoorden. Ze worden van her naar daar gestuurd, van specialist naar specialist. En die maakt zijn eigen verhaal. Of die lacht u gewoon uit van, je hebt niks. Het zit tussen twee oren.” (R18-M >2 yrs, Mild/moderate)8

Interviewees were told they were overstressed or the problem was “between their ears”.

“Een keer per week was dat een groepsessie met een psychologe wat ik zelf eigenlijk niet slecht vond ik weet dat er vanuit patiëntenverenigingen heel veel kritiek opkomt over dat psychologiseren maar dat was het bij ons niet er werd gezegd van wij kunnen u niet genezen we gaan u alleen handvaten geven om u beter te voelen om ermee om te gaan en acceptatie.” (R14-F, >2 yrs, Mild/moderate)

- Simplification, underestimation and denial

We heard in the interviews that psychologisation can also be experienced as a simplification or underestimation of patients’ suffering. The denial of ME/CFS as a legitimate disease was also reported together with the refusal to provide further medical care.

“Alsof dat daarmee de kous af is. En die patiënt door anders te gaan denken. Een beetje meer te bewegen. Ineens gaat genezen zijn. En dat is eigenlijk wel de boodschap die men geeft. Als het

al niet over psychosomatische somatiek gaat. In plaats van somatopsychische somatiek. Dat je eigenlijk gewoon wel zal genezen. Als je met een psycholoog of een psychiater gaat gaan babbelen. En ja. Helaas was het maar waar.” (R10-M, ≤ 2 yrs, Mild/moderate)

The tendency to underestimate symptoms severity was described not only within medical encounters, but also in response from people, who drew comparisons with their own experience of being tired or confused. In response to such reactions, patients reported avoiding discussion of the burden of their symptoms, expressing the view that these experiences could not be fully understood by those who do not share the same illness.

“Nee, want vorig weekend was er iemand die zei, ja, “ik denk dat ik jou begrijp”, iemand van in de zeventig. Die zei: Ik heb zojuist mijn living gepoetst en ben moe. Moet me in de zetel zetten. En ik dacht, ja, nee. Nee, dat is het niet. Dat is totaal iets anders. Maar de mensen denken dan van, ik denk dat ik weet wat dat is. Maar, ze doen een dutje en een dag daarna zijn ze weer springlevend hé. En ik heb gisteren mijn living gepoetst en moet ik recupereren voor een week. Ja. Nee, ja, ik weet het niet. Ik vind dat na al die jaren is er nog niemand tenzij dat ze het zelf hebben.” (R15-F, >2 yrs, Severe)

- Perceived lack of interest in ME/CFS

Participants perceived a lack of scientific interest in ME/CFS, which they felt had contributed to stagnation in recognition, aetiological research, and the development of effective treatments. Comparisons were made with neighbouring countries such as the Netherlands and Germany to illustrate the perceived inertia in Belgium.

“Enerzijds begrijp ik het, omdat long-covid nieuw is, en dat dat echt gewoon, ja... En dat we dan als patiënt zijn, uit geluk hebben we dat we bij zoveel zijn, dus dat we wel iets in beweging kunnen zetten daardoor. Ook al is dat relatief, als ik zie hoe weinig politiciers, en hoe weinig dat er... En hoe traag het allemaal gaat ten opzichte van onze buurlanden.” (R10-M, ≤ 2 yrs, Mild/moderate)

The emergence of long COVID has contributed to heightened interest in ME/CFS and to improved understanding of the condition. However, ME/CFS predates the pandemic and is not limited to post-COVID forms.

“C'est vrai que ça m'embête un petit peu qu'on sépare le Covid long, qui génère, entre autres, parce qu'il y a d'autres problèmes, un syndrome de fatigue chronique. Il y a beaucoup d'autres infections qui génèrent un syndrome de fatigue chronique. Ça existait avant le Covid. ” (R3-M, >2 yrs, Severe)

“En met de opmars van Corona hadden plots veel mensen langdurige vermoeidheidsklachten. En ineens werd dat allemaal wel beter herkend en beter opgevolgd. ” (R19-M, ≤ 2 yrs, Severe)

- Challenge in accessing disability support

Respondents also described major challenges in accessing disability support, mentioning length and complex procedures as well as assessment tools that do not reflect the fluctuating nature of ME/CFS or its non-visible limitations. Disability rating scales were described as overly focused on overt physical limitations, with little consideration for symptom variability or the cumulative impact of activity. For example, the ability to stand or walk briefly (even if followed by prolonged recovery) was sufficient to disqualify individuals from receiving support they strongly perceived as necessary, such as disability parking cards.

« J'avais demandé à notre généraliste de remplir un dossier de reconnaissance de handicap pour le système belge. Mais, de ce qu'on a compris, il y a une partie handicap physique, une partie handicap mental et il y a un certain seuil à dépasser pour l'un ou pour l'autre. On ne peut pas cumuler. Les critères qui sont dedans, c'est des critères fermés. Soit on a certains handicaps, soit on ne les a pas. On ne peut pas ajouter des handicaps qui ne seraient pas dans la liste. (R1-M, ≤ 2 yrs, Mild/moderate)*



« Je n'ai pas le statut de personne handicapée. Donc, j'ai passé l'entretien et je ne suis pas suffisamment handicapée selon les critères du SPF handicap. Alors, pour moi, une grande réforme, ce serait de prendre, entre autres, pas que l'autonomie comme critère, (...) ça peut paraître très bête, mais pouvoir, par exemple, me parquer sur des places réservées aux personnes handicapées si je suis à l'extérieur, donc pas près de chez moi, mais oui, parfois, ce serait vraiment intéressant pour moi parce que je suis plus près, par exemple, du magasin et donc ça me coûte moins d'énergie aussi. (...) Je me dis que par exemple, la douleur ou la fatigue pourraient être des critères à jauger aussi, ... Par exemple, concrètement, je peux me faire à manger, j'ai les capacités de le faire, mais la plupart du temps, je ne sais pas le faire, je suis trop fatiguée pour le faire, mais selon les critères uniquement de l'autonomie, la case à cocher de « est-ce que vous pouvez vous faire à manger ? » est-ce que vous pouvez vous faire à manger ? Oui, c'est ça. C'est ça qui va être coché. Même chose pour se laver. Pour moi, à 30 ans, devoir prendre une douche assise parce qu'être debout sous la douche, c'est trop énergivore, c'est un handicap. Mais à nouveau, je ne rentre pas dans la grille de « est-ce que vous pouvez vous laver seul ? » Oui. Oui, je peux me laver seule, mais ça a des conséquences après, et ça, ce n'est pas pris en compte(...) Durant cet examen, donc justement, la question de « savez-vous vous faire à manger ? », je réponds « ça dépend ». Ce à quoi le monsieur me dit « est-ce que vous savez vous faire des tartines ? » Et oui. Je pense que même les jours où c'est très compliqué, il y a un moment dans la journée où je saurais aller me faire des tartines, en général. Et donc, je lui ai dit « oui ». Et donc, sa réponse a été « vous voyez, vous savez vous faire à manger ». Ce qui m'a complètement déconcertée parce que d'abord, à nouveau, j'ai un médecin devant moi. Est-ce que ce serait OK de se nourrir uniquement de tartines 365 jours par an à tous les repas ? J'ai un petit peu du mal. Et surtout, à nouveau, non, ce n'est pas OK. Ce n'est pas parce que vous pouvez cocher la case de « OK, je ne mourrai pas de faim si je suis tout seul chez moi. », effectivement. Contrairement à d'autres handicaps où j'entends bien que, sans aide personnelle, la personne n'aurait pas la capacité physique de se faire à manger. Je l'entends tout à fait. Mais ce n'est pas OK, du coup, de me dire « pardon, vous savez vous faire à manger ». Mais non, ce n'est pas aussi simple que ça. (R6-F, >2 yrs, Mild/moderate)

◆ SECTION 2: HEALTHCARE MODELS FOR ME/CFS

KEY POINTS TABLE

Review of literature

- **There is a lack of evidence demonstrating the effectiveness of any specific ME/CFS service model, leaving insufficient basis to recommend any particular organisational configuration:**
 - In the literature search, the majority of articles were excluded because they did not address organisational aspects of ME/CFS
 - Several excluded articles offered conceptually relevant insights on integrated care, care coordination, and stepped-care approaches.
 - The limited scientific evidence reveals a gap in evaluating organisational models, highlighting the need for structured research on ME/CFS care delivery.
- **Guidance from expert groups recommends multidisciplinary specialised care, despite a lack of comparative evidence on alternative service configurations.**
- **The high-quality NICE (*National Institute for Health and Care Excellence*) guidelines (NG206- 2021), informed by systematic evidence reviews and extensive documentation, notes that evidence is insufficient to define multidisciplinary pathway composition and recommends the following service characteristics:**
 - Care should be delivered through a multidisciplinary team with specialist expertise in ME/CFS. Strengthening the expertise and skills in healthcare professionals is recommended rather than to designate a fixed core team of professionals.
 - A provision of health and social care based on the patient's needs.
 - A management shared between primary and specialist services.
 - Adaptable care pathways with access to multiple specialties.
 - Avoiding too rigid eligibility criteria for multidisciplinary services, which do not account for symptom complexity.

International comparison (UK, France, The Netherlands, Italy)

- International comparison identified notable differences in the organisation of care across the countries investigated. The UK is the only country with a structured national referral pathway, whereas in France, the Netherlands and Italy, access to specialised care largely depends on a limited number of clinician-initiated, locally organised centres of expertise.
- Across the countries analysed, priorities for reorganising ME/CFS care converged on developing or updating national guidelines (often absent), harmonising diagnostic criteria, expanding local access, and strengthening multi-level training for healthcare professionals.
- Care provision for key subgroups (notably children and young people and those with very severe ME/CFS) remains underdeveloped, with restricted access to multidisciplinary support.
- In the UK and the Netherlands, funded research programmes are underway, supporting coordinated research projects and biobank development.



- There is growing attention to societal and workplace recognition of ME/CFS, with several countries exploring potential strategies to support work participation and reasonable adjustments.
- Across the countries analysed, long COVID services have been developed. General practitioners act as the initial point of contact, providing initial management and referring patients, as appropriate, to specialist services, multidisciplinary rehabilitation, or dedicated long COVID pathways, according to clinical complexity and individual needs.

10 INTRODUCTION

Due to the complex and intertwined nature of ME/CFS, individuals often require coordinated input from a wide range of health care professionals. Depending on the severity of the illness, care may be delivered through primary, or secondary care services.

The present review was undertaken to appraise the evidence supporting the use of specific care pathways for the management of ME/CFS. Its objective is to explore the resources necessary to develop and sustain models of care fulfilling patient needs, and to identify gaps in our understanding of the organisation of healthcare services.

To this end, both the peer-reviewed and the grey literature were examined. First, we searched the peer-reviewed literature to determine whether there is evidence indicating that implementing specific healthcare service organisation for ME/CFS affects clinical outcomes (11.1). Second, we reviewed the grey literature to assess which types and configurations of healthcare systems are recommended, and the underlying evidence supporting these recommendations (11.2).

This approach was complemented by a description of practices in selected foreign countries: we described how healthcare services are organised in those settings (12). Subsequently, the analysis was extended to encompass the organisation of healthcare services also for people with long COVID, due to the overlapping characteristics of both conditions, and recognising that, in the wake of the pandemic, healthcare systems have established new care structures to manage it (12.4).

11 REVIEW OF THE LITERATURE

In this chapter, the peer-reviewed and grey literature on the use of specific care pathways to manage ME/CFS was reviewed.

11.1 Peer-reviewed literature

11.1.1 Methods

11.1.1.1 Research questions and selection criteria

The protocol for the systematic review of the literature was prospectively designed in accordance with the standard procedures used by KCE^{PP} as well as the framework 'A Measurement Tool to Assess systematic Reviews 2' (AMSTAR-2) guidance and the *Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA)* statement.¹²⁹

The research question was formulated as follows: "*Is there evidence indicating that implementing a specific healthcare service organisation for ME/CFS affects clinical outcomes?*" A systematic search

^{PP} <http://processbook.kce.fgov.be/>

was therefore conducted to identify studies comparing differences in treatment effectiveness between different types of care pathways.

The research question was then translated into a structured PICO framework (Participants-Intervention-Comparator-Outcomes) (Table 11). Only full articles published in English were included. To enhance the contextual applicability of the findings to Belgium, this review was restricted to studies undertaken in OECD countries.

Table 11 – Inclusion and exclusion criteria

PICO	Inclusion	Exclusion
Patients	Individuals aged ≥ 16 years with ME/CFS (who meet the definition criteria from CDC/IOM 2015 ¹)	Children
Intervention/exposure	Description of a healthcare structure/organisation specifically for patients with ME/CFS	Studies that do not address organisation and models of care or studies including single interventions
Comparators	Comparison with no specific healthcare structures/organisation, or comparison between different types of healthcare organisation	
Outcomes	Clinical outcomes or self-reported outcomes	
Timeframe	Studies published between 2015 – current [#]	
Setting	Studies conducted in countries belonging to OECD	Studies conducted outside of OECD countries
Type of studies	Systematic review, (randomised) controlled trials, observational studies	Case-reports, case series with limited sample size (< 30), narrative review, expert opinion
Language	English	Other language

[#] dates of search and update are presented below in this section; CDC: Centres for Disease Control and Prevention; IOM: Institute of medicine; OECD: Organisation for Economic Co-operation and Development; QoL: quality of life

11.1.1.2 Search strategy

The search strategies combined controlled vocabulary (e.g. MeSH terms, Emtree terms) with free-text keywords, using Boolean operators, truncation, and proximity operators as appropriate for each database. Full search strategies are detailed in Appendix 7.

11.1.1.3 Databases and screening process

A comprehensive literature search was conducted to identify relevant studies for this review. The following bibliographic databases were searched for relevant publications: Medline (via Ovid), Embase (via Ovid), Cochrane Library, PsycInfo, Cinahl, and Pedro, covering the publication period from 2015^{qq} to 2024. The searches were conducted between October and November 2024, with language limited to English. The search was conducted by a medical information specialist to guarantee methodological accuracy and the search was adapted to each database. If necessary, the search was complemented by the identification of additional references through various sources, including citation tracking and exploratory searches within the bibliographic databases of key references. All references were

^{qq} Although eligibility was restricted to studies published from 2015 onwards, the information specialist set the search start date to 2014 to maximise sensitivity and avoid missing studies published online ahead of print in late 2014, but assigned a 2015 publication date in print or indexing records.



imported into EndNote X20, where duplicates were identified through the software's function, supplemented by manual verification after sorting by title.

Screening for relevant articles was performed independently and in a blind way by two researchers. The initial screening of documents was performed based on titles and abstracts, followed by a full-text review of the selected articles, using the Rayyan QCRI tool^{rr}. Afterwards, a consensus meeting was held to discuss potential conflicts during the screening process.

11.1.2 Results

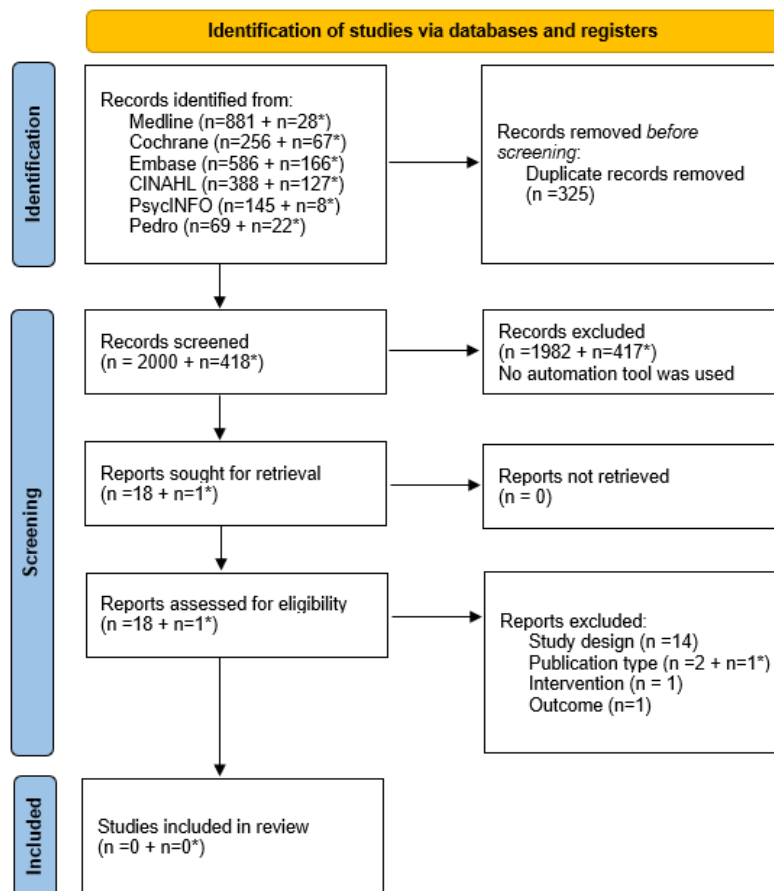
11.1.2.1 Study selection

A search across the selected databases retrieved 2 325 records. After removal of duplicates, 2 000 unique records were retained for screening. Titles and abstracts were screened against the eligibility criteria, resulting in 18 articles eligible for full-text assessment (Figure 26). Following full-text assessment, all 18 articles were excluded, and no studies met the inclusion criteria for the final analysis.

An update of the search was performed in March 2026 resulting in an additional 418 records to screen. After screening title and abstract, one article was withheld, which was not included based on full-text screening. A detailed list of excluded studies, along with reasons for exclusion, is provided in Appendix 8.

^{rr} <https://www.rayyan.ai/>

Figure 26 – Prisma flow diagram



*Results from the update of the search strategy (March 2026)

11.1.2.2 Results and discussion

During the full-text screening phase, none of the 18 (+ 1 after the update of the search strategy) retrieved studies met the predefined inclusion criteria. Specifically, no study assessed the impact of a healthcare service organisation on clinical outcomes in individuals with ME/CFS, compared to another type of healthcare organisation.

Consequently, the systematic search did not identify any eligible studies addressing the underlying evidence of organisational models for ME/CFS care. This lack of evidence likely reflects both the inherent complexity of designing and evaluating healthcare pathways for ME/CFS and the limited research activity in this area.

These excluded studies were distinguished by the following features:

- **Country:** Three studies were performed among experts of the European Network on Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (EUROMENE).^{16, 130, 131} Five studies were from the USA¹³²⁻¹³⁶ and four studies were performed in the UK.^{84, 137-139} The other studies were from Belgium (n=1)¹⁴⁰, Denmark (n=1)¹⁴¹, Latvia (n=1)¹⁴², Norway (n=1)¹⁴³, Switzerland (n=1)¹⁴⁴, and The Netherlands (n=1)¹⁴⁵.
- **Design:** One study had a quasi-experimental design¹⁴⁵, four studies were defined as a cross-sectional survey study (of which two studies surveyed ME/CFS patients, and two studies surveyed ME/CFS experts).^{130, 131, 143, 144} Two studies had a qualitative design, including interviews with patients (and GPs)^{140, 141}, and one study employed a mixed methods design resulting in a concept mapping study.¹³⁷ Six studies were labelled as narrative reviews.^{132, 134, 135, 138, 139, 142} The other



articles included a narrative description of the results of an expert consensus meeting¹⁶ or expert workshop¹³³, an opinion letter or comment^{84, 146} and a science journalism publication.¹³⁶

- **Population:** Only six studies enrolled patients with ME/CFS of which two studies included patients with severe or very severe ME/CFS.^{141, 143} The four other studies included ME/CFS patients from a Swiss ME/CFS association¹⁴⁴, ME/CFS patients diagnosed based on the CDC criteria¹⁴⁵ or patients diagnosed by a multidisciplinary team together with GPs¹⁴⁰, and for the fourth study physicians and patients with chronic fatigue visiting the *Clinics for Research & Service in Themed Assessment* (CRESTA) fatigue clinic.¹³⁷ The other studies were secondary studies and did not enroll participants.^{84, 132, 134-136, 138, 139, 142}
- **Intervention:** Only one study employed a (quasi)- experimental design investigating a specific intervention, being a stepped-care approach for mental health care (consisting of a first step with guided self-instructions based on CBT for ME/CFS using a booklet and a second step with addition of face-to-face CBT sessions with a trained therapist).¹⁴⁵ One study examined the (un)fulfilled needs of patients visiting the CRESTA fatigue clinic,¹³⁷ while another study investigated the experiences and expectations of patients (and GPs) regarding the inclusion of an advanced nurse practitioner as a care coordinator in the diagnostic care path of ME/CFS.¹⁴⁰ The other studies or articles did not investigate a specific intervention.
- **Comparator:** Only the quasi-experimental study of Janse *et al.* (2019) compared the effectiveness of their intervention (stepped-care approach for mental healthcare) to a reference group, consisting of a combination of samples from other studies where patients received individual or group CBT.¹⁴⁵
- **Outcome:** The quasi-experimental study of Janse *et al.* (2019) evaluated the effect of their intervention on fatigue severity (subscale Checklist Individual Strength) and physical health (subscale SF-36).¹⁴⁵ The cross-sectional patient surveys reported on the burden of the disease, the burden, quality and access of care, therapies and coping mechanisms.^{143, 144} The qualitative studies reported on the needs, experiences and expectations of patients with ME/CFS and GPs.^{140, 141} The other studies described general information, expert consensus, reflections, recommendations, clinical guidelines, and conceptual outcomes regarding the history of ME/CFS, diagnosis, management, research funding and policy related to ME/CFS.^{16, 84, 130-136, 138, 139, 142}

Nonetheless, three of the excluded studies contained insights warranting brief discussion.^{140, 142, 145} Although these studies did not meet the predefined inclusion criteria, they provide information that may offer valuable perspectives on the development and organisation of care services for individuals with ME/CFS. While they do not directly evaluate the impact of service organisation on clinical outcomes, they address aspects of care delivery that are conceptually relevant to the research question. For this reason, a concise narrative overview of these studies is provided below, without detailed methodological appraisal or inclusion in the main synthesis. In addition, two excluded studies focused specifically on individuals with severe or very severe ME/CFS. Since they provide insights into the challenges and considerations associated with care delivery for this subgroup, which is often underrepresented in research, key-elements of these studies were included in Appendix 9.

CONCEPTS POTENTIALLY RELEVANT TO ORGANISATIONAL MODELS OF CARE

Three articles described concepts and components that may contribute to an organisational model for ME/CFS care.^{140, 142, 145}

- The article from Araja *et al.* (2023) argues that ME/CFS can be effectively managed through an **integrative care model**.¹⁴² The authors examine the determinants of success within such models and explore avenues for their enhancement. They state that patients' personal resources should be explicitly incorporated into the framework, since this aspect is insufficiently addressed in current approaches. To this end, they introduce **the Patient Needs-Resources (PNR) model**, in which Health-related quality of life (HRQoL) would depend on independent variables including healthcare resources (e.g. health services, workforce capacity, reimbursement mechanisms) and

personal resources. Within this model, patient engagement would be a mediator and is modulated by patients' unmet needs (availability, accessibility, and affordability). Based on their preliminary research, the authors propose that coaching may constitute a key factor strengthening the model components for a successful approach (mutidisciplinarity, holistic view and patient engagement).

- Ryckeghem *et al.* (2016) investigated the potential **role of an advanced nurse practitioner as a central coordinator** within ME/CFS care pathways in Belgium through interviews assessing the experience and expectations of patients and GPs.¹⁴⁰ Patients frequently reported lacking a single intermediary or care coordinator. The authors proposed that an advanced nurse practitioner could strengthen interdisciplinary collaboration, improve communication, provide patient education, and support a more structured and coordinated care trajectory.
- Janse *et al.* (2019) investigated the effectiveness of a **stepped care approach in a mental healthcare centre** (pre-post comparison in 123 people with ME/CFS without control group) and compared the treatment effects of this stepped care approach to the treatment effects observed in patients who were treated with CBT in one ME/CFS specialist centre (individual or group CBT) (n=583).¹⁴⁵ The stepped care approach started with a minimal intervention including a booklet with guided self-instructions based on a CBT for ME/CFS protocol, provided and evaluated by a psychiatric nurse. The second step (in case patients were still severely fatigued) included referral to a trained CBT therapist for additional face-to-face CBT sessions. The pre-post comparison showed an improvement in fatigue severity and physical functioning after implementation of the stepped care approach and these effects were sustained during long-term follow-up (one to six years). However, people with ME/CFS showed significantly less improvement in fatigue severity and physical functioning after a stepped-care approach, when compared to patients treated with CBT in an ME/CFS specialist centre. This study was not selected in the review because it evaluated CBT-focused strategies without a specific comparison of healthcare pathways.
- Next to this, based on the predefined criteria focusing on healthcare service organisation, we excluded the trial by Vos-Vromans *et al.*¹⁴⁷ that compared two intervention strategies (including a multidisciplinary rehabilitation pathway versus a with CBT alone), rather than directly examining different healthcare pathways.

11.2 Review of the grey literature

The objective of this section is identify sources providing recommendations on how to structure the management of ME/CFS through models of care. Specifically, we included grey literature that outlines national or regional frameworks, strategic plans, or implementation proposals aimed at organising healthcare services for ME/CFS. The scientific evidence supporting specific recommendations was analysed as well.

11.2.1 Methods

The search for grey literature on the organisation of care for ME/CFS was informed by the checklist "*Grey Matters: a practical tool for searching health-related grey literature*" developed by the *Canadian Agency for Drugs and Technologies in Health* (CADTH, now Canada drug's Agency).^{ss} In order to align the search strategy with the research question, only selected document categories were applied: specifically, we focused on health technology assessment (HTA) agencies, clinical practice guideline repositories, and targeted internet searches (Google and Google Scholar). Other categories listed in Grey Matters were not included, as they were considered less relevant to the scope of this review^{tt}.

^{ss} <https://greymatters.cda-amc.ca/>

^{tt} The following categories were not searched: health economics, drug and regulatory approvals, advisories and warnings, drug class reviews, clinical registries, databases and health statistics.



A core search string was developed based on the main concepts of the research question and applied consistently across the browsed document type categories. It included “*chronic fatigue syndrome*” and “*myalgic encephalomyelitis*”. Regarding searches conducted in Google and Google Scholar, the strategy was adapted into more elaborate free-text queries. The following different strings were applied, each time combining them: “*chronic fatigue syndrome*”, “*myalgic encephalomyelitis*”, “*guidelines*”, “*recommendations*”, “*healthcare pathway*”, “*(health)care organization*”, “*consensus*”. The first 10 pages of search results generated from these specific keywords were reviewed. Bibliographies of the selected publications were also screened to identify additional pertinent documents, following a snowballing approach. Eligible documents include general guidance on care pathways, multidisciplinary coordination, service delivery models, and policy-level recommendations intended to structure the healthcare system for ME/CFS. The search was conducted in September 2025 with restrictions of language to English, French and Dutch, and publication year from 2020 to 2025. Details for each source are provided in Table 12.

Table 12 – List of the sources searched

Abbreviation	Institute	Country	Date of search
HTA agencies			
ACPF	The Alberta College of Family Physicians	Canada	23 September 2025
AETS	Agència de Evaluación de Tecnologías Sanitarias	Spain	24 September 2025
AGENAS	The Agency for Regional Healthcare	Italy	24 September 2025
AHRQ	Agency for Healthcare Research and Quality	USA	23 September 2025
AIHTA	Austrian Institute for Health Technology Assessment	Austria	23 September 2025
AQuAS	Agència de Qualitat i Avaluació Sanitàries de Catalunya	Spain	23 September 2025
CADTH	Canadian Agency for Drug and technology in Health	Canada	23 September 2025
HAS	Haute Autorité de Santé	France	23 September 2025
HIQA	Health Information and Quality Authority	Ireland	23 September 2025
HIS	Healthcare Improvement Scotland	Scotland	23 September 2025
HQCA	Health Quality Council Agency	Canada	23 September 2025
HQO	Health Quality Ontario	Canada	23 September 2025
INAHTA	International Network of Agencies for Health and Technology Assessment	International	23 September 2025
INESSS	Institut national d'excellence en santé et en services sociaux	Canada	23 September 2025
IQWiG	Institute for Quality and Efficiency in Health Care	Germany	23 September 2025
MUHC	McGill University Health Centre	Canada	23 September 2025
NICE	National Institute for Health and Care Excellence	UK	23 September 2025
NIHR	National Institute for Health Research	UK	23 September 2025
NIPH	Norwegian Institute of Public Health	Norway	24 September 2025
SBU	Swedish Council on Health Technology Assessment	Sweden	23 September 2025
WHO HEN	World Health Organization Health Evidence Network	International	24 September 2025
ZIN	Zorginstituut Nederland	Netherlands	23 September 2025

ZonMw	The Medical and Health Research Council of The Netherlands	Netherlands	23 September 2025
Clinical Practice Guidelines			
BC Guidelines	British Colombia Ministry of Health	Canada	23 September 2025
CDC	Center for Diseases Control and Prevention	USA	23 September 2025
CMA	Canadian Medical Association	Canada	23 September 2025
ECRI	ECRI Institute Guidelines Trust	International	23 September 2025
GIN	Guidelines international Network	International	23 September 2025
HAS	Haute Autorité de Santé	France	23 September 2025
NHMRC	National Health and Medical research Council	Australia	23 September 2025
NICE	National Institute for Health and Care Excellence	UK	23 September 2025
PHAC	Public Health Agency Canada	Canada	23 September 2025
SIGN	Scottish Intercollegiate Guidelines Network	Scotland	23 September 2025

11.2.2 Results

11.2.2.1 Description of identified documents

The search yielded 4 documents published between 2021 and 2025.^{1, 3, 12, 148} Three reports originated from national health organisations of which two were a clinical practice guidance released in the UK³, and Canada¹⁴⁸, and one was a web-document providing comprehensive information on ME/CFS and guidance for clinicians published in the US.¹ The other identified report was a clinical decision support information document.¹² Of the four documents assessed, three described their research methods^{3, 12, 148}; the fourth had a narrative approach.¹ These documents are described in Table 13.

Table 13 – Description of the identified documents

Type of document document , year	Source and country	Description of the content	Methodology	Recommendations on organisation of care
HSR report (clinical guidance)¹⁴⁸, 2023	Institut national d'excellence en santé et en services sociaux (INESSS)	Clinical guidance asked by the Ministry of Health for healthcare providers and social services on the diagnosis and management of ME/CFS	Systematic review (Medline, Embase, EBM Reviews)	The organisation of care and services in relation to the care pathway for people with ME/CFS is not addressed per se in the research questions but some guiding concepts are provided.
Clinical guidelines³, 2021	National Institute for health Care Excellence (NICE)	Guidelines covering the diagnosis and management of ME/CFS in children, young people and adults	Systematic review (Medline, Embase, Cochrane, CINAHL, EBSCO, PsycINFO, Epistemonikos)	• Coordinated multidisciplinary approach (involving health and social care professionals based on patient needs) • ME/CFS specialist team support referral from primary care (diagnosis/ care and support plan) • Regular review of care and support plan at least 1x/year, in primary care (adaptation of the frequency depending on the severity/complexity of symptoms)



Clinical guidance¹, 2024	Centers for Disease Control and Prevention (CDC)	Health guidance web-document (reference to the Mayo Clinic Proceedings document) ¹³	Narrative review	Referral of complex cases to ME/CFS specialty clinic for additional evaluation and management
Review of best practices¹², 2024	Publication from BMJ	Evidence-based, clinical decision support information	Systematic search process with regular updates	The document does not address the organisation of care and services but provides key-concepts and emphasizes the key-role of primary care physicians for: • Long-term care and monitoring in partnership with secondary care ME/CFS specialists • Coordination and communication with other health, social care, and educational professionals • Facilitation of access to resources

11.2.2.2 Recommendations on healthcare structure

One document from NICE addressed considered the organisation of care as part of a research question.³ Two other recent documents, from INESSS (2023) and published by the BMJ group (2024) provide general information on how to deliver care but were not dedicated on the analysis of the structural and overall organisation of care.^{12, 148} Both derive information from comparable sources which were identified in our search and including the two other retrieved documents i.e. the guidelines from NICE (2021)³ and the website from CDC.¹ An overall comprehensive summary is presented in this section.

- The report produced by INESSS (2023) aimed to support the development of clinical tools to improve the diagnosis and management of ME/CFS within the Québec healthcare system.¹⁴⁸ It provides evidence-informed recommendations based on scientific literature, expert consensus^{uu}, and contextual factors. Although it did not address the issue of organisation of care, it summarises the recent documents addressing that question. Among the selected documents^{vv}, the guidelines from NICE 2021 and the recommendations from CDC^{ww} were included (see Table 13).^{1, 3} The other included documents comprised recommendations from the United States ME/CFS Clinician Coalition (US ME/CFS CC^{xx}), the EUROMENE group (see Text box 3)¹⁶ and narrative reviews on the management of severe forms of ME/CFS and of children/adolescents with ME/CFS.^{135, 149}
- The clinical guidance from BMJ provided the overall consideration of the management and based its guidance on the similar documents from NICE and EUROMENE.

^{uu} A consultative committee composed of clinicians from various specialties and individuals living with ME/CFS validated the contextual relevance of the recommendations.

^{vv} The search strategy encompassed three major biomedical databases (PubMed, Embase, and EBM Reviews) and was supplemented by a manual search of grey literature and relevant websites of health technology assessment agencies, governmental and professional organisations. Timeframe was limited from 2015 to 2022.

^{ww} The CDC website refers also to the recommendation from the Mayo Clinic.⁵

^{xx} <https://mecfscliniciancoalition.org/>

Textbox 3 – Organisations of experts on ME/CFS

- The **U.S. ME/CFS Clinician Coalition** is a network of expert clinicians established to improve the clinical care of individuals affected by ME/CFS. Including physicians from diverse specialties, the Coalition promotes evidence-based practices, develops educational resources for healthcare providers, and contributes clinical expertise to research initiatives. Its activities focus on enhancing diagnostic accuracy, expanding provider knowledge, and supporting the integration of ME/CFS care into mainstream medical practice.
- **EUROMENE (European Network on Myalgic Encephalomyelitis/Chronic Fatigue Syndrome)** is a European research and clinical collaboration, which ran from 2016 to 2020, supported by the *European Cooperation in Science and Technology (COST)*.¹⁵⁰ Its main aim was to improve understanding, diagnosis, and management of ME/CFS by connecting researchers and clinicians across Europe, reducing fragmentation in research, and developing shared standards and recommendations. EUROMENE's research projects focused on several key areas: mapping the epidemiology and socioeconomic impact of ME/CFS in Europe, reviewing and coordinating biomarker research, developing consensus guidelines for diagnosis and care, and standardising research methods to make studies more comparable across countries.

These documents set out recommendations regarding the organisation of care provision (specific recommendations of NICE are provided in Table 14 – see also 11.2.2.2):

- The provision of specialist ME/CFS services, incorporating multidisciplinary teams with expertise in health and social care, is recommended. An early diagnosis by a specialist team allows early management and follow-up that may prevent deterioration in symptoms and wellbeing.
- If there is uncertainty about the interpretation of clinical manifestations (particularly in children), or if the case is complex, consultation with an appropriate specialist or referral to an ME/CFS specialty clinic should be considered for further evaluation and management. The EUROMENE network recommends that specialised services should confirm the diagnosis and establish a treatment/management plan for delivery by the multidisciplinary team. This specialist input should be aligned with a formal training for primary care providers, thereby enhancing first-line management and ensuring continuity across levels of care.¹⁶
- The patient should be referred to other health and social services professionals according to their needs for advice and services.
- A collaboration between primary care teams and ME/CFS specialist services should be encouraged to ensure continuity and quality of care. In addition, NICE guidelines underscore the importance that patients should be offered a review of their care and support plan in primary care at least once a year (see Table 15).³
- The health professionals should be familiar with the specificities of managing ME/CFS.

Table 14 – Summary of the NICE guidance on multidisciplinary care

Recommendations	Complement
• <i>Providing care for people with ME/CFS using a coordinated multidisciplinary approach.</i> • <i>Giving access to health and social care professionals with expertise, based on the patient's needs</i>	Expertise should include as a minimum (with additional expertise depending on symptoms): • Medical assessment and diagnosis • Developing personalised care and support plans • Self-management strategies, including energy management • Symptom management, including prescribing and management of medication • Managing flare-ups and relapses • Activities of daily living, including dental health



	• Psychological, emotional and social wellbeing, including family and sexual relationships • Diet and nutrition • Mobility, avoiding falls and problems from loss of dexterity, including access to aids and rehabilitation services • Social care and support • Support to engage in work, education, social activities and hobbies
People whose ME/CFS care is managed in primary care should be supported by advice and direct clinical consultation from an ME/CFS specialist team	ME/CFS specialist teams consist of a range of healthcare professionals with expertise in assessing, diagnosing, treating and managing ME/CFS. They commonly include medically trained clinicians from a variety of specialisms: • rheumatology • rehabilitation medicine • endocrinology • infectious diseases/ immunology • neurology • general practice • paediatrics They have also access to other healthcare professionals specialising in ME/CFS, which may include: • physiotherapists • exercise physiologists • occupational therapists • dietitians • clinical or counselling psychologists
Provide adults, children, and young people with ME/CFS (and their families or carers, if appropriate) a named contact person in primary care or the specialist ME/CFS team	Aims are: • to coordinate their care and support plan • to help them access services • to support them during relapses • to allow contact with any concern about the child or young person's health, education or social life

Source: NICE guideline NG2026, 2021³

Table 15 – Summary of the recommendations from NICE guidance on review in primary care

Recommendations	Complement
• Offering adults a review at least once a year, and children or young people at least every 6 months • Ensuring access to the support plan and communication from the ME/CFS specialist team • More frequent reviews according to the severity/complexity of symptoms and the effectiveness of the management • Referring to the ME/CFS specialist in case of : ○ Worsening or change ○ Uncertainty about interpreting signs and symptoms	As part of the review, the following elements should be recorded: • Condition, new symptoms and any changes in their illness and its impact • Self-management: energy management plan and (if relevant) physical activity or support to physical function and mobility • Psychological, emotional and social wellbeing • Potential future plans (ask if the person is considering any changes or if they have any challenges ahead)

Source: NICE guideline NG2026, 2021³

11.2.2.3 Description of the underlying evidence

Overall, the scientific evidence underpinning recommendations on the organisation of healthcare services is limited. Most guidance documents aim to support clinical practice but do not address the evidence relating to care organisation^{1, 16, 22, 129}

Only the NICE guidelines were based on scientific evidence, incorporating a systematic review on the organisation of care.³ The latter identified two RCTs assessing multidisciplinary care interventions^{yy} in adults with ME/CFS of mixed or unclear severity, diagnosed according to the 1994 CDC criteria.¹⁵¹ Evidence quality was judged to be low to very low, reflecting methodological limitations including risk of bias, imprecise effect estimates, lack of blinding, heterogeneity, and limited external validity.

The committee recognised that evidence on the effectiveness of multidisciplinary ME/CFS care models is scarce, and of low quality, and therefore insufficient to recommend a defined care pathway. However, optimal care for individuals with ME/CFS was considered to require access to a network of health and social care professionals with relevant expertise, supported by a flexible multidisciplinary framework tailored to individual needs.

Key-points from the reflection of the NICE committee and their recommendations include the following elements:

- The committee noted that key clinical outcomes (e.g. quality of life, fatigue, physical and cognitive functioning, psychological outcomes, and PEM) reflect the direct impact of ME/CFS on patients, and are essential for clinical decision making and evaluating effectiveness of a multidisciplinary care pathway.¹⁵¹ Effectiveness may also be assessed using additional indicators including diagnostic status, identification of comorbidities, adverse events and patient satisfaction, which are also essential to ensure that individuals receive appropriate management. Failure to adequately capturing these outcomes within a multidisciplinary care pathway could lead to unmet clinical needs and negatively affect patient care.
- The committee judged it would be more appropriate to strengthen the expertise and skills in the field of ME/CFS in healthcare professionals rather than to designate a fixed core team of professionals. This included expertise in diagnosis, self-management, and symptom management, together with support for daily living, nutrition, mobility, and participation in occupational, educational, and social activities.
- The heterogeneous and fluctuating trajectory of ME/CFS necessitates adaptable care pathways with access to multiple specialties as needs evolve over time.
- Delivery and oversight of certain interventions (e.g. activity planning or pacing, management of complex symptoms such as POTS, nutritional interventions, assessment of mobility aids or adaptations to avoid overexertion or harm, coping strategies, sensory sensitivity management, etc.) should be restricted to healthcare professionals who are members of a specialist team.
- Referral should be criterion-based but not constrained by too “rigid” checklists, which do not account for symptom complexity and may result in missed opportunities for early diagnosis. Early specialist diagnosis enables prompt management and may avert worsening of symptoms.
- The management should be shared between primary and specialist services (with a single point of contact to facilitate the access/guidance to health and social services).

^{yy} O'Dowd *et al.*, 2006 compared a multidisciplinary care pathway involving a clinical psychologist, physiotherapist, and occupational therapist to primary care and to a control group receiving education and support. Vos-Vromans *et al.*, FatiGo trial compared a comprehensive multidisciplinary rehabilitation program (including physical therapy, occupational therapy, psychology, and social work) to CBT delivered by a psychologist. This study from Vos-Vromans *et al.* was excluded from the review as it compared two intervention strategies and did not directly examine healthcare pathways (see 2.1.2.2).



11.3 Discussion and conclusion

Addressing our question about evidence for the organisation of care in ME/CFS, our search of peer-reviewed studies in the last decade revealed no demonstration of effectiveness to justify a particular service model. By contrast, all recommendations identified in our search of grey literature consistently and pragmatically advocate for a multidisciplinary care approach.

Our systematic literature search identified no eligible studies, whereas the NICE systematic review included two studies. This discrepancy may be explained by the broader inclusion timeframe covered by the NICE search compared with our search, as well as by our exclusion of the study by Vos-Vroman et al., which we considered to compare intervention strategies rather than the structure of healthcare pathways.³

NICE concluded that the evidence was insufficient to specify the optimal composition of a multidisciplinary care pathway, but, consistent with other expert bodies such as EUROMENE, recommend multidisciplinary care delivered by ME/CFS specialists in close collaboration with primary care.^{3, 16} This approach is justified by the complex, fluctuating nature of ME/CFS and its frequent association with comorbidities, which require management by a multidisciplinary team.

The limited evidence base is unsurprising, given the still-incomplete understanding of ME/CFS, the lack of effective treatments, and underinvestment in research.¹⁵² A recent evidence mapping- review by the *National Institute for Health and Care Research* (NIHR) examined studies published between 2018 and 2023.¹⁵³ Earlier, a UK working group identified the top 10 research priorities for ME/CFS using the *James Lind Alliance* process^{zz,aaa}, bringing together patients, informal caregivers, and health and social care professionals.¹⁵⁴ These priorities centred on biomedical mechanisms, diagnosis, and treatment. However, the evidence-mapping review showed that much current research still focuses on areas that do not reflect what patients consider most important, often emphasising psychological interventions or descriptive lived experience. The review therefore stressed the need to involve patient representatives from the earliest stages of research design. Within research on care organisation, the review found five systematic reviews addressing specific care topics, but none examined the overall structure or organisation of care. This gap aligns with the deficiency identified in our own systematic review. The review also highlighted important reporting concerns. Demographic information was often missing, and diagnostic criteria were applied inconsistently across studies. Addressing these issues should become part of methodological standards for future research, whether focused on aetiology, therapeutic evaluation, or service-delivery models.¹⁵³

When designing tailored and personalised care models, illness severity is essential to consider. The evidence mapping found that 81% of identified studies did not report ME/CFS severity. People who are severely or very severely affected are often too unwell to travel to a multidisciplinary centre, and they can face complications with potentially deleterious consequences such as malnutrition^{3, 155} Their needs differ markedly from standard outpatient care, yet research on this subgroup remains limited, leaving service design and accessibility insufficiently explored. Reports indicate that people with severe ME/CFS may deteriorate and can develop distrust following contact with healthcare professionals.^{141, 143} For this subgroup, measures to increase accessibility such as home visits, teleconsultations, and alternative modes of communication (written communication, audio/video recordings) are particularly important. Multidisciplinary care teams have adapted service provision such as through mobile teams delivering individualised care also including technically complex interventions such as enteral feeding.¹⁵⁶ Such initiatives require substantial mobilisation of resources and staff. Among recent efforts, the *British Association of Clinicians in ME/CFS* (BACME) has issued

^{zz} The James Lind Alliance is an initiative in the UK that brings patients, carers, and clinicians together in priority setting partnerships to identify and prioritise the research questions that matter most to them.

^{aaa} <https://www.jla.nihr.ac.uk/jla-guidebook-2025>

practical clinical guidance, aligned with the NICE recommendations, outlining care approaches and strategies to improve service accessibility for people with severe ME/CFS.⁶²

Finally, our literature search identified several concepts that may inform the development of a Belgian organisational model for ME/CFS care: an integrated, complex-adaptive system built on a holistic and multidisciplinary approach, and active patient engagement, with potential tailored support such as a care coordinator; flexible stepped-care pathways allowing adaptation and efficient use of resources. These elements already exist within the Belgian Interfederal Plan for Integrated Care, which promotes integrated, multidisciplinary care for patients with complex needs and includes case managers to support coordination.^{bbb} This alignment suggests that Belgium has a favourable policy foundation on which ME/CFS specific-care models could be developed.

12 ORGANISATION OF CARE IN SELECTED COUNTRIES

In 2016, the EUROMENE Network conducted a study that characterised the situation regarding national guidelines and recommendations for case definitions, diagnostic procedures, and clinical management of patients with ME/CFS across the countries of the EUROMENE network.¹³⁰ Results showed substantial differences among the countries, with only a few of them having a multidisciplinary team for diagnostic assessment^{ccc}, and a majority of them not having national guidelines for ME/CFS management^{ddd}. Therefore, the authors called for harmonised recommendations across European countries stressing the need for clear guidance and standards.

In this section, we provide a description of the (organisation of current healthcare services for ME/CFS across a panel from European countries, aiming to draw insights for the organisation of healthcare services in Belgium.

The central research question was: *“Does the country have a dedicated, structured organisation of care for individuals with ME/CFS?”* More specifically, the following subquestions were raised:

- Are there specific structures involved in the management of ME/CFS that are organised and funded by the health system in your country? If this is not the case, what is the healthcare services offered to patients living with ME/CFS and are there any reference centres developed by initiatives other than a national initiative?
- Are there national guidelines used for diagnosis and managing ME/CFS?
- Are there current reforms to adapt or improve the care for patients living with ME/CFS?
- Are there governmental research projects dedicated to ME/CFS?

12.1 Methods

Four countries were selected for the international comparison on the basis of their comparability and differences to the Belgian health care system regarding the services for ME/CFS: France, the Netherlands, Italy, and the UK^{eee} (the description of Belgium is provided in the introduction – see Introduction section 5).

^{bbb} <https://www.riziv.fgov.be/nl/professionals/info-voor-allen/het-interfederaal-plan-voor-geintegreerde-zorg>

^{ccc} The Fukuda diagnosis criteria were the most frequently used.

^{ddd} The recommended approaches for symptom and illness management most commonly consisted of GET, CBT, pacing strategies, or broader rehabilitation programs.

^{eee} Germany and Norway were also selected as comparator countries; however, no responses were obtained from the experts consulted.



To understand the structure of healthcare services for ME/CFS in the selected countries, experts were consulted to retrieve information^{fff}. Additional country-specific information was retrieved via national websites, health institutes, or (research) reports from professional organisations.

The consulted experts in each country were identified based on their authorship of leading scientific publications in the field and were also selected based on their availability and willingness to participate in the study (the list of experts is provided in Appendix 10). They were invited to take part in a video conference or alternatively in email exchanges between September 2025 to January 2026. A questionnaire was designed on the organisation of health services for ME/CFS in their country (Appendix 11). It should be noted that the experts did not participate in the drafting of this chapter and therefore cannot be held responsible for its content. No remuneration was provided for their contribution.

The comparison was restricted to ME/CFS care structures that possess official recognition from national health authorities and are integrated (at least partially) within the publicly reimbursed national health insurance system.

12.2 Results

12.2.1 General findings

- Across the countries surveyed, our analysis shows that the UK is the only country with a formally structured, nationally organised referral pathway for ME/CFS, with services available in most places and integrated within secondary and tertiary care levels. In contrast, ME/CFS services in the other countries examined are not formally organised, nor are they embedded within any national strategy for service provision. In these countries, the configuration of care pathways varies considerably and relies predominantly on clinician/expert-driven initiatives (i.e. healthcare professionals with specific expertise deliver dedicated consultations and management within their own centres). Consequently, access to care is limited, with marked geographical disparities and gaps in service provision across many regions. In addition, the composition of the multidisciplinary therapeutic and management teams differs considerably, both between countries as within a country. A summary of findings is provided in Table 16.
- Our analysis also indicates that the UK is the only country with regularly updated national clinical guidelines, including diagnostic criteria. Other selected countries either lack national guidelines or have outdated documents.
- Finally, countries differ with respect to ongoing reforms, emerging policy initiatives, and dedicated (national) research funding in the field of ME/CFS. These aspects are detailed in the following sections.

12.2.2 Countries without structured referral pathway for ME/CFS

12.2.2.1 France

A. Organisation of care and service provision

In France, no official state-organised care pathway or national plan for ME/CFS currently exists. There are consequently no formally designated structures within the healthcare system dedicated to the management of ME/CFS. The number of clinicians with specific competence in ME/CFS remains very limited and specialist services are only available in few major urban centres. These specialised

^{fff} We sought expert input from additional countries: we contacted two experts in Germany and one expert in Norway, but we did not receive any responses despite sending several invitation messages and follow-up reminders.

services were independently developed on a clinician-driven initiatives who have developed expertise and set up dedicated consultations within their own institutions. These centres are located in secondary or tertiary-level hospitals (Centre Hospitalier Universitaire). They offer diagnostic confirmation, development of treatment plans, and follow-up care, although the scope of clinical services varies. Where available, multidisciplinary care may involve internal medicine, infectious diseases/clinical immunology, clinical psychology, psychiatry, rehabilitation medicine, or pain specialists. Associated comorbidities can also be addressed within these settings.

B. National guidances

France currently has no national clinical guidelines for the diagnosis or management of ME/CFS, nor any formal recommendations defining a standardised care pathway. As a result, clinical practice varies substantially between the centres that do offer services for ME/CFS and mainly depends on local expertise and individual clinical judgement.

C. Financing and reimbursement

There is no dedicated national funding scheme for ME/CFS services. Patients covered by the national health insurance system receive reimbursement for consultations and hospital-based care under the standard provisions applicable to each clinical specialty. No specific financial mechanism exists to support ME/CFS management as a standalone condition, although the condition is recognised by the national health insurance^{ggg}.

D. Information dissemination

Some expert centres organise educational sessions and e-learning^{hhh} activities that are aimed to improve knowledge of ME/CFS among healthcare professionals. In addition, patient associations play a key role in disseminating scientific and administrative information, supporting patients in administrative procedures, and helping identify clinicians with expertise in ME/CFSⁱⁱⁱ.

12.2.2.2 The Netherlands

A. Organisation of care and service provision

In the Netherlands, there is no nationwide structured organisation, and the overall number of healthcare professionals with specific ME/CFS expertise is limited. There is no formally designated national referral pathway for ME/CFS. Several centres have been established by clinicians with specific expertise (and several smaller private institutes also offer care).

The GP is typically the first point of contact (gate-keeper system) and may refer patients to a specialist (most commonly a general internist) for diagnostic confirmation. Once the diagnosis is established, patients may be referred to the reference centres for treatment planning and follow-up. Notably, the full range of therapeutic interventions is not fully delivered within these centres; patients often receive components of care outside these centres, depending on local availability. Tele-consultations are used to facilitate access.

^{ggg} <https://www.ameli.fr/assure/sante/themes/asthenie-fatigue/un-cas-particulier-le-syndrome-de-fatigue-chronique>

^{hhh} Given as example: <https://hopital-europeen.fr/hopital/nos-actualites/journee-du-syndrome-de-fatigue-chronique>

ⁱⁱⁱ <https://www.asso-sfc.org/syndrome-fatigue-chronique-references-medicales.php>



B. National guidances

The Netherlands has outdated national clinical guidance (dating from 2013) available on the official national guidelines platform^{jjj}. The *Ministry of Health, Welfare and Sport (Ministerie van Volksgezondheid, Welzijn en Sport-VWS)* has noted that Fukuda criteria remain widely used, and that IOM criteria are not yet consistently applied in practice. Its website mentions that an update of the national guidelines is underway, with finalisation expected in 2026^{kkk}. The new guideline is intended for medical specialists, GPs, occupational health physicians, insurance physicians, and other healthcare providers. Its development is supported by the VWS, with the Dutch Society of Internal Medicine (*Nederlandse Internisten Vereniging*) acting as project coordinator, and the Knowledge Institute of the Federation of Medical Specialists (*Kennisinstituut van de Federatie Medisch Specialisten*) providing methodological support.

C. Financing and reimbursement

There is no disease-specific reimbursement mechanism for ME/CFS. Instead, health services are reimbursed under the general Dutch health insurance framework, in accordance with the usual provisions applicable to the disciplines involved.

D. Policy Reforms and Research

Following a citizens' initiative launched in 2013, the House of Representatives commissioned the Health Council (*Gezondheidsraad*^{lll}) to review the scientific evidence and what developments can be expected in ME/CFS^{mmm}. A Committee included experts from various fields as well as representatives of patients. The resulting 2018 report concluded that research was essential to improve patient care and that it is crucial to establish the diagnosis, take patients' symptoms seriously, provide optimal management, and recognise their functional limitations.¹⁵⁷ Consequently, the Committee issued recommendations, including that:

- The Minister of Health should commission *ZorgOnderzoek Nederland Medische Wetenschappen (ZonMw)* to run a long-term research programme on ME/CFS, prioritising diagnostic validation, disease mechanisms, and evidence-based treatments.
- The institutions in charge of undergraduate and continuing professional development should ensure their educational programme addresses ME/CFS as a serious, chronic, multisystem disease, with practical guidance on appropriate clinical care.
- The *Nederlandse Federatie van Universitaire Medische Centra (NFU)* and health insurers should designate selected university medical centres to establish ME/CFS outpatient clinics, developed in collaboration with patient representatives and regional providers (hospitals, GPs, rehabilitation and sleep clinics), with linked care networks and research groups.
- Medical assessors operating within disability and social support frameworks should recognise ME/CFS as causing substantial functional limitations and should not interpret a patient's decision to decline CBT or GET as evidence of inadequate recovery behaviour.

Following the publication of this report, *ZonMw* launched a ten-year biomedical research programme for biomedical research on ME/CFS (funding: €28.5 million). It was initiated in December 2021ⁿⁿⁿ. The programme included several disciplines (researchers, health professionals, policymakers), involving

^{jjj} <https://richtlijnen database.nl/>

^{kkk} <https://demedischspecialist.nl/nieuwsoverzicht/nieuws/ontwikkeling-richtlijn-mecvs-van-start>

^{lll} <https://www.gezondheidsraad.nl/>

^{mmm} <https://www.gezondheidsraad.nl/documenten/2018/03/19/me-cvs>

ⁿⁿⁿ <https://www.zonmw.nl/en/program/mecfs-research-programme>

international collaboration, and patients as partners. Its principal objectives are to generate biomedical knowledge on the onset, diagnosis, and management of ME/CFS; to translate evidence into clinical practice; and to strengthen collaboration among research institutions, patients, and healthcare professionals. A key component is the development of a national ME/CFS biobank^{ooo}. Projects are conducted across the Netherlands, with topics and locations listed on ZonMw's website^{ppp}.

Regarding care and support, the *Ministry of Social Affairs* asked municipalities to take the *Gezondheidsraad's* advice on ME/CFS into account when determining reimbursements under the "*Participatiewet*"^{qqq} and the "*Wet maatschappelijke ondersteuning*"^{rrr}. The advice states that ME/CFS is a serious illness associated with substantial functional limitations.^{sss} The website of the Ministry of Health also refers users to a support and advocacy association.

E. Information dissemination

The Dutch Ministry of Health provides general information on ME/CFS^{ttt,uuu} on its website, and public communication has announced the national guideline update. Practical indications on care are disseminated, including references to NICE guidance, fact sheets for primary care^{vvv}, how to find tools and facilities for life adaptations^{www} and information brochures produced in collaboration with patient associations^{xxx}.

12.2.2.3 Italy

A. Organisation of care and service provision

No state-organised care pathway or national plan for ME/CFS is currently in place. Specialist expertise is scarce^{yyy}: two dedicated reference centres accept referrals from GPs, and additional private initiatives also exist. These specialist centres focus primarily on diagnostic work-up and confirmation, alongside the development of therapeutic plans, and follow-up. The delivery of treatment is generally undertaken outside these centres, via care networks established by the centres and involving healthcare professionals with specific ME/CFS expertise. In addition to clinical care, these centres participate in research projects and occasionally offer investigational therapeutic options (e.g. use of antiviral agents).

^{ooo} <https://projecten.zonmw.nl/en/project/netherlands-mecfs-cohort-and-biobank-nmcb-consortium>

^{ppp} <https://www.zonmw.nl/en/program/mecfs-research-programme>

^{qqq} Dutch social-assistance law that helps people who can work but need support to obtain or keep employment (implemented by municipalities).

^{rrr} A law giving responsibility on municipalities to provide support that enables people with impairments to live at home and participate in society.

^{sss} <https://www.regelhulp.nl/onderwerpen/me-cvs/zorg--ondersteuning-bij-me-cvs>

^{ttt} <https://www.regelhulp.nl/onderwerpen/me-cvs/over-me-cvs>

^{uuu} <https://www.regelhulp.nl/onderwerpen/me-cvs/zorg--ondersteuning-bij-me-cvs>

^{vvv} <https://www.thuisarts.nl/chronisch-vermoeidheid-syndroom/ik-heb-chronisch-vermoeidheidssyndroom>

^{www} <https://www.regelhulp.nl/onderwerpen/me-cvs/zorg--ondersteuning-bij-me-cvs>

^{xxx} <https://www.regelhulp.nl/documenten/2024/05/13/mecvs-informatie-voor-behandelaars---patientenorganisaties>

^{yyy} <https://www.senato.it/show-doc?id=1358210&leg=18&tipodoc=Sindisp>



B. National guidance

Italy has a national guidance developed by the National Agency for Regional Healthcare Services (*Agenzia Nazionale per i Servizi Sanitari Regionali- AGENAS*), that is currently outdated (2014).¹⁵⁸ In current practice, specialist centres commonly align their management with the updated NICE guidelines and recommendations from EUROMENE.

C. Financing and reimbursement

There is no dedicated national funding scheme specific to ME/CFS: consultations and complementary investigations are covered within the national health service under standard arrangements applicable to the relevant specialties. It is noteworthy that the hospital system relies on ICD-10 coding (prior to the introduction of a specific code for ME/CFS^{zzz}, it was difficult to arrange hospital admissions specifically for the condition).

D. Policy Reforms

While no implemented national reform specific to ME/CFS is in place to date, two legislative initiatives are under parliamentary consideration: a draft bill in the Chamber of Deputies proposes to officially recognise ME/CFS as a disabling disease in Italy, thereby granting exemption from co-payments for related health services^{aaaa}. The proposal would also mandate the Ministry of Health to define diagnostic and management criteria, designate reference centres, issue national guidelines, and establish health data registration to support standardised care protocols. A separate proposal in the Senate is likewise under review^{bbbb}.

E. Information dissemination

The reference centres collaborate with patient associations to disseminate information on ME/CFS (e.g. public information sessions and educational resources). The *Istituto Superiore di Sanità* also produces explanatory materials, supporting both public awareness and professional understanding of ME/CFS^{cccc}.

12.2.3 Presence of a structured referral pathway: United Kingdom

A. Organisation of care and service provision

In the UK, a referral pathway specific for people with ME/CFS is available. The management of the condition is embedded within the *National Health Service* (NHS) with a GP as the first point of contact. The patient is referred to specialised services for diagnostic confirmation, care planning and multidisciplinary personalised plan of multidisciplinary input (the referral can also be done by a specialist physician). There are numerous NHS specialist clinics that are spread throughout the country and range from large level 2 and level 3 centres in big cities, to smaller facilities in other regions, so that the geographical coverage (and the scope of provision) may vary^{dddd}.

These specialised centers include a network of specialist in ME/CFS and the type of allied health professionals working in the centres may vary between centers (i.e. physiotherapists, occupational therapists and clinical psychologists are present most of the time but not all centers have medical experts), and all members of the multidisciplinary team have a wide range of knowledge, skills, and experience in ME/CFS. Treatments are offered either in a group or individual format. Most of the

zzz ICD-10 : G93.3 ; ICD-11 : 8E49

aaaa <https://www.camera.it/leg19/126?leg=19&idDocumento=1397>

bbbb <https://www.senato.it/leggi-e-documenti/disegni-di-legge/scheda-ddl?did=56782>

cccc <https://www.issalute.it/index.php/la-salute-dalla-a-alla-z-menu/s/sindrome-da-fatica-cronica>

dddd NHS is currently looking for solutions for this inequity since the place of living determines the access to services.

services have referral criteria, including a preliminary complementary exam that needs to be performed before being referred (e.g. blood tests).

The duration of the follow-up depends on the service and the severity of complaints from the patient. Most services are organised for moderately affected patients, but some are also equipped for patients with severe forms of ME/CFS. Online or phone consultations are now also possible. There are also services for children, but they are fewer in number.

In many places, other conditions such as long COVID and fibromyalgia are being treated within the same service.

B. National guidance

The UK has regularly updated national clinical guidance through which it provides national standards for diagnosis and management. The specialised services follow the national guidance developed by NICE.

C. Financing and reimbursement

Specialised services are funded within NHS structures as part of routine commissioning (community/secondary care).

D. Information dissemination and professional training

Information on how to find health services is available on the NHS website^{eeee}, and specifically on the websites of each specialised service across the country. In addition, an up-to-date database of the availability of NHS specialist services is provided by the BACME^{ffff}; an interactive map is available on their website to find the nearest ME/CFS service. BACME is a multidisciplinary organisation for health professionals who are involved in the delivery of clinically effective services for patients with ME/CFS in the UK and advocate for equity of access and excellence in service provision. This is an additional initiative to support health services and professionals (signing-up is done on a voluntary basis). The association supports professionals disseminating up-to-date evidence clinical practice, and providing education and training grounded in up-to-date evidence. Information and practical management guides are provided on their website.⁶² The associations aim also at facilitating networking amongst clinicians, researchers, and patients, and promoting research.

The members of ME/CFS specialised services team generally attend regular training events to update their knowledge and skills within this specialist area, and to network with other ME/CFS services.

E. Policy Reforms and research

In 2025, the UK has recently adopted a new cross-sector delivery plan for ME/CFS. The plan responds to concerns of health authorities about variation in care, disease recognition and access to services.¹⁵⁹ The delivery plan was developed by the department of Health and Social Care in partnership with NHS England, the Department for Work and Pensions, and the Department for Education. It aims to improve outcomes for people with ME/CFS and it covers three domains in which changes and improvements are encouraged⁹⁹⁹⁹: (1) research, (2) attitudes and education, and (3) living with ME/CFS (including health services, adult social care, quality of life, support for children and young people, welfare, and employment support).

^{eeee} <https://www.nhs.uk/conditions/chronic-fatigue-syndrome-cfs/>

^{ffff} <https://bacme.info/>

⁹⁹⁹⁹ Quote of the Ministerial foreword of the delivery plan: “We know that more research, better services and a better understanding of the condition all have the potential to make a huge difference to the quality of life of people with ME/CFS, whether that be those with symptoms on the milder end of the spectrum or those with very severe ME/CFS, and everyone in between.”¹⁵⁹



The plan was developed in two steps (interim plan in December 2024 followed by the final version in July 2025) and was informed by public consultation and a cross-sector consultations of stakeholders including researchers, clinicians, people with lived experience and charities. It sets out actions, the people responsible for, and includes measurements of the results and progress. Although the plan is organised by distinct topics, many actions are cross-cutting^{hhhh}. A brief description, with examples of actions, is provided in this section:

Research

The plan first outlines the key challenges for ME/CFS research: too few biomedical studies, inadequate capacity to meet needs, and limited stakeholder trust in research prioritization. Accordingly, actions aim to build researcher skills and capacity, and to align proposals with patient -prioritised questions (workshops with funders and the community to develop proposals aligned with *James Lind Alliance*ⁱⁱⁱⁱ priorities were organised- see 11.3). In parallel, the *NIHR* produced a formal evidence review and an infographic to reveal gaps and guide high value- studies^{jiii}. A researcher toolkit was also developed^{kkkk}. In addition, the plan proposes exploration of overlaps with post-acute infection syndromes such as long COVID.

NIHR therefore actively encourages funding applications for translational research across all aspects of ME/CFS, including, for example, pathophysiological mechanisms, diagnostic tests, the effects of repurposed drugs or clinical interventions, as well as projects to improve service efficiency and cost effectiveness.

Between 2019 and 2024, the UK Government invested around £6.5 million in research into ME/CFS, including several projects such as the DecodeME study for example^{llll}.⁵³

Attitudes/education/information

Consultation feedback on the interim plan identified priority issues regarding healthcare professionals' knowledge, inconsistent adoption of up-to-date guidance, and limited disease awareness. The plan therefore prioritised and initiated several actions (that are completed or still ongoing). Among the actions undertaken, a few examples include^{mmmm}:

- Professional training: development of e-learning for professionals; online *Clinical Knowledge Summary* for healthcare professionals intended to align practice with the 2021 NICE guideline; and co-designed training with people with lived experience to ensure education is practical and aligned with real-world needs.
- Discussion on medical education with the *Medical Schools Council* to strengthen undergraduate and postgraduate medical training.

hhhh Overlap with other conditions such as Long COVID were recognised but the plan was maintained to be focused on ME/CFS.

iiii <https://www.jla.nihr.ac.uk/>

jiii https://www.journalslibrary.nihr.ac.uk/sites/journalslibrary/files/journal_data/BTBD8846/NIHR136262_su_pp7.html

kkkk <https://psp-me.co.uk/researcher-toolkit/>

llll DecodeME aimed to find genetic factors contributing contributing to the development of ME/CFS in individuals (using a control group of healthy volunteers).

mmmm The plan publishes delivery tables detailing action owners, progress updates, indicators, and indicative timelines across the three domains. For the list of all completed and ongoing actions, we refer to the delivery plan.¹⁴⁰



- The *Department of Health and Social Care* is developing a public awareness initiative to promote broader understanding of ME/CFS and to improve support for people with the condition and their families and carers.
- The *British Psychological Society* is developing good-practice guidelines for working with people with ME/CFS.

Living with ME/CFS

The plan addresses 6 main topics including: provision of health services, adult social care, quality of life (including family members and carers), support for children and young people, welfare, and employment support. Overall, it seeks to improve access to services, and to ensure provision is matched to illness severity or patient's age. Among the ongoing actions, examples include:

- Strengthened measures to improve the quality and accessibility of health services and social care.
- Development of a standard service template for mild-moderate ME/CFS, neighborhood multidisciplinary teams for children and young people and exploration of a specialised service for very severe ME/CFS.
- Actions relevant to wider health services improvement for people with broader long-term conditions (existing initiatives).
- The Department for Education will build ME/CFS considerations into wider education policy for children and young people, pursue opportunities to raise awareness and promote reasonable adjustments to help pupils stay engaged in school.
- There are no ME/CFS specific welfare changes; instead, ME/CFS is included within ongoing Department for Work and Pension reforms to disability benefits and employment support for people with long-term conditions.



Table 16 – Summary on main characteristics of health services offer for ME/CFS in each selected country

Item / country	UK	France	The Netherlands	Italy
1. <i>Existence of a nationally organised care pathway</i>	Nationally organised referral pathway embedded in the NHS, and available across most regions and integrated within secondary and tertiary care.	No national structured care organisation; services depend on clinician-initiated specialist consultations	No national structured care organisation; services depend on clinician-initiated specialist consultations	No national structured care organisation; services depend on clinician-initiated specialist consultations
2. <i>Availability and distribution of reference centres</i>	Extensive network of specialist ME/CFS clinics; geographical coverage and scope of provision vary by region.	Very limited number of clinician experts; centres concentrated in major urban centres; significant geographical disparities	Limited specialist capacity; marked geographical variation; (use of tele-consultations)	Two dedicated reference centres, alongside additional private initiatives
3. <i>Multidisciplinary composition of services</i>	Generally multidisciplinary teams with inter-centres variability	Multidisciplinary care composition variation between centres	Therapeutic delivery often occurs outside reference centres (multidisciplinary involvement varies locally)	Treatment delivered through centre-led care network with appropriate expertise (multidisciplinary composition context-dependent)
4. <i>National guidances</i>	National guidelines updated (2021)	No national guidelines for diagnosis or management.	Yes, but outdated (2013) update in progress, expected in 2026	Yes, but outdated (2014)
5. <i>Financing</i>	Specialist services funded within NHS structures as part of routine commissioning (community/secondary care).	No dedicated funding; standard reimbursement through national health insurance according to each clinical specialty	No ME/CFS-specific reimbursement; services covered under the general Dutch health insurance framework	No dedicated disease-specific funding; standard reimbursement covered by the national health service
6. <i>Ongoing reforms or policy initiatives</i>	Cross-sector Delivery Plan (2025): • Department of Health and Social Care • Department of Work and Pensions • Department for Education Domains include: • Research • Attitudes/education	No national-level reforms specifically targeting ME/CFS identified.	Policy direction shaped by <i>Gezondheidsraad</i> advice: encouragement to develop research, to designate University medical centres (UMC)-based outpatient clinics and linked care networks and	Legislative initiatives under consideration

	• Living with ME/CFS		to review reimbursement rules and recognise ME/CFS as condition that causes significant functional impairments	
7. National Research Funding	New Government/NIHR funding opportunities for ME/CFS and post-acute infection conditions	No dedicated national research programme; research occurs through isolated academic initiatives	ZonMw 10-year biomedical program launched in 2021	No dedicated national research programme; research occurs through isolated initiatives.
8. Professional training and knowledge dissemination	Professional support structures exist: • E-learning Clinical Knowledge Summary, engagement with the Medical Schools Council • BACME resources • Public-awareness work ongoing (Department of Health and Social Care)	Training activities depend on expert centres; patient associations play a key role in information dissemination	• Relevant information on the Ministry website • National guideline update publicly announced	Educational materials from the <i>Istituto Superiore di Sanità</i> and initiatives from collaboration between experts and patient associations



12.3 Discussion and conclusion

In this international comparison, we found that, of the four countries examined, only the UK has a nationally structured referral pathway for ME/CFS. Although regional variations in coverage and services may exist, the existence of a nationally organised care pathway provides a clearer framework for access, via referrals from GPs to specialist services, with provision for both adults and children. By contrast, in the other three countries, access largely depends on a limited number of clinician-led centres of expertise, which may exacerbate geographical inequalities and increases the risk of delayed diagnosis. Moreover, in these countries the composition of the healthcare teams delivering care varies substantially, which may affect the quality and continuity of follow-up care.^{16, 160}

Our analysis also indicates that **several policy changes** are under way or planned across different countries: among the most salient developments are initiatives to promote broader understanding and knowledge of ME/CFS among health professionals with the aim of improving support for affected individuals. This knowledge gap has been documented in both primary and secondary care.^{131, 161, 162} Accordingly, knowledge-dissemination programmes have been developed in the Netherlands and the UK to enhance clinicians' competencies and recognition of the condition, with the dual aim of facilitating early referral and aligning practice with current scientific evidence and existing guidelines. In the Netherlands, guidance and information materials have recently been issued following an advisory opinion from the *Gezondheidsraad*, highlighting the changes introduced by the NICE guideline.¹⁵⁷ The UK has implemented these measures pragmatically through the dissemination of e-learning and by engaging with the Department of Education to assess how this content could be incorporated into curricula of pre-graduate students. Moreover, this is not confined to healthcare professionals since work is under way to explore options to increase public and societal awareness of ME/CFS. In parallel with these initiatives, several countries are planning or working on adaptations of service provision. In the UK, efforts are under way to improve access to local care, particularly for children and young people. Disparities have indeed been recognised: a study published in 2025ⁿⁿⁿⁿ identified diagnostic variation by sex, age, ethnicity and place of residence, suggesting that people of non-white ethnicity, older adults, and those living in areas of multiple deprivation were disproportionately likely to remain undiagnosed.³⁵ In the Netherlands, the *Gezondheidsraad* has recommended the establishment of reference centres within University Medical Centres (UMCs), underpinned by a network of affiliated healthcare professionals. Similarly, a bill in Italy has proposed the creation of reference centres implementing standardised care pathways. However, these proposals have not yet been translated into concrete measures in those countries.

Regarding **care provision**, all experts consulted stressed that the severe and very severe forms of ME/CFS remain challenging to manage appropriately. For those patients who may be bedridden and dependent on family members for activities of daily living, access to care (and also first-line of care i.e. GP) remains challenging, and there is generally no suitably tailored multidisciplinary service provision. This situation has also been reported in countries beyond those included in our analysis and is associated with a substantial burden on family members.¹⁴³ Moreover, this subgroup is under-represented in research, and there are even fewer healthcare professionals with expertise in these severe forms of ME/CFS. In the UK, a working group on severe disease has been established within the new Delivery Plan and is examining options to adapt care provision through specialist services.¹⁵⁹ Initial discussions with the *Department of Health and Social Care* and the NHS have already taken place, and an expert group will be convened (no definitive timeline has yet been set).

Our analysis also identifies the absence of **national guidelines** in most of the countries examined. In some countries, guidelines exist but were outdated (exceeding 10 years). This suggests an inconsistent use of diagnostic criteria among health professionals. A 2019 survey conducted by

nnnn Collected data were from 100 055 individuals diagnosed with ME/CFS.

EUROMENE across 17 European countries had already shown that only five countries^{oooo} had developed guidelines over the preceding ten years, and that the Fukuda criteria were still frequently used.¹³⁰ The current situation supports the authors' recommendation to develop regulation, guidance and standards to address inconsistent diagnosis and management. This is particularly important given that the 2021 NICE guideline substantially revised care recommendations. It is noteworthy that the Netherlands, on its *Ministry of Health* website, has officially announced that national guidelines will be available shortly and already refers to the 2021 NICE guideline^{pppp}.

Several countries also report investing in coordinated **research programmes** (e.g. cohort and biobank development, pragmatic trials). Regarding biomedical research, the UK and the Netherlands have made available multidisciplinary research funding for programmes that include biomedical components. In the Netherlands, the *ZonMw* programme provides a robust infrastructure capable of accelerating diagnostic validation development and mechanistic investigation. In the UK, the NIHR funds large-scale projects such as the *DecodeME* study⁵³ and encourages translational studies. Linkages with post-infectious syndromes, including long COVID, further strengthen the relevance of these research agendas. Irrespective of the allocated budgets, the development of **translational oriented research** projects is crucial to accelerate knowledge generation and to develop more effective interventions. In this regard, efficient interaction between researchers and clinicians (and, consequently, the rapid recruitment of essential biospecimens and clinical data) is an important step, that has been emphasised by the *Gezondheidsraad* in the Netherlands.¹⁵⁷ Furthermore, involving people with lived experience at the study-design stage and throughout the whole study process helps to ensure that research objectives are aligned with real-world needs.

Beyond translational research, **health economic research**, including service efficiency studies, is also essential to assess the benefits and costs of current and future service models. The UK's new Delivery Plan likewise encourages this type of evaluation.¹⁵⁹ Notably, in the latest NICE guideline, only two studies with a relevant comparator were identified, and they provided very limited quantitative evidence.³ As a result, the cost-effectiveness of a multidisciplinary care pathway remains uncertain. Consequently, any multidisciplinary service offer should be evaluated using reproducible, validated indicators anchored in functional and clinical outcomes (e.g. health-related quality of life, fatigue severity, return to work).

Although our analysis did not include **welfare benefits or formal recognition of disability**, we noted that the UK addresses return-to-work issues for people with ME/CFS in its new Delivery Plan.¹⁵⁹ This includes support for adaptations to both the working environment and reasonable work adjustments. As part of the Department for Work and Pensions' ongoing reforms to disability benefits and employment support for people with long-term conditions, ME/CFS has been included. Parallel consultations on how to strengthen work, health and skills support for people living with a disability are in progress. The Netherlands emphasised recognition of ME/CFS as a disabling illness, integrating this into decisions on social support and reimbursements.^{qqqq}

Limitations of the analysis

Our analysis has several limitations. First, the findings derives from a review of national sources and (a limited number of) expert interviews, with a scope that focuses on services recognised by national authorities and at least partially reimbursed. Non-accredited private services were excluded, which may underestimate the true availability of care. In addition, because some countries have limited service provision, our analysis relied on the perspectives of the interviewed experts and their own

^{oooo} Countries in which guidelines were developed over the last ten years in the publication from Strand et al. (2019) and year of the publication of the guidelines into parenthesis for each country: UK (2007), Italy (2011), Netherlands (2013), Norway (2015), Spain (2014)

^{pppp} <https://www.regelhulp.nl/onderwerpen/me-cvs/over-me-cvs>

^{qqqq} <https://www.regelhulp.nl/onderwerpen/me-cvs/zorg--ondersteuning-bij-me-cvs>



experience. Preparatory work for potential reforms is not always publicly accessible, which may have led us to overlook national initiatives known only to involved experts and not identified in our review. Also, the time horizon (Dutch guideline expected in 2026; actions from the UK Delivery Plan still under way) means that developments are likely and updates of this analysis will be required. It should be mentioned that the international comparison limited its focus mainly to organisation of care and care models (in congruence with the literature research question). Therefore, other relevant topics covered in this report related to patient's needs (e.g. welfare benefits, recognition of disability, types of support, etc.), were not part of this international comparison. We will further discuss these topics in the general discussion of this report.

Conclusion

In conclusion, our comparison identified notable differences in the organisation of care across the countries investigated. The UK is the only country with a structured national referral pathway, while others rely on clinician-led expertise centers.

However, the countries do share similar priorities for a reformed care organisation for ME/CFS. The information indicates that there is a clear need to update or develop national guidelines, harmonise diagnostic criteria, and strengthen multilevel training for healthcare practitioners. Countries are also seeking to standardise care pathways, expand local access (particularly for children and young people suffering from ME/CFS) and ensure continuity from diagnosis to follow-up. However, severe and very severe ME/CFS remains inadequately served with limited access to tailored multidisciplinary care.

Societal and workplace recognition of ME/CFS is increasingly emphasised, and several countries are exploring strategies to support work participation and adjustments

Research priorities converge around coordinated cohort development, biobanking, translational studies, and pragmatic trials, supported by organisations such as NIHR and ZonMw. Stronger clinician-researcher integration and health-economic evaluation remain essential to inform effective and sustainable care models.

Similar priorities are evoked across the countries showing a shared consistent reform agenda. However, operational implementation, particularly for guideline alignment, severe-case care, and diagnosis to follow-up care remains a key challenge.

12.4 Organisation of healthcare services for long COVID

12.4.1 Introduction

The analysis was expanded to examine how healthcare services are organised for people living with long COVID to identify emerging approaches that may also be relevant to the organisation of care for ME/CFS. This extension was motivated by the substantial overlap between long COVID and ME/CFS, particularly in terms of symptom profiles, functional limitations, and the complexity of care needs over time. In Belgium, national guidelines on diagnostics, rehabilitation, follow-up and referral for patients with persisting symptoms after COVID-19 were developed for primary care in 2022. The guidelines recommend that patients, whether or not they were previously hospitalised, who have persistent symptoms for at least four weeks after an acute, confirmed COVID-19 infection consult a GP.¹⁶³

Depending on the level of clinical concern, the GP may provide information and tailored self-management advice, refer the patient for single-discipline treatment in primary care, or multidisciplinary care (when the clinical spectrum is complex, or when single-discipline care has proved ineffective). The GP plays a central role in supporting the patient, coordinating care, and providing follow-up consultations to reevaluate the patient's health status after referral to primary or secondary care. Informed by the guidelines, Belgium adopted a national guideline-based care pathway for long COVID in 2022.^{163, 164} The referral pathway is coordinated by a GP and includes the following services: physiotherapy and dietetics in the first line, occupational therapy and neuropsychology sessions.

In this section we provide a brief overview of similarities and differences with a selection of other European countries.

12.4.2 *Methods*

A search was conducted for documents reporting on the organisation of care pathways for long COVID in a selection of European countries including the Netherlands, France, UK, Italy (last search was performed on 30/10/2025). The search focused on key-publications from international institutions such as the OECD, EU, as well as national and government bodies documenting guidelines and recommendations for long COVID care organisation. The different approaches of how care pathways can be organised, especially focusing on adults, were summarised. We did not focus on whether and how these guidelines and recommendations are implemented in the specific country. No information was retrieved regarding care pathways for specific populations (such as children).

12.4.3 *Results*

Organisation of care pathways was summarised for each country including amongst others information regarding the used source, first point of contact, referral options, presence of specialised care facilities, coordination of care and services delivery ways. A summary of the care pathways can be found in Table 17.

12.4.3.1 *The Netherlands*

A care pathway guide has been developed by the ACTION research group.¹⁶⁵ The care pathway applies equally to people who were hospitalised or not. The pathway starts with consulting a GP, occupational physician, or relevant specialist for persistent symptoms.¹⁶⁵ Possible options are discussed with the patient, including tailored self-management advice, referral to diagnostic tests, and/or possible treatment options.

Care starts with a multidisciplinary consultation and the appointment of a care coordinator, such as a GP, physiotherapist, specialist, or another qualified professional.¹⁶⁵ The type of intervention (such as paramedical care, psychological care, rehabilitation, or referral to an expertise center) is determined selected according to symptom profile and severity, and treatment intensity (mono- or multidisciplinary care in primary or secondary settings) follows a stepped care principle.

Patients with persistent symptoms more than one year after a COVID-19, with functional loss of at least 25%, and who already followed a care pathway, may be referred to one of the six national long COVID centers of expertise.¹⁶⁶ These university specialised centres (i.e. Maastricht UMC, Amsterdam UMC, Erasmus MC (Rotterdam), LUMC (Leiden), UMCG (Groningen) and UMC Utrecht) opened in 2024.¹⁶⁷ An individual treatment plan is developed in line with the care pathway proposed by the ACTION research group.¹⁶⁵ In January 2026, a first version of the “Integrale Handreiking post-COVID” was published by the “Post-COVID Netwerk Nederland” with the aim of supporting healthcare professionals and social workers to implement the up-to-date scientific knowledge on long COVID and to facilitate connection and collaboration between professionals. The current version does not yet include information regarding healthcare organisation but this is planned for future updates of the document.^{168, 169}

12.4.3.2 *France*

In France, a care pathway guide has been developed covering adults and adolescents aged ≥ 15 years with long COVID and applies equally to those hospitalised or not.¹⁷⁰

(1) People with suspicion of long COVID are advised to first contact a GP who may refer the patient to other primary care practitioners (e.g. physiotherapists and psychologists, other specialists) and/or competence centers. The GP acts as the principal coordinator of care and follow-up, with support as needed from local long-COVID coordination units, competence centres, or rehabilitation teams.



(2) In case of a complicated situation (such as persistent limitations in daily life requiring multidisciplinary), patients may be referred to a competence center. Such centers can be a geographically defined place including all relevant professionals, or can be a network of different professionals working in the same territory. They deliver coordinated, multidisciplinary diagnostic and therapeutic care and monitoring within a coordinated care pathway. The GP remains the point of contact for the patient, bridging between patients and the competence center. Follow-up consultations are organised and can partly be provided virtually. The system is supported by long-Covid coordination units, which help professionals and patients to coordinate and organise complex care. Patients may be referred to these units by a care professional to streamline the care pathway, involving all regional organisations, including patient partners and associations.

(3) Patients may also be referred to multidisciplinary rehabilitation care (most severe cases), particularly those managed by specialist centres or experiencing significant physical, cognitive, or functional impairment. Following an initial functional assessment, an individualised rehabilitation plan is developed. Care may be delivered through day-hospital service or hospitalisation.

According to a recent OECD report based on the “2025 OECD Long COVID Mapping Policy Survey”, there is currently no official structure operational to coordinate the care according to the existing recommendations for the model of healthcare pathway.¹⁶⁹

12.4.3.3 *United Kingdom*

In the UK, NHS England has developed separate pathway guides for non-hospitalised patients (“Primary care/community long-COVID pathway”) and for those who were hospitalised (“Post-hospital discharge pathway”), informed by commissioning and service oversight since 2020.^{171, 172}

Individuals with new or persistent symptoms four weeks or more after a COVID-19, who were not hospitalised, are advised to first contact their GP. Cases with severe functional impact, lack of improvement or atypical presentations are referred to long COVID multidisciplinary team services, with onward referral to long COVID multidisciplinary rehabilitation (including physical, fatigue, breathlessness, psychological and vocational rehabilitation).

Care for long COVID is based on three crucial elements: (1) a patient-tailored care approach, (2) multidisciplinary support and rehabilitation, and (3) facilitation of self-care including digital supported self-management. Primary care professionals are central to the care pathway, guiding symptom management and determining when referral to long COVID services is appropriate. They are responsible for providing tailored self-management advice and signposting online resources, referring patients to vocational rehabilitation, psychological therapies and community-based support. They develop a tailored care plan together with the patient, based on the patient’s needs and priorities. In addition, they provide follow-up care after discharge from long COVID services and refer patients again to these services if necessary.

In 2021, multiple clinics across the UK were established to provide multidisciplinary care, diagnostics and rehabilitation for people with long COVID.^{171, 173} However, due to declining referrals and reduced funding, several clinics have closed, resulting in patients being referred back to primary care for case management.¹⁶⁹

12.4.3.4 *Italy*

No specific care pathway guide was identified but good practice recommendations have been published, stressing for example the need to provide multidisciplinary services with both remote and face-to-face consultations, and the importance of having a case manager to coordinate the patient’s care.¹⁷⁴ Over the years, a lot of specialised health services for people with long COVID have been initiated but the organisation of these services and the mode of delivery are heterogeneous.¹⁷⁵



Reported experiences on the characteristics of 124 long COVID centers in Italy, showed that most patients enter these centers after referral from primary care professionals or specialists, or after hospitalisation.¹⁷⁵ Most centers provide outpatient visits or day hospital services to patients. Almost all centers are in close contact with primary care physicians. Telemedicine tools are used, and a small percentage provide care at home. Most centers have a coordinator who is almost always a specialist (mostly respiratory or infectious diseases). There is a high variability regarding the number of specialisms available at these centers, ranging from one specialist to multiple specialisms. At least half of the centers have specialists in cardiology, respiratory diseases, radiology, infectious diseases, neurology and psychology.



Table 17 – Care pathways for long COVID in a selection of European countries

Country	Source	First point of contact	Referral options	Specialized care facilities	Coordination	Services delivery ways
Belgium	National guidelines on diagnostics, rehabilitation, follow-up and referral ¹⁶³	General practitioner	- Primary care professionals (mono- or multidisciplinary) - Secondary care professionals (mono- or multidisciplinary)	Several centers provide multidisciplinary services for long COVID	General practitioner	Face to face
The Netherlands	Care pathway guide developed by ACTION research group ¹⁶⁵	- General practitioner - Occupational doctor - Specialist (after hospitalisation)	- Primary care professionals (mono- or multidisciplinary) - Secondary care professionals (mono- or multidisciplinary) - National center of expertise (multidisciplinary)	Six national centers of expertise (only access after referral by general practitioner)	- General practitioner - Physiotherapist - Specialist	Face to face (Other ways not specified)
France	Care pathway guide developed by HAS ¹⁷⁰	General practitioner	- Primary care professionals - Specialists - Competence centers (multidisciplinary) - Multidisciplinary rehabilitation	Several competence centers which can be organised as: - Geographically defined place uniting all relevant professionals or - Network of different professionals working in the same territory	- General practitioner - Post-COVID coordination unit (currently not operational)	- Face to face - Teleconsultations and tele rehabilitation - Rehabilitation at home
United Kingdom	Care pathway guide developed by the NHS ^{171, 172}	- Non-hospitalised: General practice team - Hospitalised: specialist post-discharge	- Specialists - Post-COVID multidisciplinary services - Post-COVID multidisciplinary rehabilitation	Several centers provide multidisciplinary services for long COVID	Not specified	Specialist review after hospitalisation can be held virtually or face to face
Italy	Heterogeneous modes of delivery without structural organisation ¹⁷⁵	General practitioner	Long COVID centers (mono or multidisciplinary)	Long COVID centers	Coordinator long COVID centers by a specialist	20 % of long COVID centers use telemedicine tools



12.4.4 Discussion

In all selected countries, the GP was advised as a first point of contact for people with persistent symptoms after an acute COVID-19, which is in line with other OECD countries.^{176, 177} Depending on the patient's need and complexity of the clinical presentation, the GP can provide education and tailored self-management advice or can refer the patient to other primary or secondary care professionals, social support and community-based support services. Depending on their needs, patients can be referred to multidisciplinary rehabilitation or in some countries to expertise centers for long COVID.

In all countries, the importance of a multidisciplinary approach is recognised, which is in accordance with NICE, EU and WHO guidelines and expert panels.^{171, 178-180} In general, multidisciplinary teams comprise at least two distinct disciplines and may encompass a broad range of specialties, including occupational therapy, physiotherapy, neuropsychology, and medical specialties such as pulmonology, cardiology, rehabilitation medicine, and neurology. Teams also include professionals providing social support and patient partners trained in patient education.

All countries have multiple centers providing multidisciplinary long COVID services but only the Netherlands has several national centers of expertise. In the UK, a separate care pathway has been developed for hospitalised and non-hospitalised patients. The care pathway was mostly coordinated by the GP. Since long COVID management often involves a complex care pathway, a clear contact point and coordinator is advised to guide patients and professionals.^{178, 180}

According to the recent OECD report on long COVID, most surveyed OECD countries including France and The Netherlands reported the use of specialised clinics to organise care for people with long COVID.¹⁶⁹ These mostly regionally organised clinics are equipped with multidisciplinary teams, hospital medicine specialists or primary care professionals. In 2025, several countries (not including France and the Netherlands) reported that specialised clinics have been closed due to a lack of funding, integration into general healthcare pathways, or a shift towards classifying patients under broader categories such as post-acute infection syndrome. Four countries, including Belgium, reported they do not have specialised long COVID healthcare structures. While most OECD countries have developed clinical guidelines for the management of long COVID, only six countries (including Belgium, France and the Netherlands) reported the existence of an official healthcare pathway for people with long COVID.

◆ GENERAL DISCUSSION AND CONCLUSION

This discussion brings together findings from the assessment of health-related unmet needs (based on the survey and interviews) and the second part of the study, which includes the literature review and international comparison. The aim is to interpret the results, outline their implications for service design, and highlight priorities for future research. To achieve this, the discussion takes a multilevel perspective, addressing issues at the micro (patient), meso (care organisation), and macro (societal and policy) levels.

13 SUMMARY OF FINDINGS

- In the first part of the study, a mixed-methods design was used to assess health-related needs. Quantitative data were collected through a large online patient survey involving 749 individuals with ME/CFS in Belgium, complemented by 19 in-depth qualitative interviews. The survey sample was predominantly female, with a mean age of 50 years, and most respondents lived in Flanders. About half had received their diagnosis more than ten years earlier. Comorbidities were common, particularly fibromyalgia and irritable bowel syndrome.

Across all domains, the findings point to a substantial and multidimensional burden. Survey respondents reported a marked decline in HRQoL, a high symptom burden, reduced autonomy, and major impacts on social life, employment, and financial security. The qualitative interviews provided important context, illustrating how the condition affects every-day life, the coping strategies people adopt, and the emotional and social consequences of living with a condition that is often contested and insufficiently recognised.

Participants also reported major challenges related to the organisation, coordination, and accessibility of care. These organisational issues were further detailed in the interviews, where respondents described a lack of professional expertise and reliable information, limited recognition of their condition, and an absence of specialised care. Beyond organisational barriers, participants expressed a strong unmet need for effective treatments and, more broadly, for increased scientific research.

- In the second part of the study, we reviewed the literature on care organisation to determine whether specific service structures are associated with improvements in symptom burden and/or overall health status. Although a multidisciplinary approach is generally considered appropriate, given the complexity, the fluctuating and evolving course, and frequent comorbidities of the condition, no robust evidence was identified to support the superiority of any specific, predefined service model. Consistent with this, the most recent NICE guidelines (2021) also highlight the limited evidence base for defining formal care pathways. They recommend prioritising clinician expertise over fixed multidisciplinary team configurations and emphasise the need for flexible, individualised care pathways that can accommodate heterogeneous presentations and comorbidities.

This review was complemented by an analysis of care organisation in selected European countries, which revealed considerable variation in ME/CFS service provision. The UK emerged as the only country with a large nationally organised care model. In parallel, some countries have initiated policy actions aimed at improving recognition and support for people with ME/CFS, in certain cases in connection with broader post-viral illness or long COVID initiatives. Dedicated research programmes have been launched to improve the understanding of the condition, although clinical research activity and service development remain limited as some initiatives remain temporary, pilot-based, or subject to changing policy priorities across Europe.



14 PATIENT UNMET HEALTH NEEDS

Survey respondents reported a large and statistically significant decline in EQ-5D-5L utility scores, from a mean of 0.84 prior to symptom onset to 0.36 at the time of the survey, indicating a very severe deterioration in quality of life. This score is substantially lower than the mean EQ-5D-5L score observed in the Belgian general population (0.79), as reported in the 2018 Sciensano^{rrrr} Health Interview Survey.¹⁸¹ These findings are consistent with a recent multinational cross-sectional study, with a broadly comparable methodology and sample (1418 patients) which reported very low scores on the EQ-5D-3L, recorded as a visual analogue scale (mean 33.7; standard deviation 17.5), also indicative of a high disease burden.¹¹¹ Another study using the EQ-5D-5L further corroborated these findings on 198 people living with ME/CFS (mean: 0.46; 95% confidence interval: 0.42-0.50)¹⁸²

Overall, ME/CFS is associated with a very high disease burden, comparable to conditions widely recognised as severely disabling. For example, a study using the SF-36 in patients with ME/CFS showed profound impairments in functional status, with significantly worse physical and social functioning compared to individuals with multiple sclerosis.¹⁸³ Quality-of-life scores have also been shown to be markedly lower than those observed in other major chronic diseases.¹⁸⁴ Of note, while a recent systematic review suggests that the burden of illness in ME/CFS is very similar to that observed in long COVID¹¹², the 123 participants in our survey who self-reported long COVID demonstrated a more pronounced decrease in HRQoL (-0.55 vs -0.45, $P=0.007$).

Nearly all survey respondents reported on symptoms such as abnormal fatigue, PEM, sleep disturbances, and pain, which were predominantly categorised as very or extremely burdensome. Consistent with this, the EQ-5D-5L results indicated impairments across all dimensions, with the most pronounced deficits observed in usual activities, pain/discomfort, and mobility. Furthermore, sexual and reproductive health were also adversely affected.

Subgroup analyses particularly showed that greater declines in HRQoL were associated with diagnostic delays and a higher burden of symptoms. Particularly PEM is strongly associated with reduced QoL,¹⁸⁵ making day-to-day management extremely challenging. PEM was reported by nearly all participants in our sample, and participants described their lives as a series of “forced choices”, often restricted to a small number of essential activities per day (as pacing techniques are necessary to perform all the activities).

The impairment of mental health was more moderate, characterised primarily by a shift from no or minimal issues to moderate levels of distress for a majority of participants. This predominance of physical over mental health impairment has previously been reported by Nacul *et al.* using the SF-36 questionnaire¹⁸⁴, or by Vyas *et al.*, who similarly identified pronounced impairments in activities of daily living, pain management, mobility, and self-care, contrasting by a comparatively lower impact on anxiety levels.¹¹¹ The interviews conducted in our study suggest that the psychological burden of ME/CFS is secondary to the physical and social consequences of the illness. Systemic factors such as medical inadequate understanding and societal stigmatisation of ME/CFS have, indeed, been identified as primary drivers of secondary mental health complications.^{110, 186} Additionally, the participants of our study reported that the ongoing necessity to adapt activities and manage limited energy levels (‘pacing’) reflected the substantial burden associated with ME/CFS and may contribute to secondary emotional distress in some individuals. Moreover, individuals who were interviewed in our study reported experiences of psychologisation of their symptoms by others. As highlighted in the literature, the limited understanding of the aetiology and mechanisms of ME/CFS has contributed to tensions between patients and health professionals and to the reported experiences of stigmatisation, psychologisation, minimisation and underestimation.¹⁸⁷

^{rrrr} <https://www.sciensano.be/fr>

15 PATIENT UNMET HEALTHCARE NEEDS

15.1 Perceived effectiveness of management and associated burden

Survey respondents reported on the perceived effectiveness and burden of interventions received over the past five years, revealing substantial inter-individual variation. Notably, a non-negligible proportion of respondents reported worsening following several approaches. While respondents reported generally positive experiences with physiotherapy (overall and with pacing techniques), experiences with GET and rehabilitation programmes were predominantly perceived as harmful.

However, an important limitation of our analysis is that we cannot fully and comprehensively characterise the 26% of respondents who reported moderate or substantial improvement with GET, including the illness severity, intervention characteristics, implementation context, or illness trajectory. This heterogeneity limits interpretation of perceived benefit and highlights the complexity of individual responses to physical activity-based interventions. NICE therefore emphasises that any physical activity or exercise programme should only be considered on an individual basis (for those who feel ready to progress their physical activity beyond their current activities of daily living or feel able to incorporate exercises into managing their ME/CFS) and delivered by professionals with appropriate expertise and understanding of the associated safety considerations.³ It has also been shown that the exertion threshold that triggers PEM varies between patients, further underscoring the importance of personalised approaches.¹⁸⁸ Qualitative findings of our study further underscore the need for more personalised care, as symptoms tend to be managed in a uniform manner, without adequate consideration of the underlying pathology. In other words, similar symptoms appear to be treated identically across conditions, without the implementation of distinct, condition-specific approaches. Importantly, the interviews also added nuance by emphasising that the burden extends beyond direct adverse effects. For examples, strategies such as pacing, while often necessary to prevent PEM exacerbation, may also contribute to reduced spontaneity and increased social isolation in some individuals.

Beneficial effects were reported for psychological counselling (i.e. approximately half described improvement, and around one third reported no change). The experiences with CBT were more mixed: while a majority reported a feeling of improvement, nearly one in five reported a feeling of deterioration, and around one third reported no change. Although NICE guidelines do no longer routinely recommend CBT as a treatment for ME/CFS, it may be considered only as an adjunctive therapy, in the context of suffering from a chronic illness.³ In our survey, CBT was generally not associated with a high treatment burden, and a subgroup reporting perceived effectiveness was identified. However, the subgroup characteristics were not examined due to limited clinical data and the potential confounding effects such as concurrent treatments or comorbidity or moderating effects such as variations in disease stage. In this context, a recent meta-analysis indicates that the effects of CBT on fatigue severity are moderated by factors such as age, functional impairment, and physical activity patterns, underscoring the heterogeneity of treatment responses.⁹⁸ Improvements were also reported among respondents following specific dietary regimens, although our survey did not capture details of these diets. By contrast, prolonged antibiotics use was associated with a substantial treatment burden and a low self-reported effectiveness.

15.2 Diagnostic trajectory

Survey data indicated a substantial diagnostic delay, with nearly three quarters of respondents waiting more than one year, and 39% waiting more than five years, from first medical consultation to diagnosis. The qualitative analysis contextualised these findings describing repeated consultations, misattribution of symptoms (often to psychological explanations or other conditions), extensive investigations without resolution, and the diagnosis frequently made by exclusion of other potential conditions. These experiences were perceived as physically and emotionally burdensome. This finding



aligns with a recent large UK study showing substantial variation in diagnostic delay: 22.1% of participants received a diagnosis within 1-2 years of symptom onset, whereas 12.9% waited more than 10 years. That study also reported that a higher number of comorbidities was significantly associated with longer diagnostic delay, suggesting that greater clinical complexity may hinder timely recognition of ME/CFS.³³

International recommendations emphasise the importance of early diagnosis to support appropriate management.^{3, 7, 12, 16} In line with this shift, the NICE guidelines reduced the symptom duration threshold for referral to a specialist team from 6 to 3 months.³

Improving the clinical recognition of ME/CFS requires increased awareness among healthcare professionals and, more broadly, within society. In the UK, diagnosis is confirmed by specialist reference centres following referral by the GP. Our international comparison illustrated that (postgraduate) knowledge-dissemination programmes have been developed in the Netherlands and the UK to enhance clinical competencies, promote early referral, and align clinical practice with current evidence and guidelines. In the UK, this effort has been reinforced through the *2025 Delivery Plan*, which includes the rollout of e-learning resources and collaboration with the Department for Education to explore integration into undergraduate curricula.¹⁵⁹ The need to improve professional knowledge has also been indicated by an international survey conducted within the EUROMENE network.¹³¹ In the UK, the BACME association provides materials to support dissemination of information regarding diagnosis and management.⁶²

A further contributor to diagnostic delay is the inconsistent application of diagnostic criteria, which represents an additional barrier to early recognition.¹⁶ This lack of standardisation also has downstream consequences for research and knowledge development.¹⁸⁹ Our international comparison revealed that most of the countries examined lack national clinical guidelines, and in case guidelines do exist, they are frequently outdated. Similar findings were reported by the EUROMENE group in a survey in 17 countries, prompting calls for strengthened regulations, guidance, and standards for the diagnosis and management of ME/CFS.¹³⁰

Finally, continued research, and particularly translational research, is also essential to advance understanding of the pathophysiology of ME/CFS and, in turn, improve clinical recognition.¹⁹⁰ Further research on diagnostic biomarkers remains essential. Although no biomarker suitable for routine clinical use has yet been identified, recent research, accelerated in the wake of the COVID-19 pandemic has begun to yield new insights into the biological hallmarks of ME/CFS. Multiple research teams are currently working on the development of potential biomarkers.^{54, 56, 191}

15.3 Coordination of care and expertise

Half of respondents reported that care was rarely or never well organised and planned over time, and two-thirds indicated that no health professional was responsible for coordinating their care. The qualitative analysis illustrated how these organisational gaps may shift responsibility onto patients, who are required to identify appropriate specialists, seek referrals, pursue investigations and assemble their own care networks.

Our findings further highlighted a widespread lack of expertise among healthcare professionals, with only a minority of participants reporting that they received clear information about the disease and its treatments. Poorer care experiences were associated with lower HRQoL (see 9.2.1.1). Interviewees corroborated perceived gaps in clinicians' knowledge of ME/CFS and diagnostic skills, which contributed to limited information provision, use of ineffective management strategies, and scepticism or prejudicial attitudes. This pattern is well documented in the literature and includes the tendency to attribute ME/CFS symptoms to psychological conditions such as depression or burnout, resulting in inappropriate management.¹⁹²

Indeed, a literature review conducted by the EUROMENE expert group identified limited acceptance of the ME/CFS diagnosis among GPs and low confidence in diagnosing and managing the condition, even when it was recognised.^{161, 193} These shortcomings have important downstream consequences,

contributing to underdiagnosis, diagnostic delay, and challenge in estimating disease prevalence and the burden of ME/CFS.¹³¹

15.4 Forgone care and barriers

Nearly half of participants reported having forgone care in the previous 12 months, most commonly due to a lack of qualified staff, insufficient recognition, long waiting times, and limited information. Access barriers were further compounded by restricted specialist provision, particularly language and distance constraints related to the geographical location of specialist services. This is reflected by the fact that Belgium has a single ME/CFS centre offering services reimbursed by the NIHDI, which is located in the Dutch-speaking part of the country. Other centres based in large hospitals exist but have not adhered to the NIHDI agreement, meaning that services are not fully reimbursed. The interviews additionally highlighted limited flexibility in care delivery, including limited possibilities to reschedule appointments and an unmet need for teleconsultations or home visits. A recent survey conducted in Germany indicated comparable findings, with participants describing medical underservice, citing geographical and financial barriers, low satisfaction with physician care, and a scarcity of dedicated ME/CFS centres.¹⁶⁰

This situation is generally similar to that observed in several of the countries included in our international comparison. Indeed, in France, Italy, and the Netherlands, access to care largely depends on a limited number of clinician-led centres of expertise, as an organised national service structure is absent. Such models may exacerbate geographical inequalities, increase the risk of delayed diagnosis, and adversely affect the quality and continuity of follow-up care. Similarly, a Norwegian qualitative study based on in-depth interviews with 48 participants identified similar structural shortcomings, including the absence of appropriate services after diagnosis, long waiting times, and a lack of personalisation (i.e. services not tailored to patients' specific physical limits and symptom profiles), and limited knowledge about ME/CFS among healthcare professionals.¹⁹⁴ This scarcity of services may also drive patients to seek alternative care, increasing the risk of exposure to fraudulent or exploitative practices which is an issue that was also reported in our interviews (see 9.3.3.7).

15.5 Ethical aspects

From an ethical point of view, there are two considerations to be made. First, on the individual level, it is important to balance the risks of undertreatment and overtreatment. Achieving this balance can be facilitated by a process of informed shared decision-making between healthcare professionals and patients. Patients should be provided with transparent information about available treatment options, including their potential benefits and harms and the level of scientific knowledge about the different treatment options. Given the heterogeneous nature of the disease, treatment should be tailored to the individual. This necessitates a high level of expertise among healthcare professionals, including up-to-date knowledge of the scientific evidence on available treatments, their expected effects, and their potential risks.

Secondly, on the societal level it is important to balance the use of treatments without solid scientific evidence base but potential (yet unknown or uncertain) benefits for individual patients with the opportunity costs within the healthcare system. Because not all therapeutic options are supported by robust evidence, non-evidence-based interventions (e.g. emerging or promising treatments that have not yet been validated through sound clinical trials) may entail societal costs and generate issues of distributive justice. Experienced healthcare professionals may be able to identify clinical patient subgroups that are more or less likely to benefit from specific interventions, thereby reducing potential inefficient use of scarce healthcare resources. This underscores the need for continued research into the effectiveness of treatment strategies, as well as the importance of strengthening education and training opportunities for healthcare professionals to provide them tools in managing this complex condition.



16 PATIENT UNMET SOCIAL NEEDS

Our analysis suggests that ME/CFS is a profoundly socially disabling condition, marked by disruption of family and social networks, adverse effects on education and employment trajectories, and substantial financial consequences.

16.1 Social isolation

A substantial burden on social relationships was reported, with approximately three-quarters of respondents indicating that ME/CFS significantly affects relationships with family members, partners, and friends. Qualitative findings further suggested that relationships may deteriorate following misunderstandings, difficulties in mutual understanding, and challenges in maintaining or forming relationships, leading to progressive social isolation. This is consistent with findings from a recent international survey underscoring the contribution of social isolation to reduced self-perceived QoL.¹¹⁰ In line with these findings, Wotherspoon *et al.* (2023), proposed a conceptual model of social isolation in ME/CFS based on the experiences of 42 individuals¹⁹⁵ identifying three key mechanisms: (1) spatial and temporal restrictedness due to limited energy or disability; (2) communicative alienation arising from difficulties in conveying illness experiences; and (3) discreditation driven by disbelief and stigma. When suffering from a legitimised illness, patients are exempted from fulfilling social roles (mostly the professional role by means of sickness leave). However, for those suffering from 'contested' illnesses such as ME/CFS, this exemption is part of a long struggle with little support from the healthcare and social systems. From a Parsonian perspective, this reflects a form of limited access to the sick role, commonly observed in under-medicalised conditions like ME/CFS, where recognition by medical authority is a prerequisite (i.e. the rights and obligations associated with being ill).^{196, 197}

16.2 Impact on work

ME/CFS has a major impact on employment, with most respondents reducing their working hours or ceasing working altogether, accompanied by a marked shift towards disability status. Indeed, as seen in the data from NIHD (see 4.2), ME/CFS is associated with profound and long-lasting productivity losses, characterised by early sustained withdrawal from the labour market and long durations of invalidity. We identified an association between symptom burden and cessation of work, which is unsurprising. In addition, we observed an association between the impact on personal relationships and stopping work, highlighting the social dimension of employment and its role in societal participation. A notable contribution of the qualitative analysis involved employers' reactions to workers with ME/CFS ranged from supportive accommodations (e.g. flexible energy-based schedules and telework) to inflexible "all-or-nothing" expectations. Participants described these responses as critical in determining whether any form of employment could be sustained. Overall, our findings align with the existing literature indicating that many people become unable to work due to ME/CFS and that the economic burden attributable to productivity loss is substantial.^{11, 198-200} Moreover, a recent qualitative analysis illustrated the challenges of working with ME/CFS, including a weakened professional identity, communication difficulties, and limited workplace accommodations, alongside the need for continual adjustments to manage fluctuating symptoms and maintain work participation where possible.²⁰¹ It is pertinent to note that interviewed participants in our study expressed a strong willingness to return to work.

16.3 Financial consequences

Consequently, financial impacts were common and often severe, with a majority reporting difficulty meeting expenses with household income, or assistive supports. The qualitative findings suggest that financial insecurity was not solely a downstream consequence of reduced employment, but also reflects out-of-pocket expenditure for therapies, compounded by limited reimbursement or insurance coverage. These factors may amplify inequities between patients with different socioeconomic

backgrounds.²⁰² Similar concerns were highlighted by the EUROMENE socioeconomic working group.²⁰⁰

Although our international comparison did not examine welfare benefits or disability recognition, it identified emerging policy initiatives in several countries aiming at improving financial support for people with ME/CFS and facilitating work reintegration through adapted employment arrangements and guidance for employers. Notably, the UK Delivery Plan-2025 includes measures to support return to work for people with ME/CFS and aligns this with ongoing reforms to disability benefits and employment support.¹⁵⁹

16.4 Burden on caregivers

It is worth noting that our study also addressed unmet societal needs, although only with respect to the impact on informal caregivers. We found that ME/CFS is associated with a marked deterioration in family members' QoL. Although caregiver burden was assessed only in a limited manner in this study, these findings suggest substantial effects of ME/CFS on household health and functioning, consistent with the existing literature.^{203, 204}

16.5 Prioritisation of needs: research and effective treatments

In the survey question asking respondents to prioritise their needs, it is important to note the prominence given to effective treatments and to scientific research. These two categories are inherently linked, as no treatment currently exists that can modify the course of the disease or lead to sustained improvement. This suggests that unmet needs are likely to remain substantial unless both research and service capacity expands.

In our international comparison, we identified that several countries have established coordinated research programmes. Notably, the UK and the Netherlands have provided multidisciplinary funding to support programme-based research. For example, the NIHR has funded translational projects such as the DecodeME study.⁵³ The development of translationally oriented projects is essential to build effective interactions between researchers and clinicians. Advancing knowledge of the biological bases and mechanisms underlying symptom development is essential to enable the subsequent development of potential treatments. Furthermore, involving people with lived experience from the study design stage throughout the research process helps to ensure that the research agenda remains aligned with real-world needs.

Finally, identifying patient subgroups linking biological markers to symptomatology would make it possible to deliver more personalised care. In this line, linkages with post-infectious syndromes, such as long COVID may shed light on common pathophysiological mechanisms.³²

17 STRENGTHS AND LIMITATIONS

To our knowledge, this is the first study in Belgium to comprehensively assess the health-related unmet needs of people with ME/CFS using a mixed-methods design combining quantitative survey data with qualitative interviews. A key strength of this study was the use of the NEED framework, a methodology validated with both policymakers and patients. This framework enabled a comprehensive, multi-dimensional assessment of patient needs, reducing the risk of omitting relevant domains. Although the resulting presentation of findings may appear fragmented, this structure was necessary to ensure that the data collected are transparently reported and readily accessible for international sharing and comparison.

Another strength of the study lies in the active involvement of patients, represented by four patient associations, who contributed throughout all phases of the research via a structured patient-involvement process. This approach aligns with recommendations advocating patient engagement to ensure that research remains responsive to patient needs and priorities.²⁰⁵



The questionnaire was adapted from the generic NEED adult patient questionnaire, originally developed by the KCE. It was tailored to the condition under study in collaboration with patient associations. Survey dissemination occurred primarily through these organisations' networks. A potential limitation of this sampling approach, which relied predominantly on active dissemination through patient networks, is that participant inclusion may not have resulted from a random process, with implications for external validity. Indeed, individuals who are not members of patient associations may have had less opportunity to access the study, and the sample may therefore be more likely to reflect the experiences and perspectives of affiliated members rather than those of the wider patient population.²⁰⁶ Nevertheless, despite this potential sampling bias, our findings are broadly consistent with the existing literature.

An additional limitation, inherent to this type of design, is the risk of misclassification bias, as diagnoses were self-reported and not independently verified. Although eligibility required a prior diagnosis of ME/CFS, diagnostic accuracy cannot be fully assured. However, it should be noted that based on responses to several open-ended questions, we excluded some participants who indicated that their diagnosis had not been medically confirmed.

Several potential sources of measurement (information) bias should be considered when interpreting these self-reported data. Perceived treatment "effectiveness" was self-reported across heterogeneous interventions (e.g. different durations, settings, and participant clinical profiles) and is therefore best interpreted as reflecting patient experiences rather than clinical efficacy. As such, these findings should not be used to infer comparative effectiveness between treatments, which remains debated for some approaches. In addition, cognitive function was not formally assessed among a sample of patients with ME/CFS to prepare the questionnaire used for the survey. However, the questionnaire was reviewed by the patient associations.

As with all recall-based online surveys, our findings related to the retrospective questions (inquiring about the start of their symptoms) may be affected by recall bias, with respondents potentially over- or under-reporting information depending on its salience or changes in their condition over time, particularly given that more than half of participants had been living with ME/CFS for over a decade.

Our study is prone to selection bias since the study sample is not fully representative for the target population, being all people living with ME/CFS. First, the study excluded children and younger adolescents. Consequently, our findings are not generalisable to paediatric ME/CFS populations, whose clinical presentation, care pathways, and responses to interventions may differ from those observed in older adolescents and adults.

Second, survey participation may have been affected by digital exclusion, as some individuals may have lacked the digital literacy required to complete an online survey, introducing potential selection bias. In addition, the questionnaire was lengthy, included highly specific questions, and was available only in French and Dutch. It is also worth mentioning that the survey was not available in German.

Third, the sample exhibited a marked linguistic imbalance, with Dutch-speaking participants overrepresented relative to French-speaking respondents, likely reflecting the predominance of patient organisations based in Flanders. The absence of dedicated ME/CFS infrastructure in the French-speaking regions may also have contributed to lower visibility and participation in that region. Although, ME/CFS is more prevalent in women than in men, a strong imbalance in sex was observed in our survey with women being predominantly represented (89.9%).

Fourth, the study may not have adequately reached more socially isolated individuals or those from lower socioeconomic backgrounds.

Individuals with the most severe symptomatology may have been unable or unwilling to participate because illness-related limitations, resulting in the likely underrepresentation of those with the greatest disease burden and unmet needs. With respect to interview sampling, we used ME/CFS severity as a selection criterion, but we did not specifically recruit individuals with very severe disease. This subgroup remains particularly challenging to manage and is underrepresented in research. While we

planned a chat-based interview option for individuals with extreme severity, this alternative was not used, consistent with the limited inclusion of very severely affected participants. The needs of this subgroup may differ markedly and service design and accessibility are therefore insufficiently explored.²⁰⁷ Moreover, people with severe ME/CFS may develop distrust following contact with healthcare professionals.^{141, 143}

Finally, because interview participation was voluntary, self-selection may have favored individuals with more pronounced experiences. As we did not compare interviewees with non-interviewees, we cannot assess whether the interview subsample differed systematically. Nevertheless, the interview findings broadly align with the existing literature on unmet needs in ME/CFS, which supports the credibility of our findings.^{194, 208-210}

18 POTENTIAL SUGGESTIONS TO TACKLE UNMET NEEDS OF PEOPLE LIVING WITH ME/CFS

Within our framework for assessing unmet needs, equity was treated as a transversal dimension, across health, healthcare, and social dimensions. Equity is understood here as the absence of unfair and preventable differences in health outcomes, access to and quality of healthcare, and social impacts between population groups defined by social, economic, demographic, geographic, or other characteristics.¹²²

Applying this to the online survey and interview data highlighted several recurrent mechanisms through which people living with ME/CFS perceived inequity. This equity-focused synthesis provides a coherent summary of the study's overall findings, as the issues identified clinical, institutional, and societal levels of experience and is in line with the overall literature on experiencing ME/CFS.^{194, 208-210} First, participants emphasised a substantial need for societal and institutional recognition of ME/CFS. Its perceived invisibility ('invisible disease') and the lack of a coordinated healthcare response were described as contributing to unequal care pathways and access to appropriate support. Second, respondents situated their experiences within a perceived societal climate towards chronic illness, including pressures to resume work and productivity, which were experienced as socially unjust and as exacerbating vulnerability. Third, reports of stigmatisation in encounters with health professionals, within social networks, and in society were prominent. Moreover, psychologisation was experienced as inequitable when compared with more consistently biomedical diagnostic and therapeutic trajectories for other conditions. Finally, interviewees described simplification, underestimation, or denial of illness severity, sometimes accompanied by refusal of further medical care, as well as a perceived lack of scientific interest in ME/CFS.

The results of our study demonstrate that misrecognition of ME/CFS is not merely a social or clinical misunderstanding but a structural determinant that shapes health outcomes, quality of life, and socioeconomic stability. Misrecognition at several levels (clinical, institutional, and societal) induces cascading consequences across other levels. Breaking this cycle requires coordinated efforts across macro, meso, and micro domains, grounded in updated scientific knowledge, integrated care structures, and respectful, patient-centered approaches. Informed by the study findings, the following suggestions proposed below highlight policy-relevant implications to strengthen care and support for people living with ME/CFS:

- Strengthen recognition and visibility of ME/CFS among healthcare professionals and the general public in order to reduce stigma and its downstream consequences. Awareness should also extend to the impact on patients' relatives and informal caregivers.
- Improve early diagnosis of ME/CFS by enhancing clinical knowledge on ME/CFS by including and expanding this topic in the training and lifelong learning of healthcare professionals and ensuring regular updates in line with emerging evidence.
- Advance scientific understanding of disease mechanisms, evaluate the effectiveness of management strategies, and identify therapeutic avenues. This requires a research agenda



combining translational approaches (bringing together clinicians, researchers, and patients), with experimental methodologies, including clinical trials. Potential overlaps with other post-infectious syndromes, including long COVID, should be explicitly considered to promote synergies and joint projects.

- At the level of care organisation, there is a clear need for structured care pathways, including reference centers aligned with current scientific knowledge and ensuring multidisciplinary collaboration with clearly defined coordination between primary and secondary care. This need is reinforced by qualitative findings showing that patients often have difficulties to identify healthcare professionals with adequate knowledge of the condition, experience insufficient medical guidance, and may sometimes receive inappropriate care. In this context, identifiable care trajectories and access to specialised expertise are essential to support the development of personalised care plans. These pathways could be developed within, or anchored by, specialist reference centres that consolidate the expertise required for holistic management, facilitate access to relevant specialties according to symptom profiles, and incorporate the necessary psycho-social support services. Although we found no evidence to endorse any single service model, current recommendations emphasise the importance of strengthening ME/CFS-specific clinical expertise and delivering personalised care.¹⁶ Given the overlap with post-infectious conditions such as long COVID, including shared features like PEM and dysautonomia, opportunities to develop shared expertise and, where appropriate, coordinated service provision should be explored.
- When feasible, support job retention workplace adaptation, by promoting flexible working arrangements. Employers should be provided with recommendations for reasonable adjustments that explicitly accommodate patients' energy limitations.

19 CONCLUSION

This study highlights the burden of ME/CFS and the multifaceted associated unmet needs of individuals with ME/CFS in the health, healthcare and social domains. It reveals the broad spectrum of unmet needs across health, healthcare, and social domains. The findings show that these unmet needs span multiple interconnected areas, where difficulties in mental health and social participation often arise as consequences of severe physical and functional limitations. Limited recognition and understanding of ME/CFS at societal, clinical, and institutional levels remain major contributors to persistent unmet needs. Fragmented care structures and the lack of coordinated national pathways and reference centers further exacerbate patient vulnerability and complicate access to appropriate services. A holistic approach is therefore essential to adequately address the multidimensional nature of ME/CFS in clinical practice. Strengthening system-wide capacity through integrated, patient-centered care models is crucial to improve outcomes and reduce long-term burden for individuals with ME/CFS.

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